CUI

Lab Manual

CSE494-Software Project Management

Department of Computer Science

Islamabad Campus

Lab Contents:

The topics include: Introduction to various Project Management Software; Getting started with MS Project professional and Trello; Generating Project Professional Views and Reports; Applying Project Professional Filters and Creating Tasks List; Develop Work Break Down structure using MS project professional and Trello; Developing the schedule and Establishing Task Dependencies using MS project professional and Trello; Developing Gantt chart, Network Diagrams, and perform Critical Path Analysis using MS Project Professional; Applying different Cost Estimation Techniques; Applying Quality Management techniques; Establishing Communication Management techniques in Trello; Project Risk management Techniques; and Setting-up Resources for Project Human resource Management.

Student Outcomes (SO)

S.#	Description
1	Apply knowledge of computing fundamentals, knowledge of a computing specialization, and mathematics, science, and domain knowledge appropriate for the computing specialization to the abstraction and conceptualization of computing models from defined problems and requirements
2	Identify, formulate, research literature, and solve complex computing problems reaching substantiated conclusions using fundamental principles of mathematics, computing sciences, and 3 relevant domain disciplines
3	Design and evaluate solutions for complex computing problems, and design and evaluate systems, components, or processes that meet specified needs with appropriate consideration for public health and safety, cultural, societal, and environmental considerations
4	Create, select, adapt and apply appropriate techniques, resources, and modern computing tools to complex computing activities, with an understanding of the limitations
5	Function effectively as an individual and as a member or leader in diverse teams and in multi-disciplinary settings

Intended Learning Outcomes

Sr.#	Description	Blooms Taxonomy Learning Level	SO
CLO -5	Develop a comprehensive software project plan.	*Creating*	2,4
CLO -6	Build and implement a project using different project management software's.	*Creating*	2-5

Lab Assessment Policy

The lab work done by the student is evaluated using rubrics defined by the course instructor, viva-voce, project work/performance. Marks distribution is as follows:

Assignments	Lab Mid Term Exam	Lab Terminal Exam	Total
25	25	50	100

Note: Midterm and Final term exams must be computer based.

List of Labs

Lab 01

Introduction to various Project Management Software

Objective:

The objective of this lab is to introduce students to various project management tools which will help equipping them with the knowledge and skills necessary to effectively plan, execute, monitor, and control projects using modern technological solutions.

Activity Outcomes:

After the successful completion of this lab, students will become familiar with a variety of project management software tools, enabling them to adapt to different workplace environments that may utilize different tools.

Instructor Note:

As pre-lab activity, read Chapter 1 from the book PMBOK: Ch1.

1) Useful Concepts

Various project management tools offer a wide range of features and concepts that enhance project planning, execution, monitoring, and collaboration. All project management tools offer different features, approaches, and strengths. Some of the tools offering different features, which are extensively used in software industry, are discussed:

1. Trello:

Concept: Trello uses Kanban boards to organize tasks into customizable columns.

Strengths: It's simple and intuitive, ideal for smaller teams and agile projects.

Focus: Visual task management and collaboration.

2. Microsoft Project Professional:

Concept: Microsoft Project is a comprehensive project management tool for planning, scheduling, and resource management.

Strengths: Provides advanced scheduling, resource allocation, and reporting capabilities.

Focus: In-depth project planning and management.

3. Jira:

Concept: Jira is tailored for software development teams using agile methodologies.

Strengths: Offers customizable workflows, issue tracking, and integration with development tools.

Focus: Agile project management and issue tracking.

4. Asana:

Concept: Asana focuses on task tracking, project planning, and team communication.

Strengths: Offers robust collaboration features, timeline views, and detailed task management.

Focus: Team communication and work tracking.

The Project Management tools are not limited to the ones discussed above, there are several other tools available. But in this course we will focus on MS Project Professional and Trello because Trello and Microsoft Project are both highly regarded project management tools, but they serve different purposes and excel in distinct areas.

2) Solved Lab Activities

Sr.No	*Allocated Time*	*Level of Complexity*	*CLO Mapping*
1	*50*	*Medium*	

Activity 1:

Compare why MS Project Professional and Trello are considered best project management tools?

Solution:

Here's a comparison of why Trello and Microsoft Project are considered among the best tools in their respective categories:

Trello:

Simplicity and Ease of Use: Trello's simplicity and visual interface make it easy for individuals and small teams to quickly set up and manage projects without a steep learning curve.

Agile and Visual Task Management: Trello's Kanban board layout is ideal for agile methodologies, enabling teams to visualize work stages, track progress, and manage tasks collaboratively.

Flexibility and Adaptability: Trello's customizable boards and cards allow you to tailor the tool to match your workflow, making it suitable for various types of projects and teams.

Collaboration and Communication: Trello fosters real-time collaboration through card comments, attachments, and team discussions, enhancing communication and teamwork.

Integration with Other Tools: Trello integrates with various third-party tools, such as Slack and Google Drive, allowing users to centralize project-related information.

Personal and Small Projects: Trello is excellent for personal organization, small teams, startups, and simpler projects where a lightweight tool suffices.

Microsoft Project:

Advanced Project Planning: Microsoft Project's robust planning features enable detailed project scheduling, task dependencies, resource allocation, and timeline management.

Resource Management: It excels in managing resource workloads, ensuring optimal allocation and preventing over commitment, making it suitable for larger projects and teams.

Complex Project Management: Microsoft Project is designed for managing complex projects with intricate task relationships, dependencies, and resource constraints.

Enterprise-Level Capabilities: For organizations with multiple projects and portfolios, Microsoft Project can be integrated with Project Server or Project Online, offering advanced portfolio management features.

Reporting and Analysis: Microsoft Project provides advanced reporting capabilities, allowing you to generate customized reports and analyze project data in-depth.

Professional Presentation: It's well-suited for presenting project plans and progress to stakeholders, clients, and management due to its comprehensive reporting capabilities.

Project Portfolio Management: Microsoft Project supports managing multiple projects within an organization and provides tools for resource pooling and allocation.

In summary, Trello is excellent for agile, small-scale projects with simple workflows and a focus on visual task management. On the other hand, Microsoft Project shines when dealing with complex projects, resource-intensive tasks, and the need for advanced planning, reporting, and enterprise-level features.

3) Graded Lab Task:

Note: The instructor can design graded lab activities according to the level of difficulty and complexity of the solved lab activities. The lab tasks assigned by the instructor should be evaluated in the same lab.

Lab Task 1:

Explore about different Software Project Management Tools and compare and contrast about their features.

Lab 02

Getting started with MS Project Professional and Trello

Objective:

The objective of this lab is to familiarize students with the software's essential features and enable them to create and manage project plans effectively. It will also help students to organize tasks and collaborate effectively using the Kanban approach through Trello.

Activity Outcomes:

After successful completion of this lab, the students will become proficient in Software Navigation and will be able to create new projects, define project details (name, start date, etc.), and choose appropriate templates if available.

Instructor Note:

As a pre-lab activity, decide on the project management methodology you'll follow for your semester project, whether it's agile, waterfall, hybrid, or another approach. Also, read Chapter 2 from the book PMBOK: Ch2 for Project teams composition and Software Project Life cycle.

1) Useful concepts

Microsoft Project Professional is a project management software application developed by Microsoft. It is part of the Microsoft Office suite of productivity software and is designed to help project managers and teams plan, execute, and track projects of various sizes and complexities.

MS Project is feature rich, but project management techniques are required to drive a project effectively. A lot of project managers get confused between a schedule and a plan. MS Project can help you in creating a Schedule for the project even with the provided constraints. It cannot Plan for you. As a project manager you should be able to answer the following specific questions as part of the planning process to develop a schedule. MS Project cannot answer these for you.

- What tasks need to be performed to create the deliverables of the project and in what order? This relates to the scope of the project.
- What are the time constraints and deadlines if any, for different tasks and for the project as a whole? This relates to the schedule of the project.
- What kind of resources (man/machine/material) is needed to perform each task?
- How much will each task cost to accomplish? This would relate to the cost of the project.
- What kind of risk do we have associated with a particular schedule for the project? This might affect the scope, cost and time constraints of your project.

Trello is the visual work management tool that empowers teams to ideate, plan, manage, and celebrate their work together in a collaborative, productive, and organized way.

Trello adapts to any project whether a team is starting something new or trying to get more organized with existing projects. It helps you simplify and standardize your team’s work process in an intuitive way. But don’t let its simplicity fool you! Trello is user-friendly, yet still able to handle your team’s most robust projects.

2) Solved Lab Activities

Sr.No	*Allocated Time*	*Level of Complexity*	*CLO Mapping*
1	***20***	***Medium***	***CLO-5***
2	***30***	***Medium***	***CLO-5***

Activity 1:

How to create a new project in MS Project Professional.

Solution:

To create a blank project in MS Project Professional:

1. *Open Project Professional 2016*. There are slightly different methods for opening Project Professional depending on your operating system. For example, in Windows 10, click the **Start** button on the taskbar, All Apps, and then click **Project 2019 or type it in the search bar.** Alternatively, a shortcut or icon might be available on the desktop; in this case, double-click the icon to start the software.
2. *Start a Blank Project*. Click on **Blank Project**, the first option as shown in Figure 2. The left part of the screen shows recent files (if you have any) and allows you to open other projects as well. The current date is the default project start date.

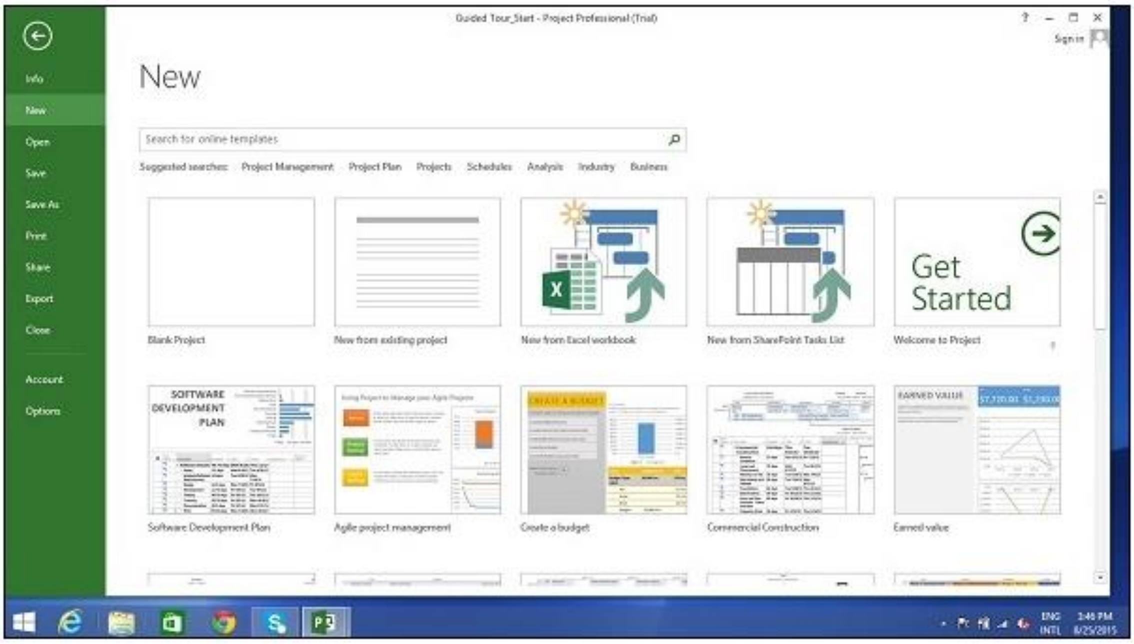


Figure 2.1: MS Project professional main menu

Activity 2:

Getting familiar with Ms Project UI.

Solution:

After clicking the Blank Project Tab, the following screen pops up.

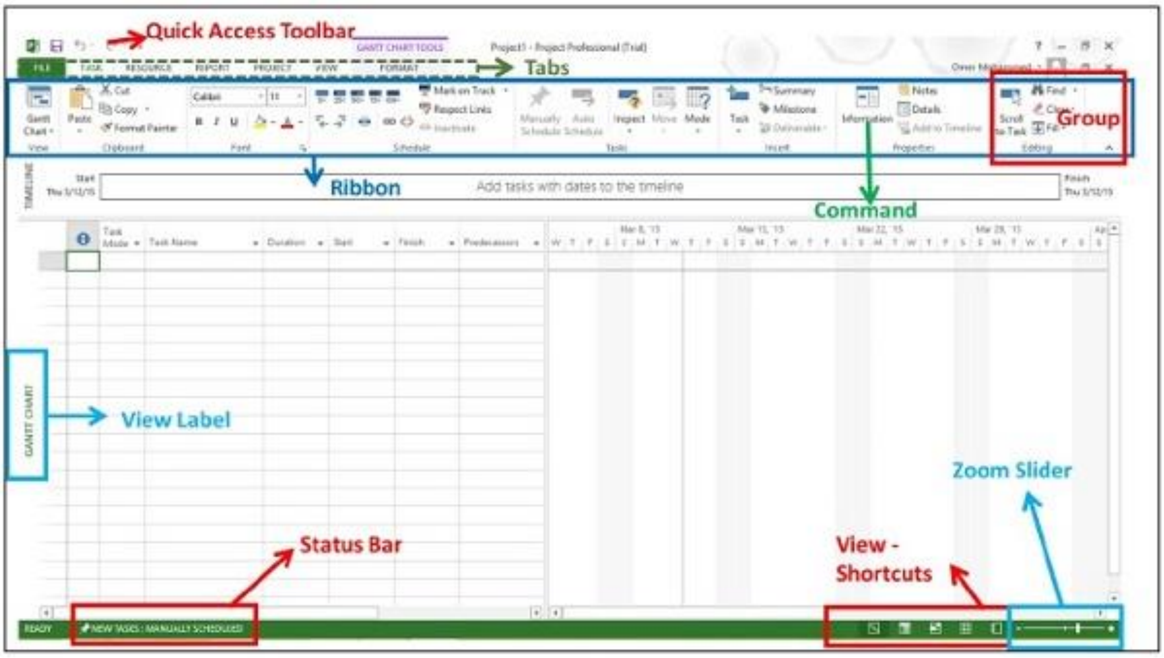


Figure 2.2: MS Project UI

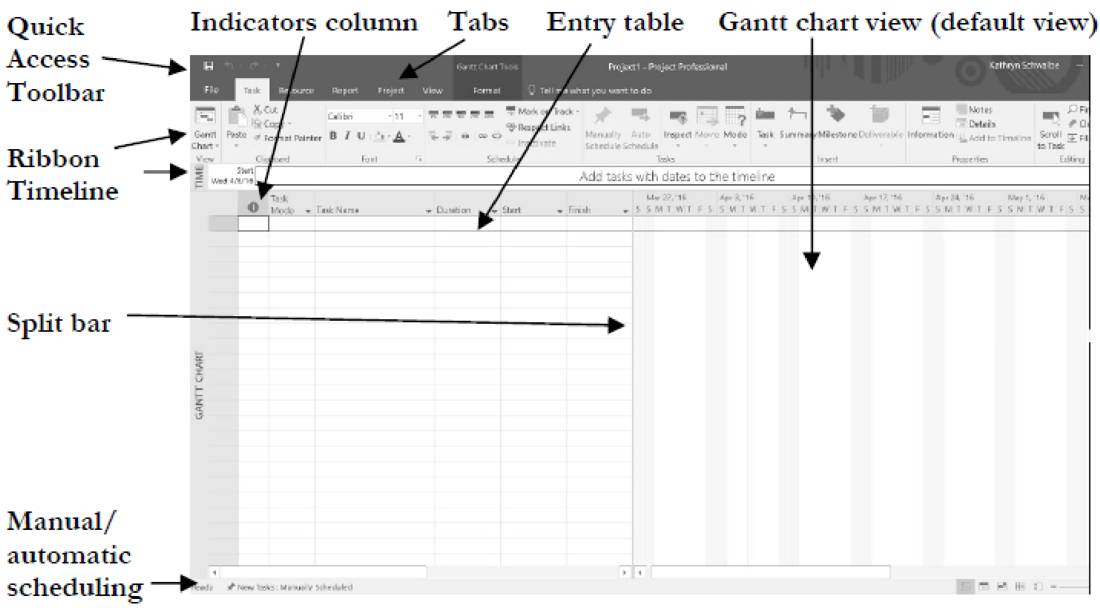


Figure 2.3: Parts of MS Project Interface

The screen should have the MS Project interface displayed. The major parts of this interface are:

Quick Access Toolbar: A customizable area where you can add the frequently used commands.

Tabs on the Ribbon, Groups: A "quick-access toolbar" came to be known as the ribbon having multiple tabs, each holding a toolbar bearing buttons and occasionally other controls. Toolbar controls have heterogeneous sizes and are classified in visually distinguishable Groups. Groups are collections of related commands. Each tab is divided into multiple groups.

Commands: The specific features you use to perform actions in Project. Each tab contains several commands. If you point at a command you will see a description in a tooltip.

View Label: This appears along the left edge of the active view. Active view is the one you can see in the main window at a given point in time. Project includes lots of views like Gantt Chart view, Network Diagram view, Task Usage view, etc. The View label just tells you about the view you are using currently. Project can display a single view or multiple views in separate panes.

View Shortcuts: This lets you switch between frequently used views in Project.

Zoom Slider: Simply zooms the active view in or out.

Status bar: Displays details like the scheduling mode of new tasks (manual or automatic) and details of filter applied to the active view.

Default view: Default view is the Gantt chart view, which shows tasks and other information in a calendar display. You can access other views by clicking the View icon on the far-left side of the ribbon.

Entry Table: The areas where you enter information in a spreadsheet-like table are part of the Entry table. For example, you can see entry areas for Task Name, Duration, Start, Finish, and Predecessors.

The Split bar: You can make the Entry table more or less wide by using the Split bar. When you move the mouse over the split bar, your cursor changes to the resize pointer. Clicking and dragging the split bar to the right reveals columns for Resource Names and Add New columns.

The indicators column: The first column in the Entry table is the indicators column. The indicators column displays indicators or symbols related to items associated with each task, such as task notes or hyperlinks to other files.

Activity 3:

How to create a project from already existing templates using MS Project Professional.

Solution:

To create a project from a template

1. Open Project.

Or select **File > New** if you're already working in a project plan.

2. Select a template or type in the Search for online templates box and then press Enter.

Or, choose **Blank Project** to create a project from scratch.

3. When you **select** a template, select the options you want, and **select Create**.

4. Change the resources, tasks, and durations in the template so they are right for your project.

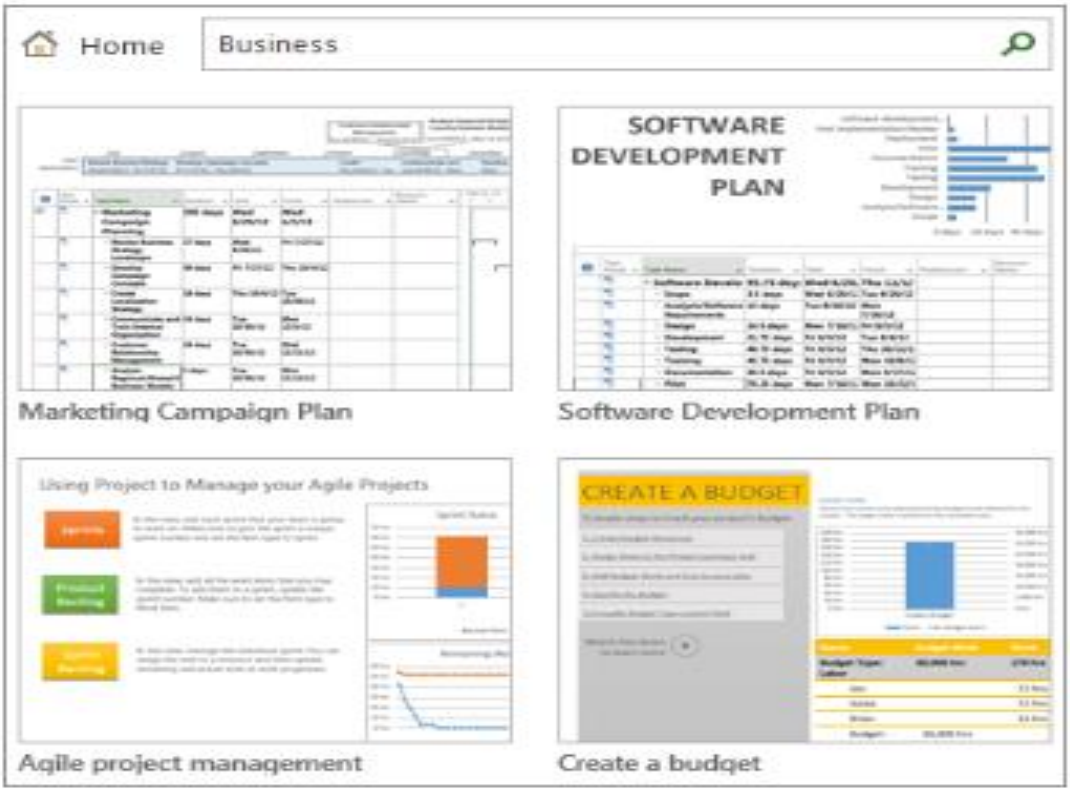


Figure 2.4: Basic tasks

Note: To create a new project from an existing project, change the project start and finish dates, and then save the project file with a new name or in a different location.

Activity 4:

How to create a Trello account.

Solution:

Go to Trello's Website

Getting started with Trello is simple. Head to Trello's website, and click the "get Trello for free" button in the top-right corner. From here, you'll be asked to enter your email address.

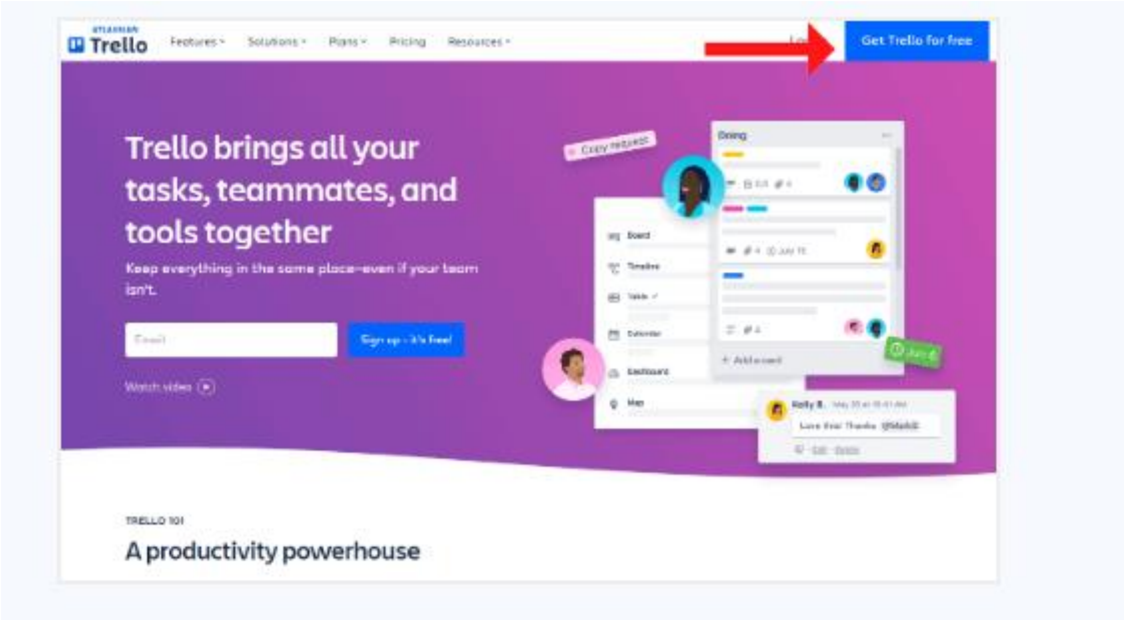


Figure 2.5: Trello Getting started

Enter Your Information

Next, name your workspace and add your teammates to the board. The wonderful thing about Trello's free account is that there are no limits on the number of team members.

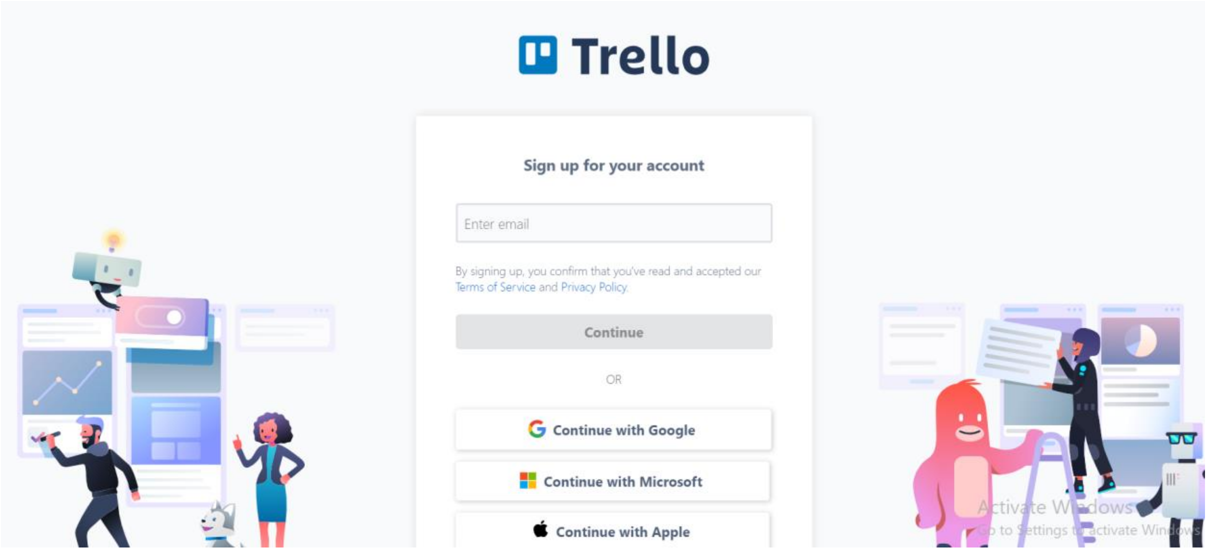


Figure 2.6: Trello account creation

Activity 5:

Create a project in Trello.

Solution:

Click the “create your workspace” button.

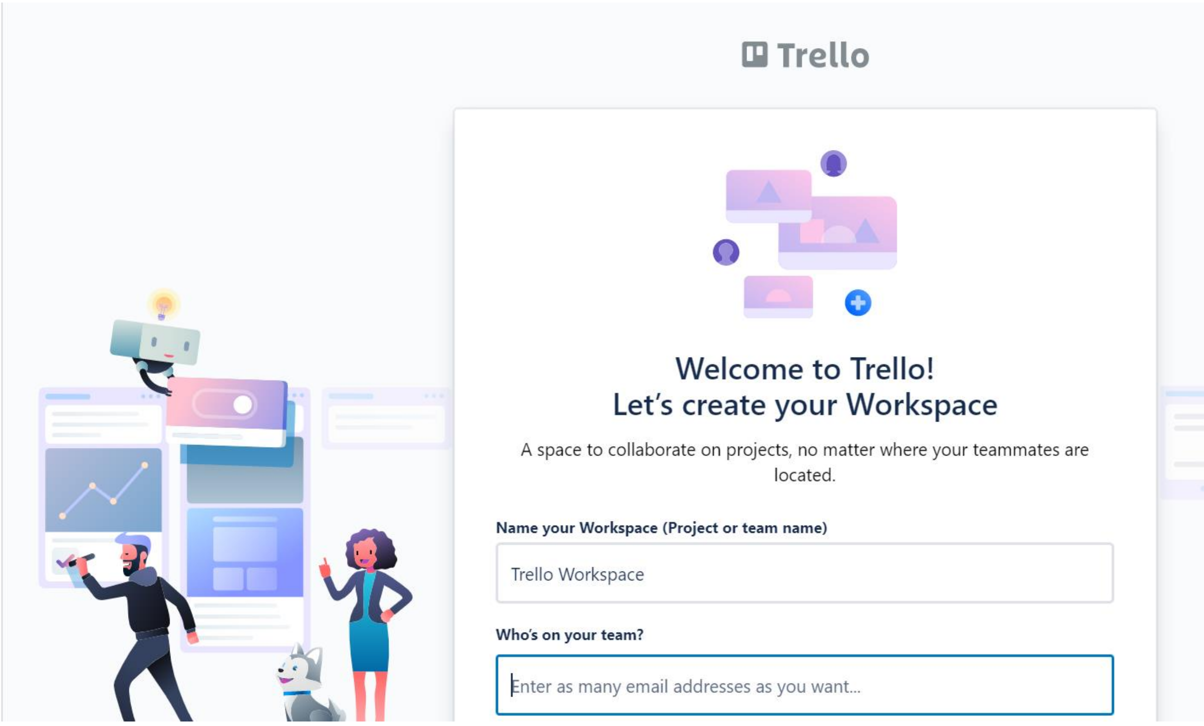


Figure 2.6: Trello workspace creation

Activity 6:

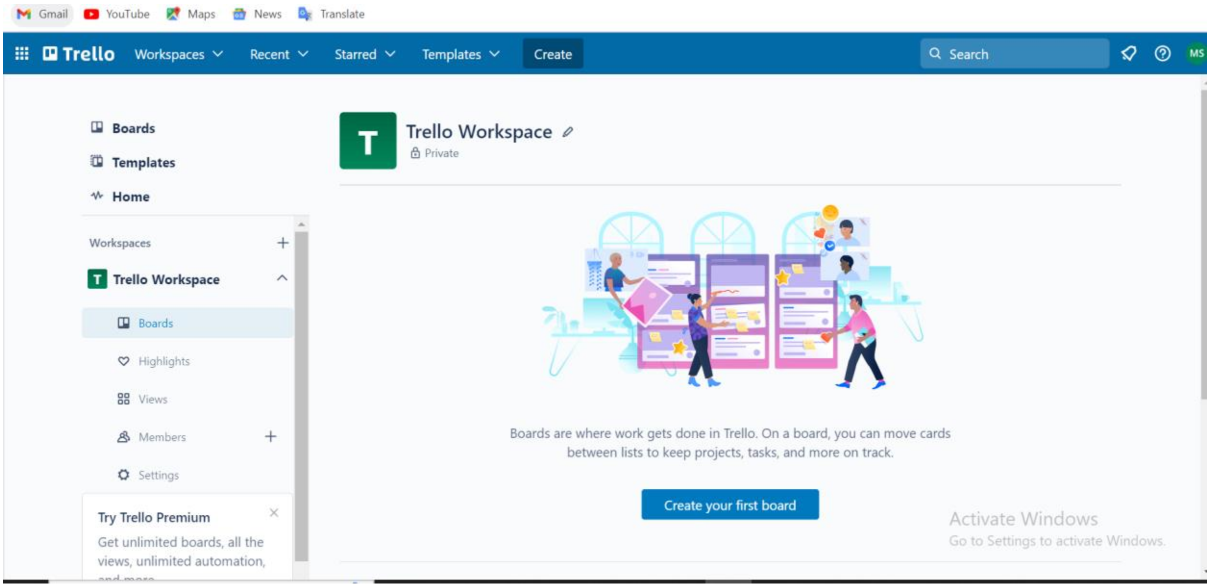
Create your first board.

Solution:

A board (A) represents a place to keep track of information — often for large projects, teams, or workflows. Whether you are launching a new website, tracking sales, or planning your next office party, a Trello board is the place to organize tasks, all the little details, and most importantly—collaborate with your colleagues.

Steps:

Click on create your first board.



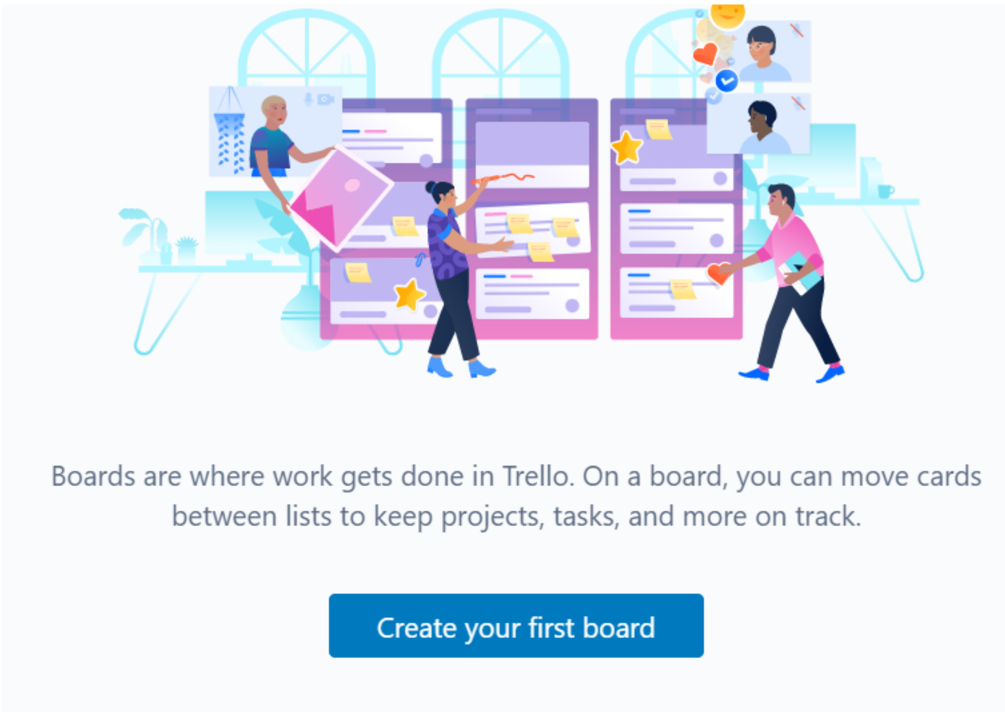


Figure 2.7: Trello board creation

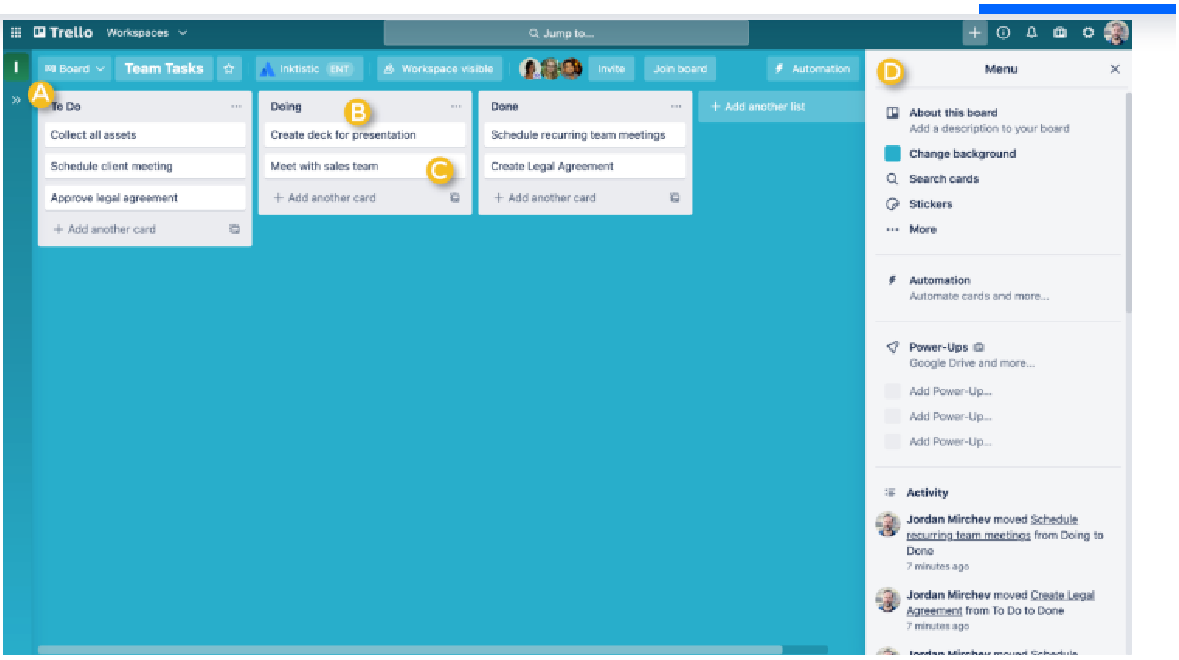


Figure 2.8: An example of board

Creating a board another example.

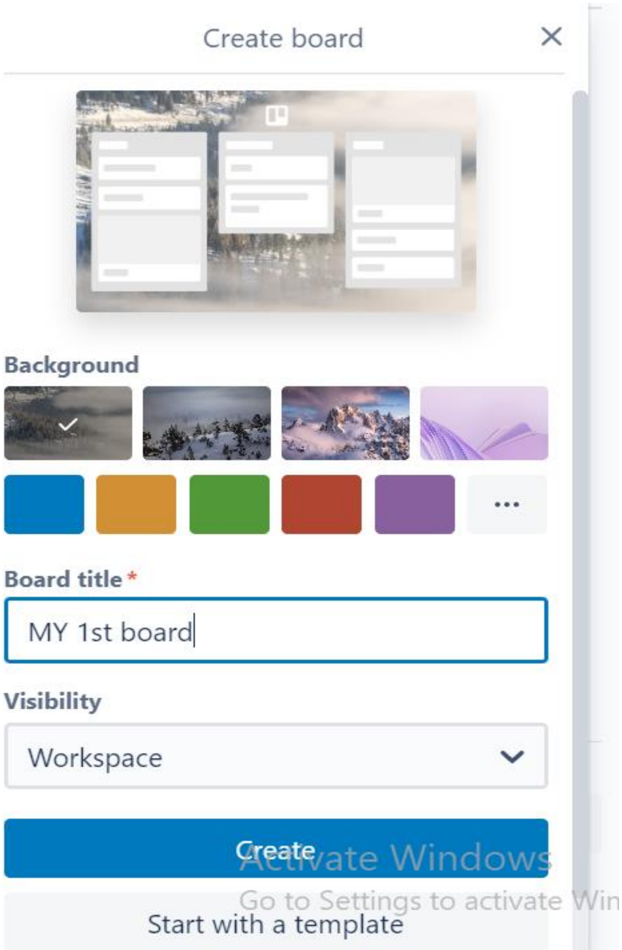


Figure 2.9: First board creation

3) Graded Lab Task

Note: The instructor can design graded lab activities according to the level of difficulty and complexity of the solved lab activities. The lab tasks assigned by the instructor should be evaluated in the same lab.

Lab task 1

Install Ms project Professional and create new Semester project.

Lab task 2

Get started with trello and create your own workspace and create boards. Also explore different features of Trello.

Lab 03

Generating Project Professional Views and Reports

Objective:

The objective of this lab is to make students learn about how to switch to different views and generate multiple automated reports.

Activity Outcomes:

After successful completion of this lab, the students should be able to analyze project information in different ways and will be able to generate reports.

Instructor Note:

As a pre-lab activity, read Chapter 3 and 4 from the book PMBOK.

1) Useful concepts:

Project Professional software provides many ways to display or view project information. In addition to the default Gantt chart, you can view the network diagram, calendar, and task usage views, to name a few. The View tab also provides access to different tables that display information in various ways. In addition to the default Entry table view, you can access tables that focus on data related to areas such as the Schedule, Cost, Tracking, Variance, and Earned Value tables.

Project Professional also provides many ways to report project information as well. In addition to traditional reports, you can also prepare visual reports, with both available under the Report tab. Note that the visual reports often require that you have other Microsoft application software, such as Excel and Visio.

2) Solved Lab Activates

Sr.No	***Allocated Time***	***Level of Complexity***	***CLO Mapping***
1	***20***	***Low***	***CLO-***
2	***20***	***Low***	***CLO-***
3	***20***	***Medium***	
4	***20***	***Medium***	
5	***20***	***Medium***	
6	***20***	***Medium***	
7	***20***	***Medium***	
8	***20***	***Medium***	

Activity 1:

How to access and explore different views in MS Project?

Solution:

To access and explore different views:

1. *Explore the Network Diagram for the Market Research Schedule file. Open the* **Market Research Schedule** *file again.* Click the **View** tab, the **Outline** button, and **All Subtasks** again. Click the **Network Diagram button**, and then move the **Zoom slider** on the lower right of the screen all the way to the left. Criticaltasks automatically display in red in the Network Diagram view.
2. *Explore the Calendar view*. Click the **Calendar** button (under the Network Diagram button). Notice that the screen lists tasks each day in a calendar format.
3. *Change the table view*. Click the **Gantt chart** button on the ribbon, click the **Tables** button, and then click **Schedule**.

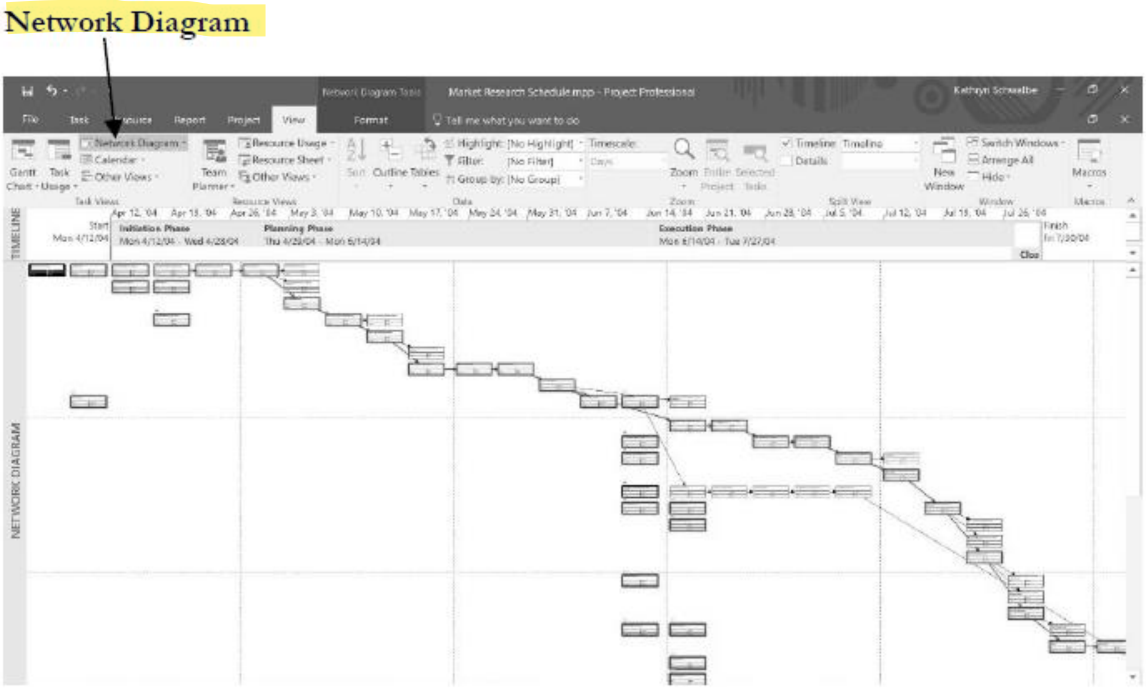


Figure 3.1: Network diagram view

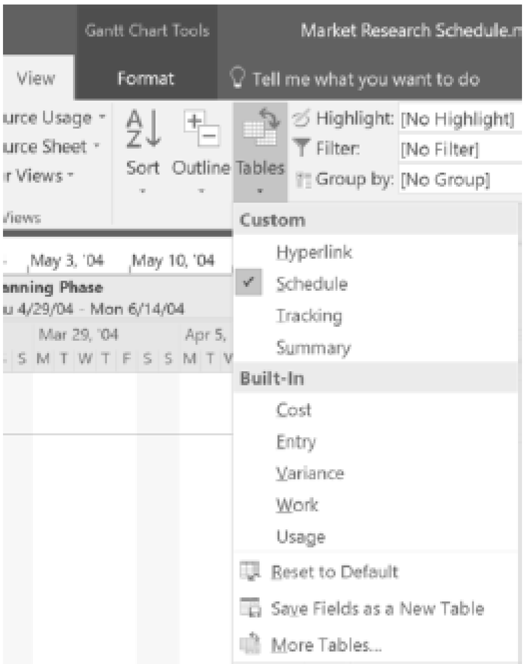


Figure 3.2: switch view

Activity 2:

Examine the other tables and views

Solution:

Move the **Split bar** to the right to review the Total Slack column. Notice that the columns in the table to the left of the Gantt chart, as shown in Figure 7, now display more detailed schedule information, such as Late Start, Late Finish, Free Slack, and Total Slack. Remember that you can widen columns by double-clicking the resize pointer to the right of that column. You must do this for the Start, Finish, and Late Start columns to remove the #### symbols. You can also move the split bar to reveal more or fewer columns. Experiment with other table views, then return to the **Entry** table view.

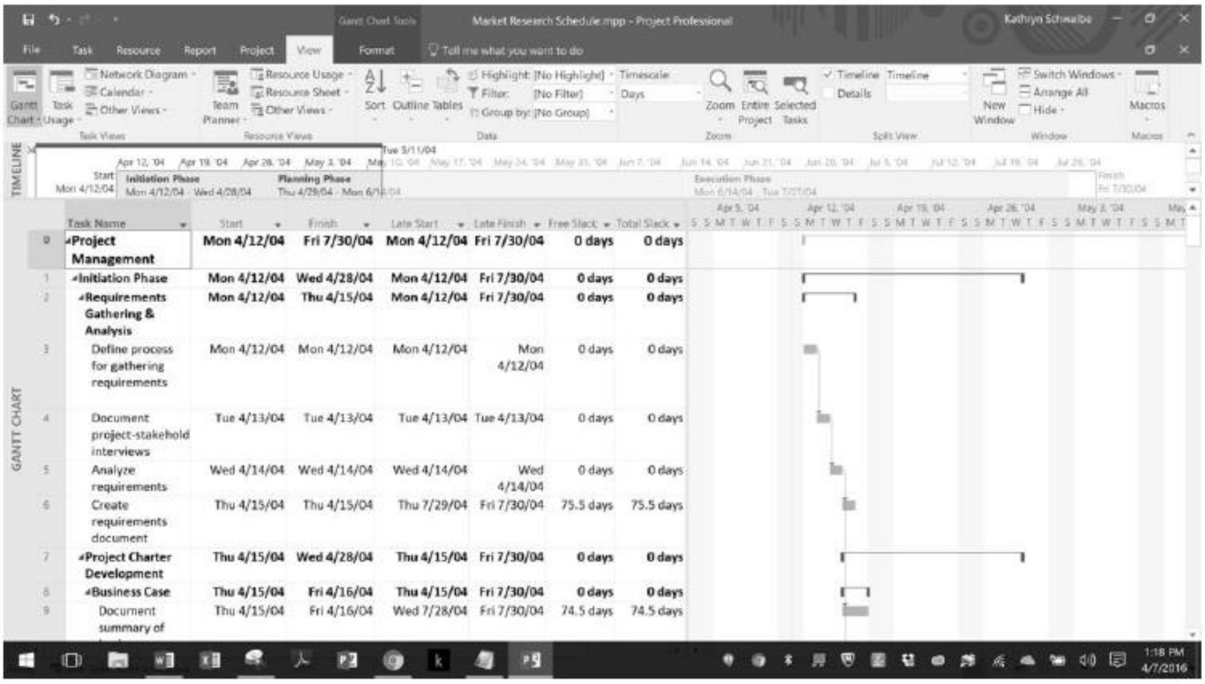


Figure 3.3: Detailed View

Activity 3:

How to switch to different views? For example, Gantt chart view, resource view, or Network diagrams view.

Solution:

1. Click on view tab
2. On top right there are options to switch the different view.
3. Clicking on desired view, so you can switch to that view.

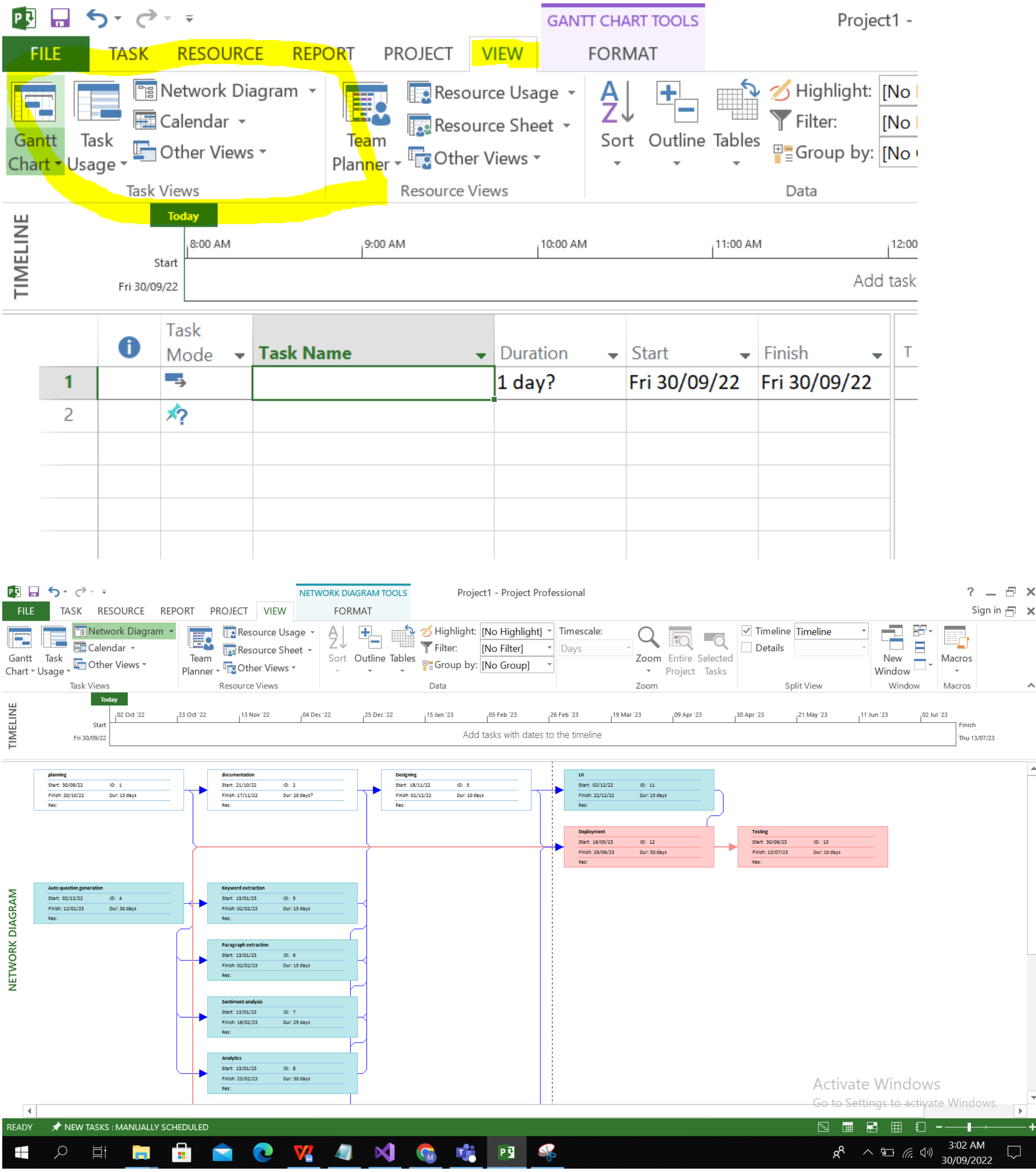


Figure 3.4: Switching between views

Activity 4:

How to create a visual report in MS project?

Solution:

1. Click on reports
2. Then click on visual reports.
3. Select desired report
4. Then click ok

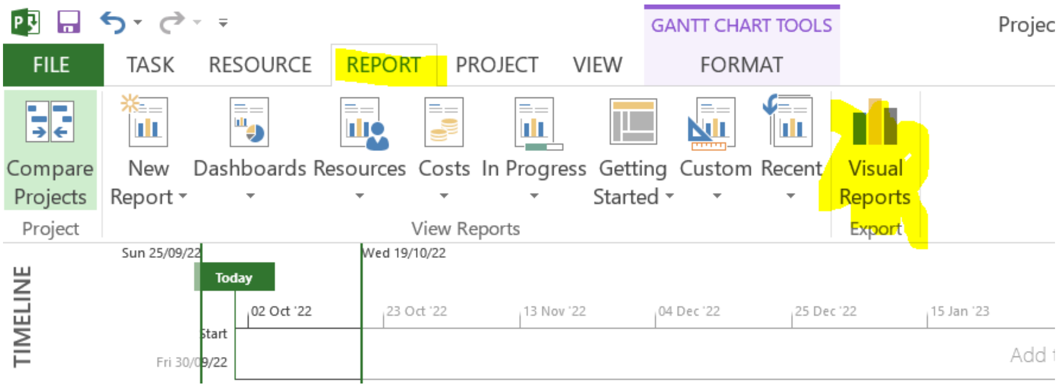


Figure 3.5: Visual reports

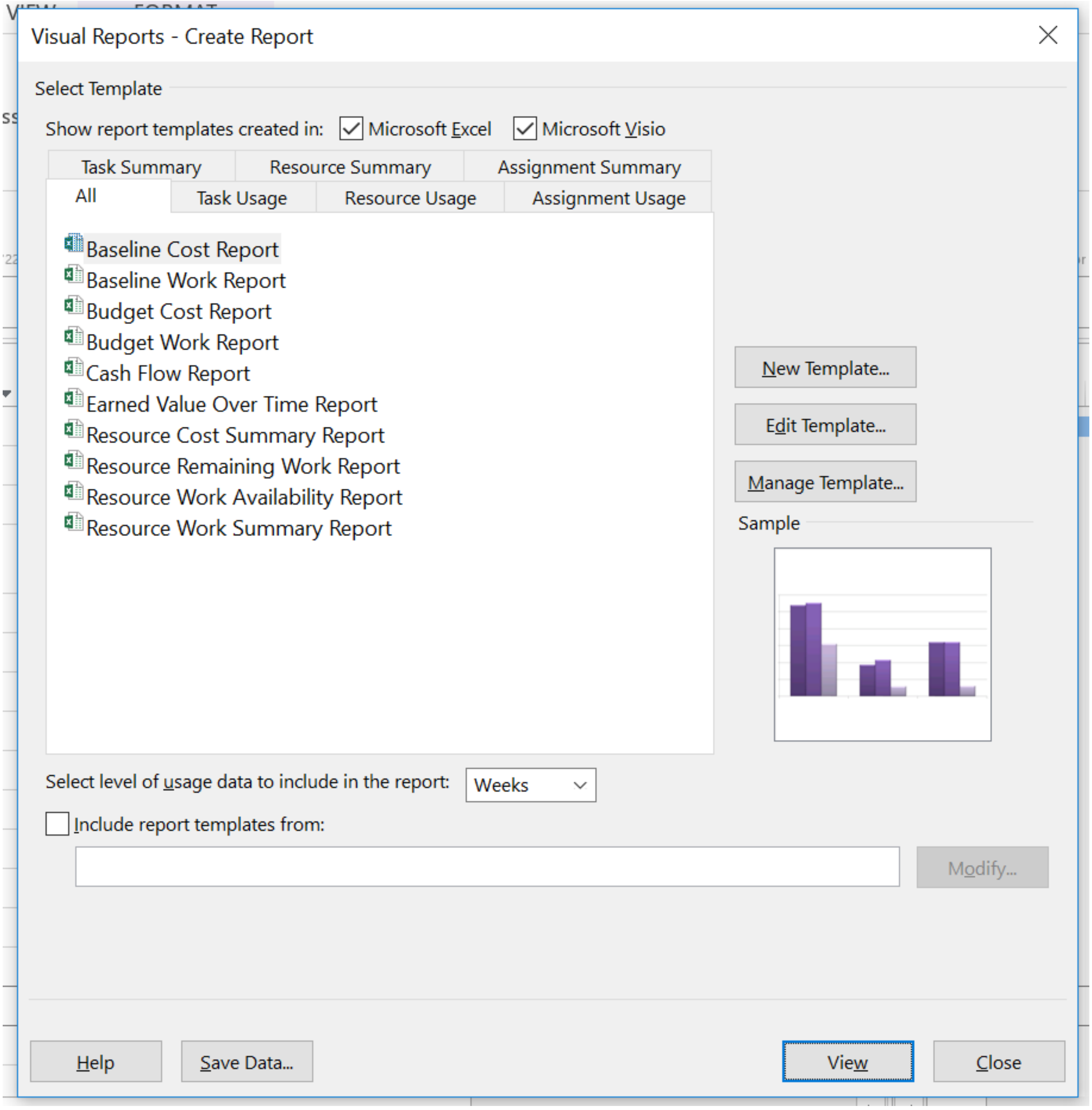


Figure 3.6: Selecting different reports

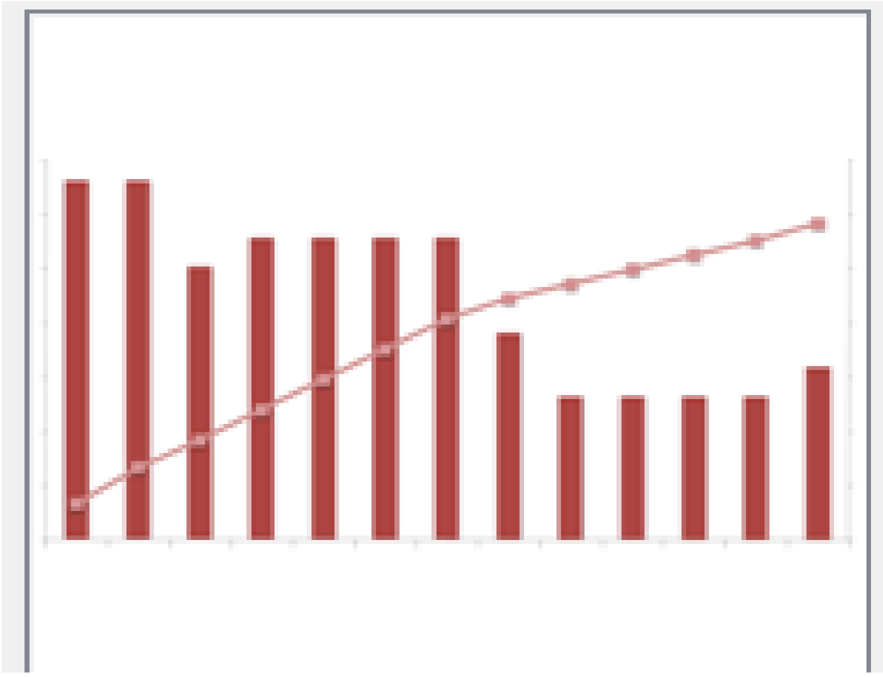


Figure 3.7: Report view

Activity 5:

How to display the project summary task / Title.

Solution:

1. Click on the Format tab.
2. Click Show/Hide group
3. you will find the Summary Tasks listed in the bottom of the group
4. Check the box in front of it, you can choose to show the Summary Tasks.
5. If not, just click the box again to uncheck it.

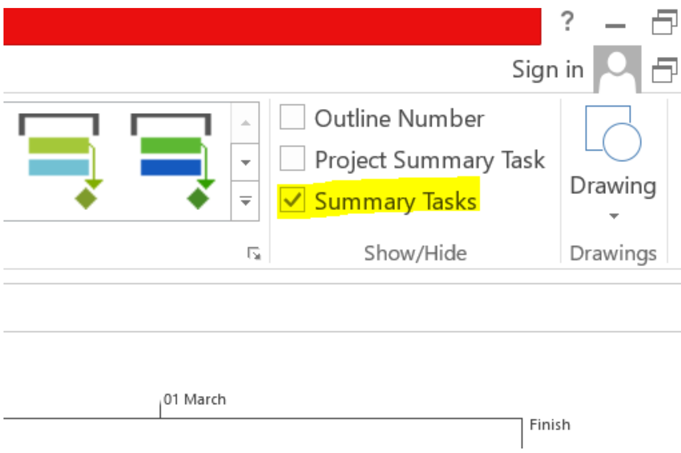


Figure 3.8: Displaying summary tasks

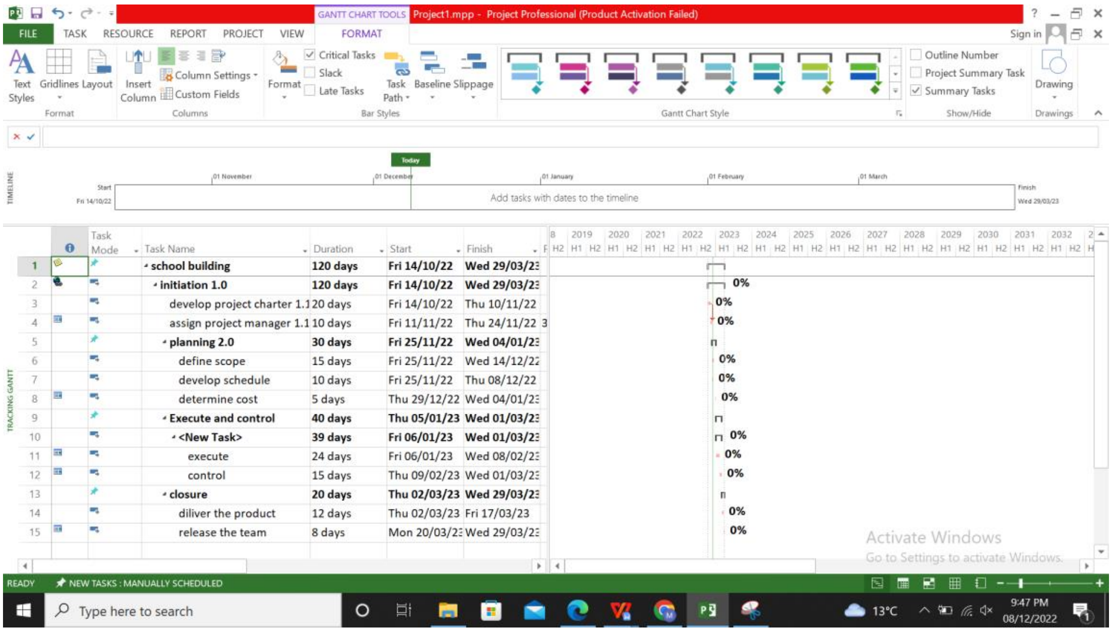


Figure 3.9: Displaying summary tasks

Click **File** > **Options** > **Customize Ribbon**

In the right column, click the tab **(View)** and then click new group

Activity 6: From Helping Material

How to create a new view based on an existing view?

Solution:

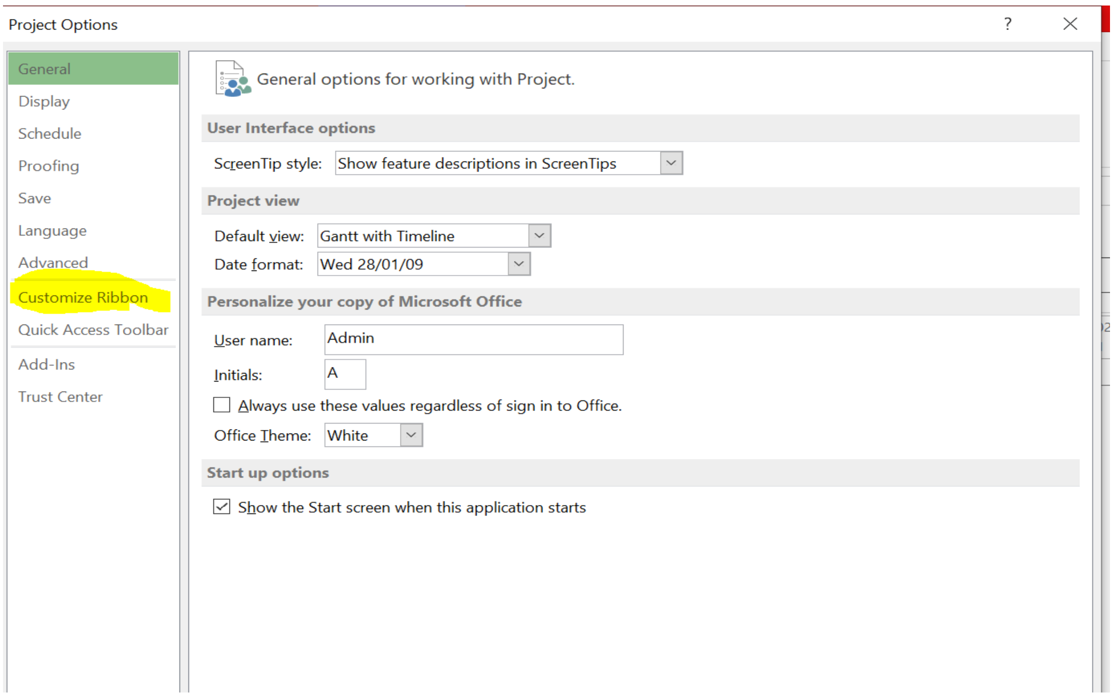


Figure 3.10: Creating New view

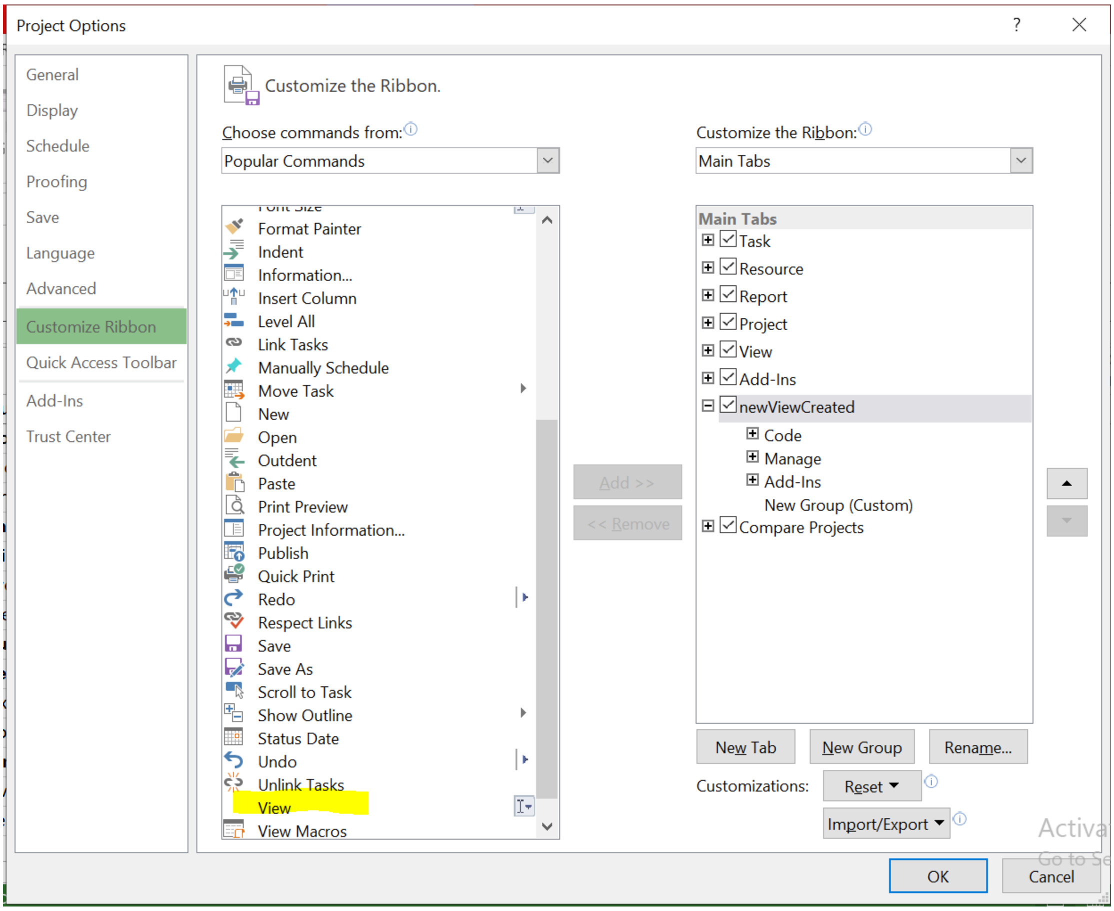


Figure 3.11: New view customization

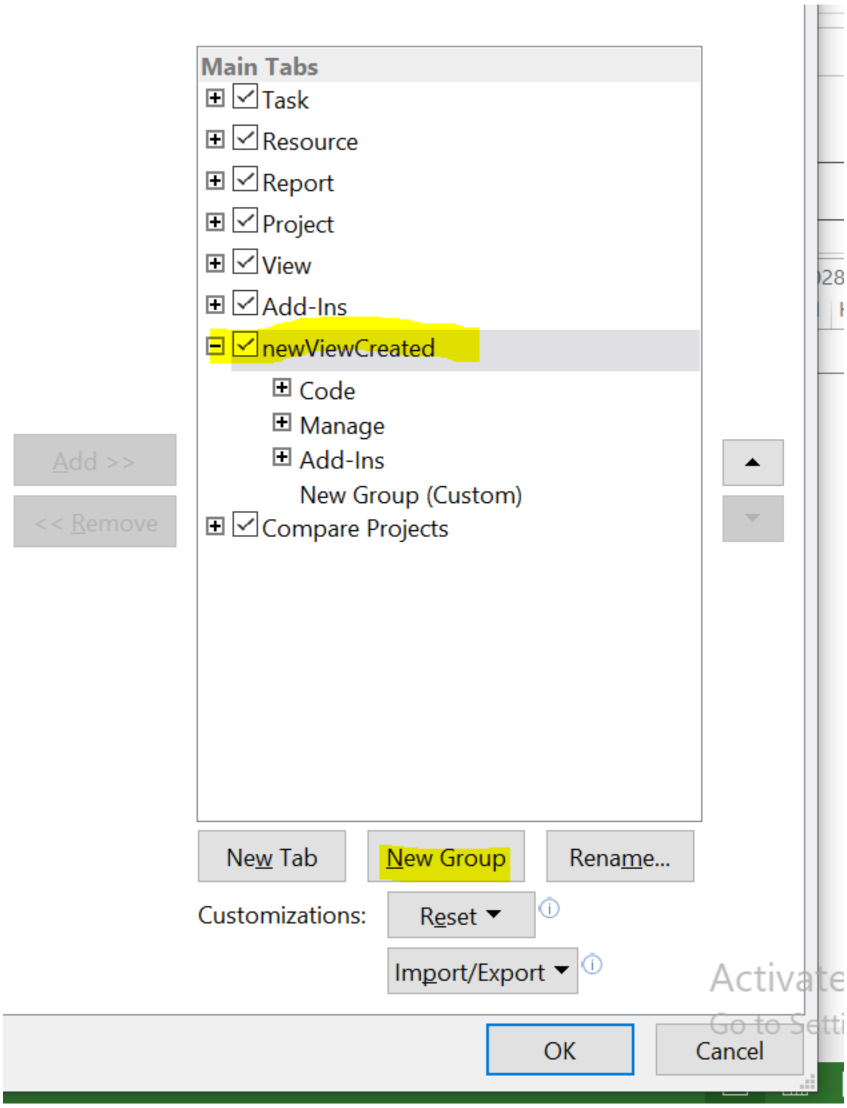


Figure 3.12: New view creation

Activity 7:

How to format in a view? For example, menu font header and footer.

Solution:

1. In the **File** tab, click **Print**, then click **Page Setup**
2. On the **Header**, click the **Left**, **Center**, or **Right** tab.
3. In the text box, type or paste the text, add the document or project information, or insert or paste a graphic.
4. To add page numbers, click **Insert Page Number** ,**Insert Total Page Count**.
5. Click Ok

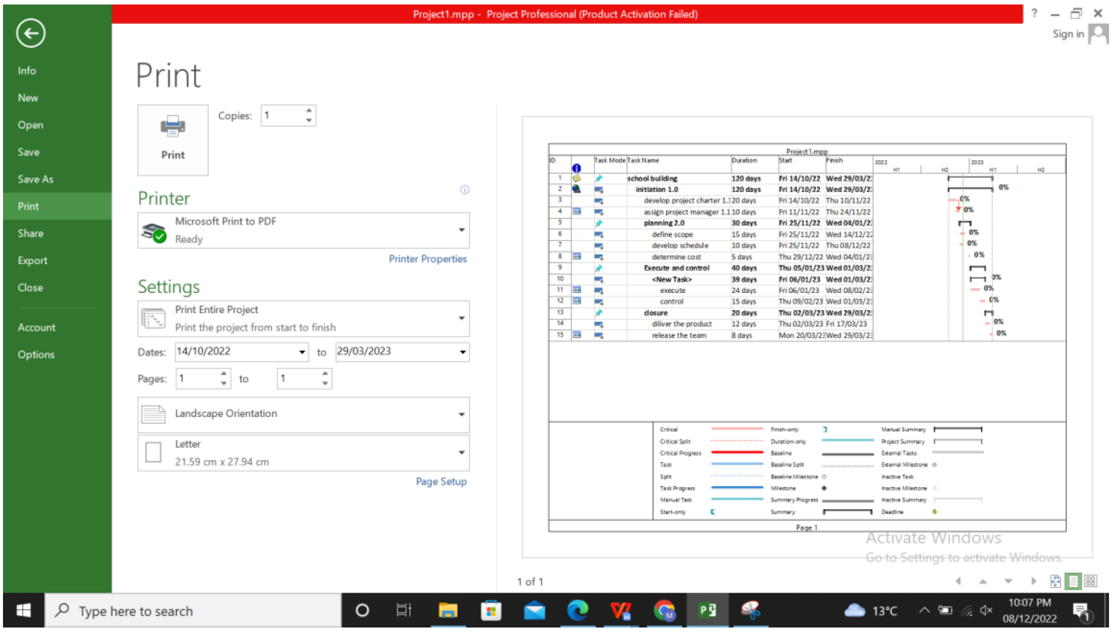


Figure 3.13: Format in a view

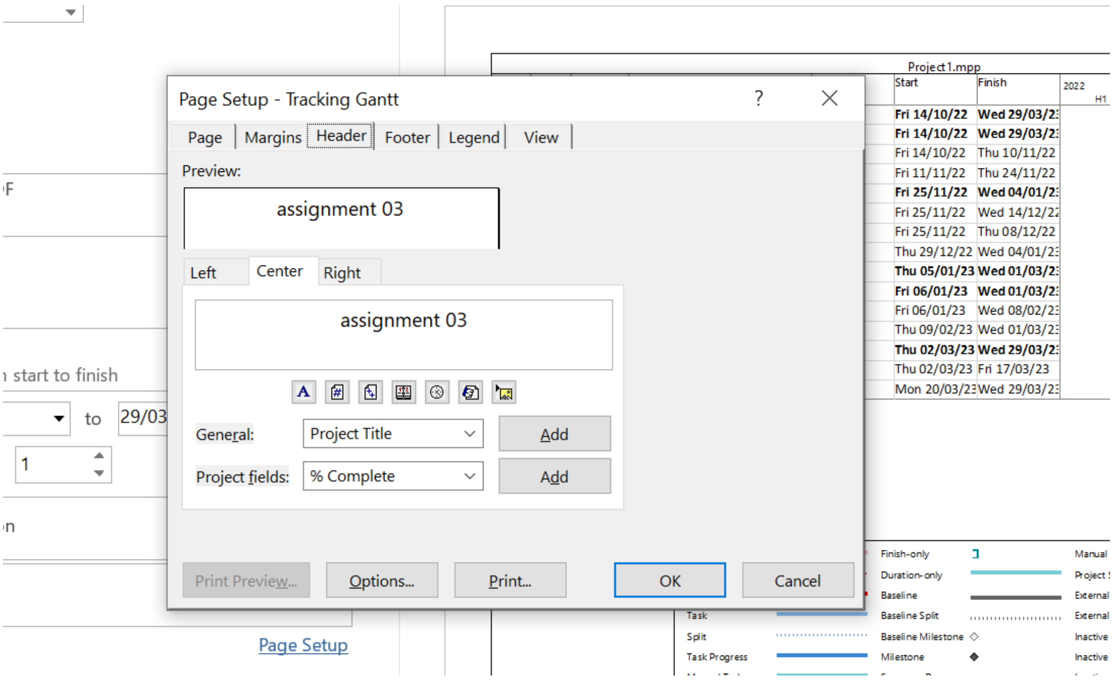


Figure 3.14: Header setup

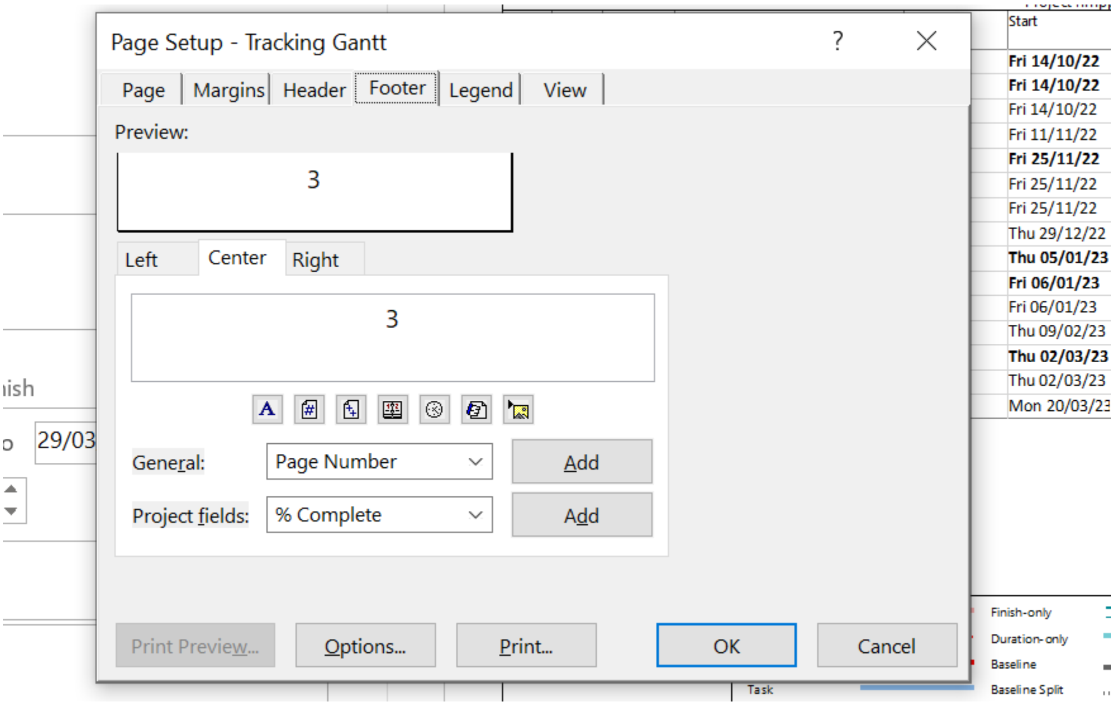


Figure 3.15: Footer setup

Activity 8:

Learning and analyzing all the details of reports projection.

Solution:

To access and explore different reports:

1. *Open another template file*. With Project 2016 open, click on the **File tab,**click **New**, then double-click on the template called **Earned value**. Read the information on the initial screen to learn more about the concept of earned value. Click on the **View tab**, then click the **Gantt chart** icon.

2. *Explore the reports feature*. Click the **Report tab to see the variety of reports available in Project 2016**, as shown in Figure 8.

Figure 3.16: Report selection

3. *View the Project Overview report*. Click **Dashboards**, and then double-click **Project Overview**. Review the report, noting that the project is 17% complete. You must have entered some actual information and cost data to take advantage of this report. Also notice the new options on the ribbon, as shown in Figure 9.

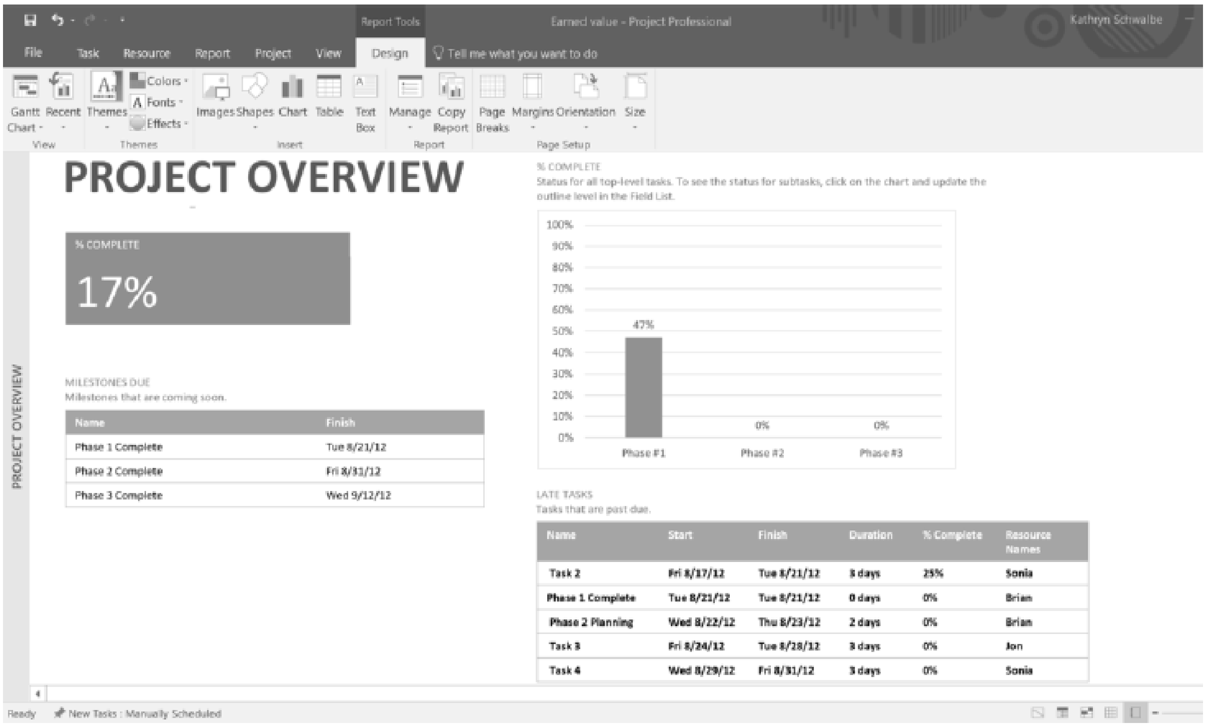


Figure 3.17: Project overview

4. Open the Resource Overview report. Click the **Report tab again**, click **Resources**, and then click **Resource Overview**. Review the report, as shown in Figure 10.

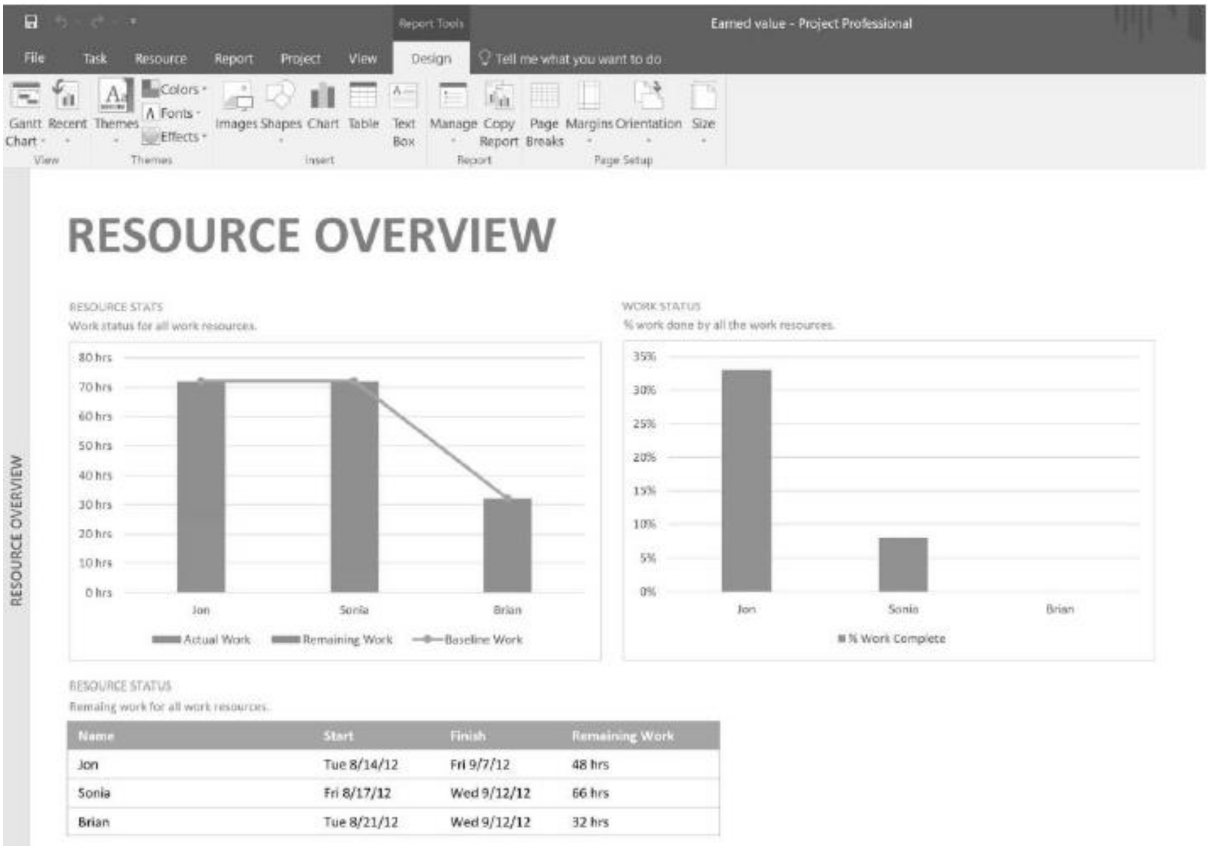


Figure 3.18: Resource overview

5. Examine other reports. Click on the **Report tab**, click **In Progress**, and then click **Critical Tasks** to display the Critical Tasks report, as shown in Figure 11. Examine other reports.

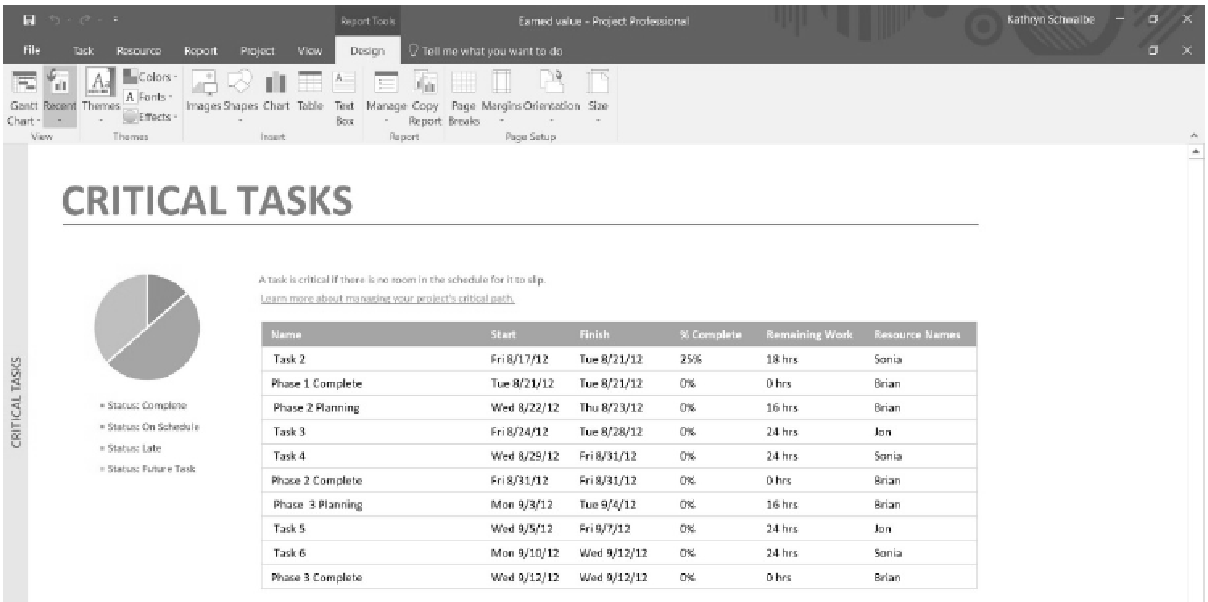


Figure 3.19: Critical tasks Report

6. Return to the Gantt chart. Click the **View tab, and then click on Gantt Chart** to return to the Gantt chart view. Next, you will use this same file to explore filters.

3) Graded Lab Task

Note: The instructor can design graded lab activities according to the level of difficulty and complexity of the solved lab activities. The lab tasks assigned by the instructor should be evaluated in the same lab.

Lab Task 1

Explore different views of MS Project.

Lab Task 2

Generate Critical tasks reports.

Lab # 04

Applying Project Professional Filters and Creating Tasks List

Objective:

The objective of this lab is to guide participants in developing proficiency in MS Project Professional 2019 by teaching them how to effectively apply filters to view and manage specific task sets.

Activity Outcomes:

After successful completion of this lab, the students will be able to perform the following:

- Filter tasks or resources
    - Apply or remove a filter
    - Apply a color to filtered tasks using a highlight
    - Create a custom filter
    - Modify an existing filter
- Use AutoFilters
    - Apply and remove AutoFilters
    - Create a custom AutoFilter
    - Turn on AutoFilters automatically for new projects
- Task List
    - How to create a new project plan and set its start date?
    - How to set working & non-working time?
    - How to set properties about a Project Plan?
    - How to enter tasks, task duration & milestone in a Project?
    - How to organize tasks into phases?

Instructor Note:

As pre-lab activity, read Chapter 13 from the book "Microsoft Project 2019 Step by Step, Cindy Lewis, Carl Chatfield, Timothy Johnson., Microsoft Press, 2019".

1) Useful Concepts:

With Project, you can filter your view so that you only see the critical tasks, milestones, and other information that's most important to you. There are three ways to filter the tasks or resources in your project provides predefined filters for viewing specific aspects of tasks and resources. If none of these filters meet your needs, you can create a new filter or modify an existing filter.

- **Predefined filters** These are filters that ship with Project. These are used to quickly filter for tasks or resources, such as tasks that are incomplete, or resources that are overallocated.
- **Custom filters** These are filters that you design for you own project needs.
- **AutoFilters** When these are turned on, they appear as arrows at the top of each column in a sheet view. Use these to quickly filter the items in a column.

Filter tasks or resources

- There are times when you want to view only a particular type of information in your project. For example, you may want to view only the milestones of your project or see tasks that haven't started. If your project has many tasks and involves many resources, filters can be very useful for viewing a specific range of information. Filters allow you to display only the information you are interested in and hide the rest.

- You can filter task or resource data using the pre-defined Project filters. If none of the filters meet your needs, you can create a new filter or modify an existing filter.

2) Solved Lab Activities

Sr.No	Allocated Time	Level of Complexity	CLO Mapping
1	20	Low	CLO-6
2	20	Low	CLO-6
3	20	Low	CLO-6
4	20	Low	CLO-6

Activity 1:

How to Apply or Remove a Filter?

Solution:

1. On the **View** tab, in the **Data** group, choose a filter in the filter list.

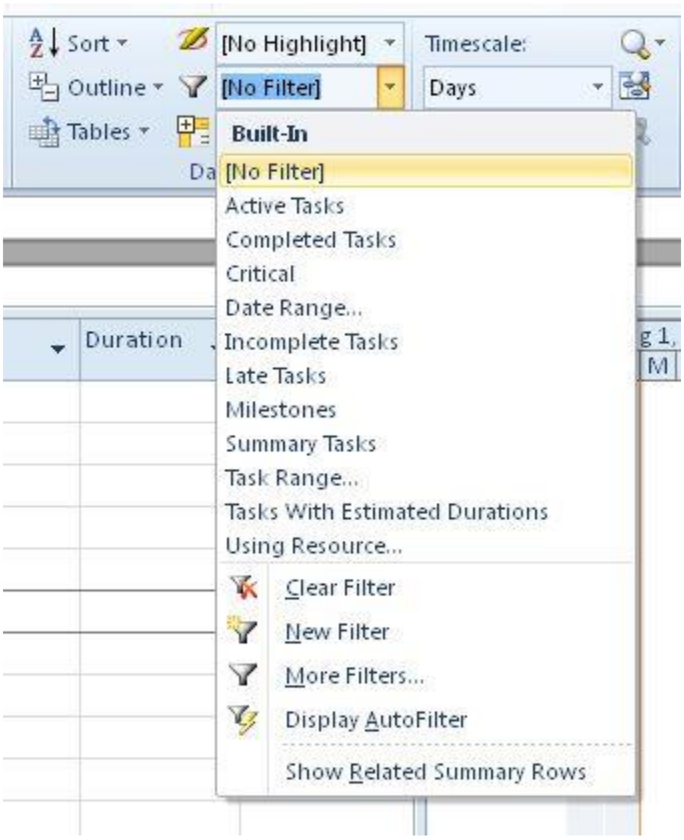


Figure 4.1: Filter List

To apply a filter that isn't on the list, choose **More Filters**, and then do one of the following:

- To select a task filter, choose **Task**, choose a filter name in the **Filters** list, and then choose **Apply**.
- To select a resource filter, choose **Resource**, choose a filter name in the **Filters** list, and then choose **Apply**.

  Note: You cannot apply task filters to resource views or resource filters to task views.

2. If you apply an interactive filter, type the requested values, and then choose **OK**.
3. To turn off a filter, choose **No filter** in the list of filters.

Apply a color to filtered tasks using a highlight

When filtering tasks or resources, you can apply a highlight. Highlighted tasks or resources appear with non-filtered tasks, but with a different color.

1. On the **View** tab, in the **Data** group, select a filter in the filter list., and then choose **More Filters**.
2. Select a filter in the filter list, and then choose **Highlight**.

Tip: To apply a different color to the highlighted tasks, use a different text style. Choose the **Format** tab, and then choose **Text Styles**. In the **Item to Change** list, select **Highlighted Tasks**, and then select formatting options.

Create a custom filter

1. On the **View** tab, in the **Data** group, choose the arrow next to **Filter**, and then choose **More Filters**.

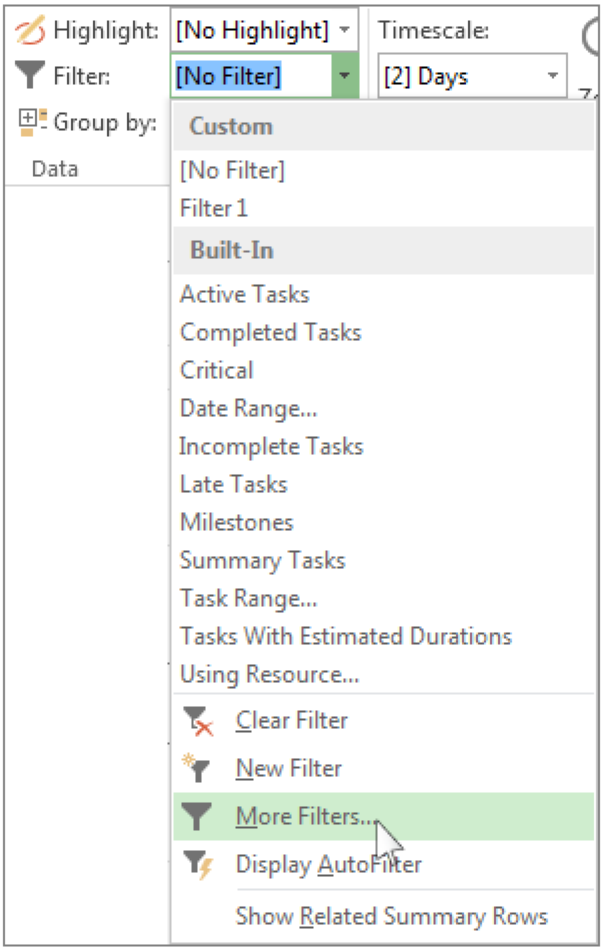


Figure 4.1: More Filters

2. Select **Task** or **Resource** (depending on which type of filter you want to create), and then choose **New**.
3. Type a name for your new filter. Select **Show in menu** if you want to include this filter in the **Data** group list.
4. In the **And/Or** column, choose **And** to show results that meet more than one of your filter criteria. Choose **Or** to show rows that meet one or the other.

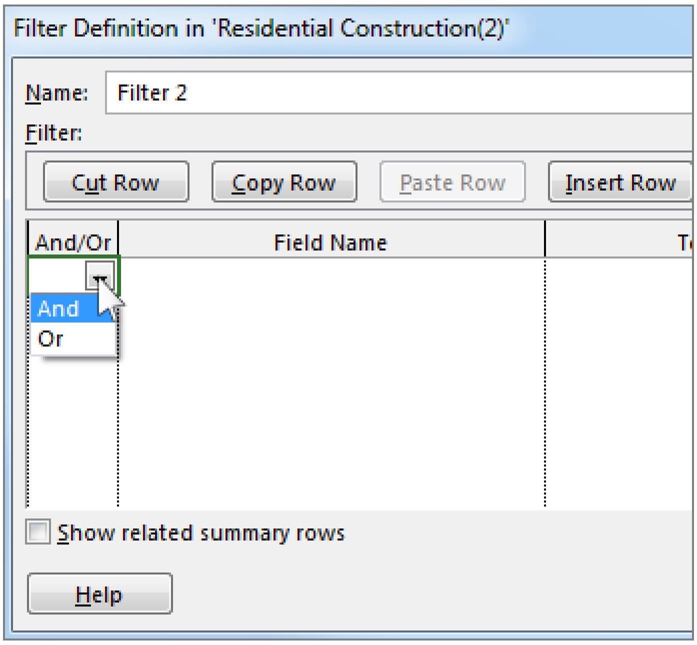


Figure 4.3: Filter Definition

5. In the **Field Name** column, choose which field you want to filter.
6. In the **Test** column, decide how you want to match what you chose for the **Field Name** column to the **Value(s)** column in the next step.
7. In the **Value(s)** column, choose the value you want, or type a new one.

   If you chose **Equals** or **Does Not Equal** in the **Test** column, you can type a wildcard character in the **Values** column (instead of choosing an option from the list). For example, typing a question mark (?) would find single characters. Typing an asterisk (*) finds any number of characters. If you use wildcards, be sure that your choice for the **Field Name** column involves text (like **Name**), instead of numbers (like **Duration**).

8. To add another row to your filter (and to choose another field name to filter on), choose **Insert Row**. You can group rows by adding a blank row. Add an **And** or **Or** in the blank row to filter one group against another.
9. When you're done, choose **Save**.

Tip: To quickly remove all of your filters, use the F3 key. Keep in mind that task filters only work with task views, and resource filters only work with resource views.

Modify an existing filter

1. On the **View** tab, in the **Data** group, select the filter list, and then choose **More Filters**.
2. Do one of the following:
   - To modify a task filter, choose **Task**, choose the filter you want to modify, and then choose **Edit**.
   - To modify a resource filter, choose **Resource**, choose the filter you want to modify, and then choose **Edit**.
3. Modify the settings for the filter using the controls in the **Filter Definition** dialog box.

ACTIVITY 2:

How to Use auto filters?

Solution:

You can apply an AutoFilter to fields in any sheet view. In addition to standard filters, project provides AutoFilters, visible at the top of each column in sheet views.

Apply and remove auto filters

1. On the **View** tab, in the **Data** group, choose the arrow for the filter list, and then choose **Display AutoFilter**.

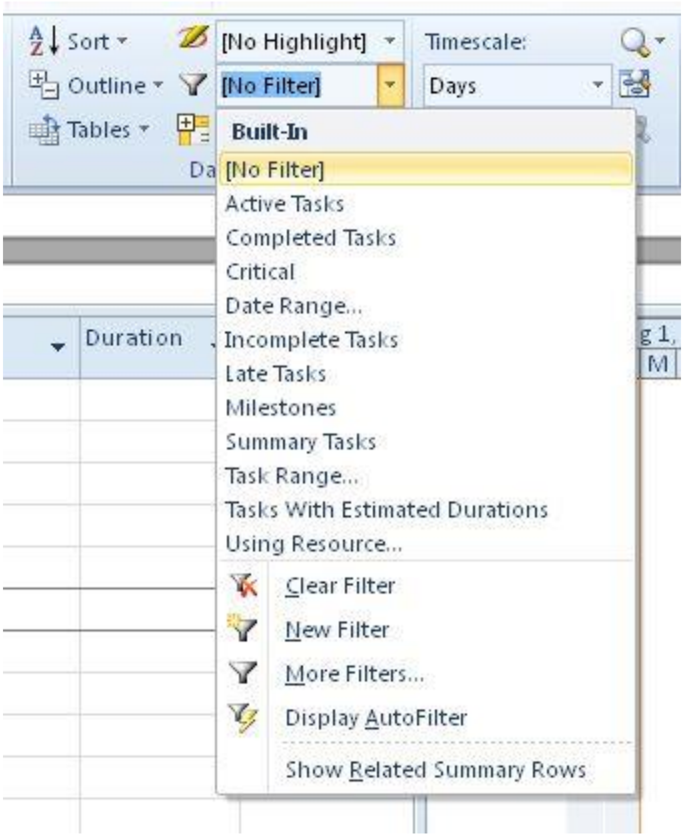


Figure 4.4: Auto Filter

2. Choose the AutoFilter arrow next to the column heading that contains the information you want to display, and then choose a value to filter the table.
   The AutoFilter arrow and field heading turn blue.
3. To apply an additional condition based on a value in another column, repeat step 2 for the other column.
4. To remove the filtering on a specific row, choose **Clear All Filters** in the AutoFilter list for that field.
5. To turn off AutoFilters, choose **Display AutoFilter** again.

Note: If information changes in one of the rows in your view, you can refresh your AutoFilter settings by choosing the arrow and re-selecting the filtering values.

Create A Custom Autofilter

1. Display AutoFilters.
2. Choose an AutoFilter arrow, point to **Filter**, and then choose **Custom**.
3. Choose the operator you want to use in the first box, and then type or select the value you want to match in the second box.

   For example, to match a specific date in a date field, choose the **equals** operator in the first box, and then select the date you want to match in the second box.

4. To apply two conditions to the AutoFilter, do one of the following:
   - To display rows in your view that meet both conditions, choose the operator and value you want in the second row of boxes, and then choose **And**.
   - To display rows in your view that meet either one condition or another condition, choose the operator and value you want in the second row of boxes, and then choose **Or**.
5. To save your AutoFilter settings, choose **Save**.

   If your view already has a predefined filter applied, then the conditions you set for the AutoFilter are included as additional conditions to the current filter. When you save an AutoFilter setting, the filter is saved with other filters in your file and is available only through the **More Filters** dialog box.

Turn On Autofilters Automatically For New Projects

1. On the **File** tab, choose **Options**.
2. Choose **Advanced**, and then in the **General** section, select the **Set AutoFilter on for new projects** check box.

ACTIVITY 3

How to create a new project plan and set its start date?

1. Click on file
2. Click on option
3. Then click on schedule
4. Set starting date of project

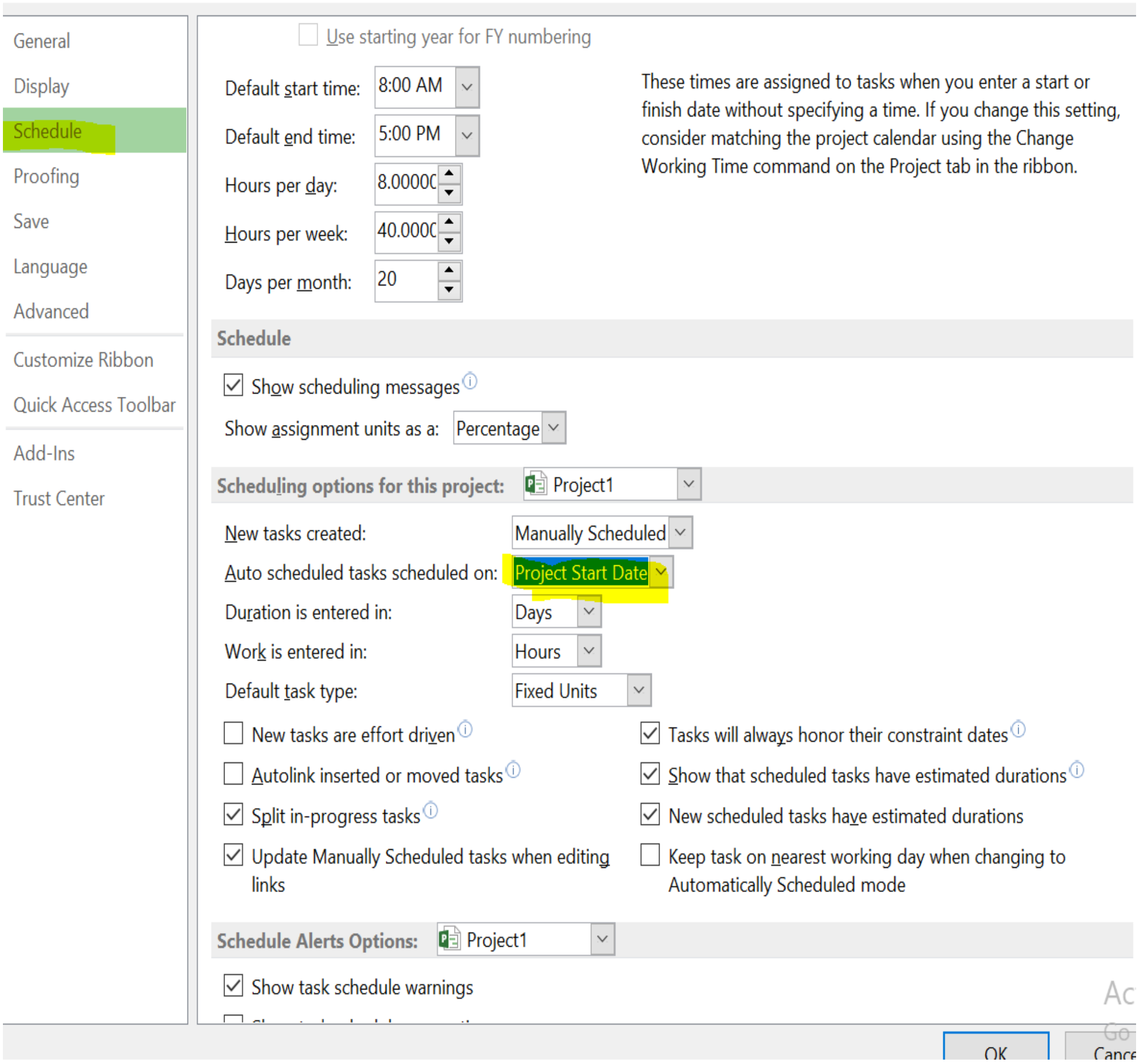


Figure 4.2: Project Schedule

HOW TO SET WORKING & NON-WORKING TIME?

Click Project > Properties > Change Working Time.

With the calendar marked as (project calendar) selected from the For calendar list, click the Work Weeks tab, and then click Details.

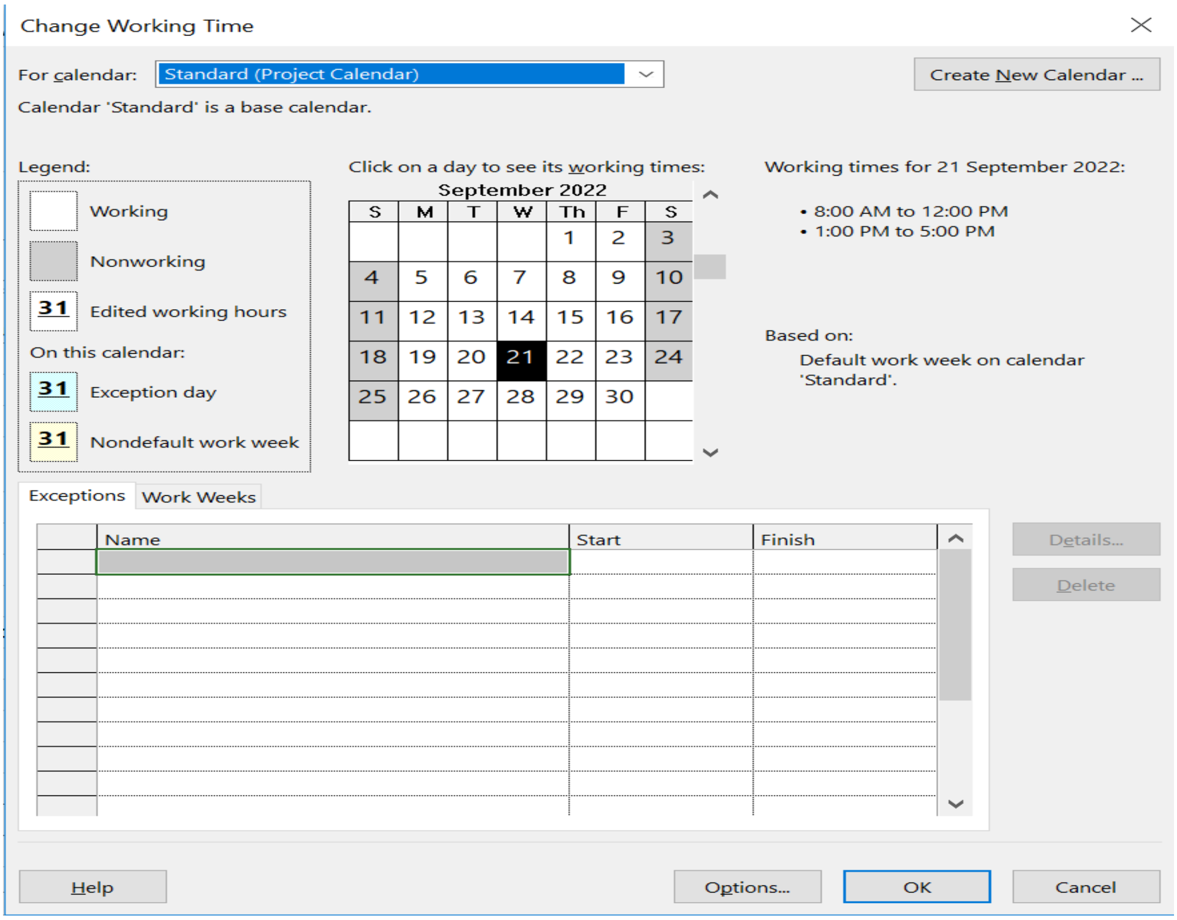


Figure 4.3: Working Time

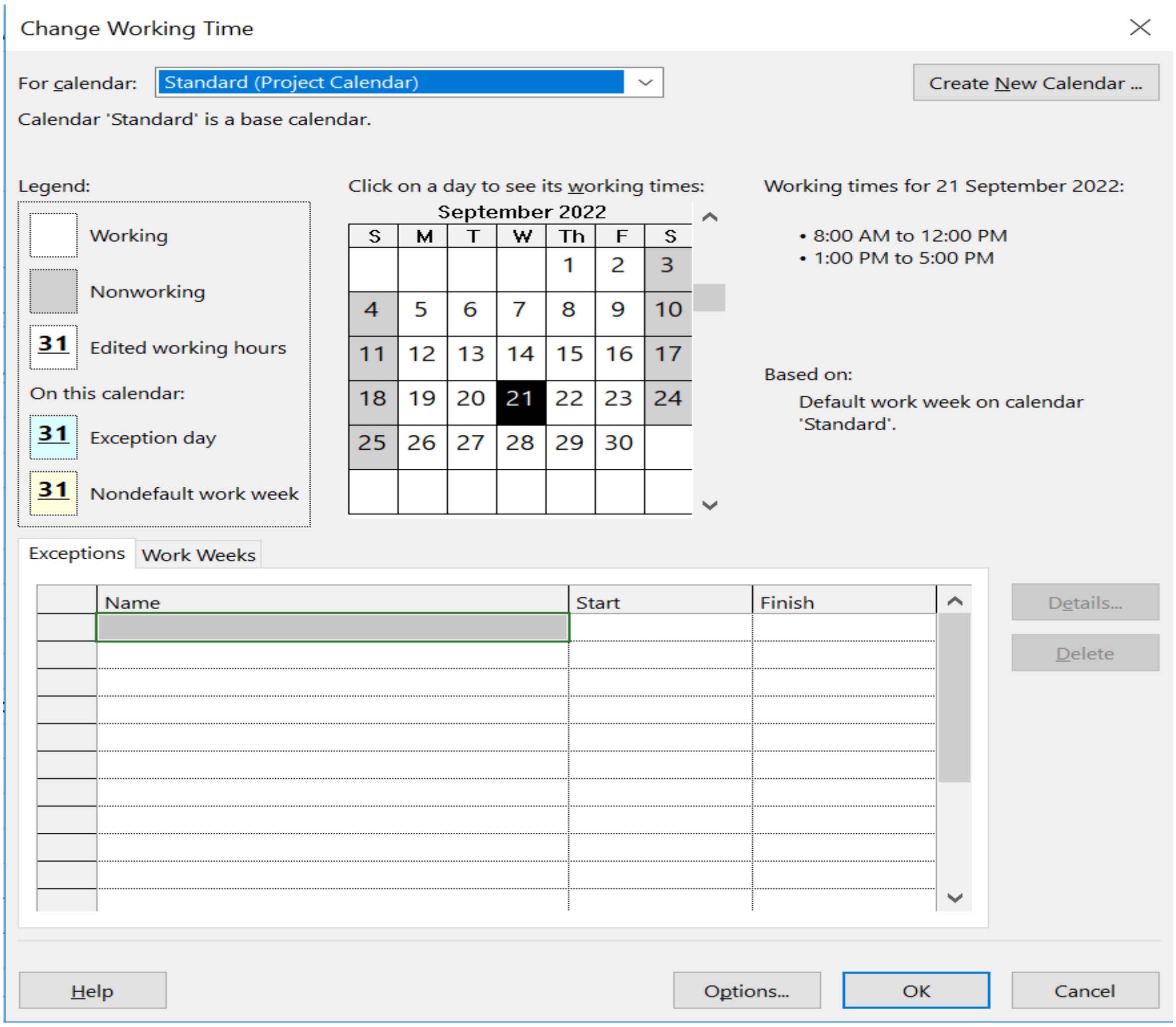


Figure 4.4: Change Working Time

HOW TO SET PROPERTIES ABOUT A PROJECT PLAN?

1. Click on file
2. Click on info
3. Then you can change properties of project here

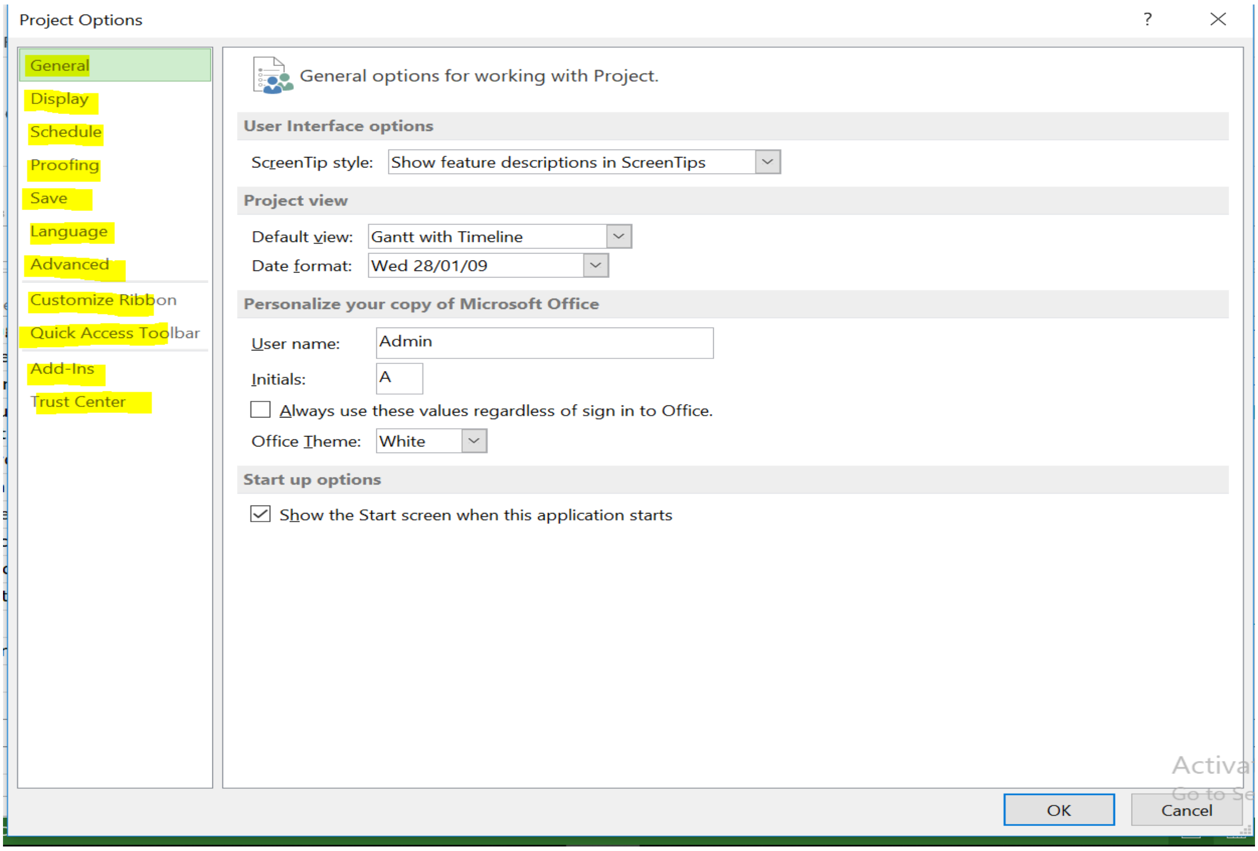


Figure 4.5: Project Options

HOW TO ENTER TASKS, TASK DURATION & MILESTONE IN A PROJECT?

1. Select manually or auto scheduling
2. Add starting date and ending date
3. Add predecessor

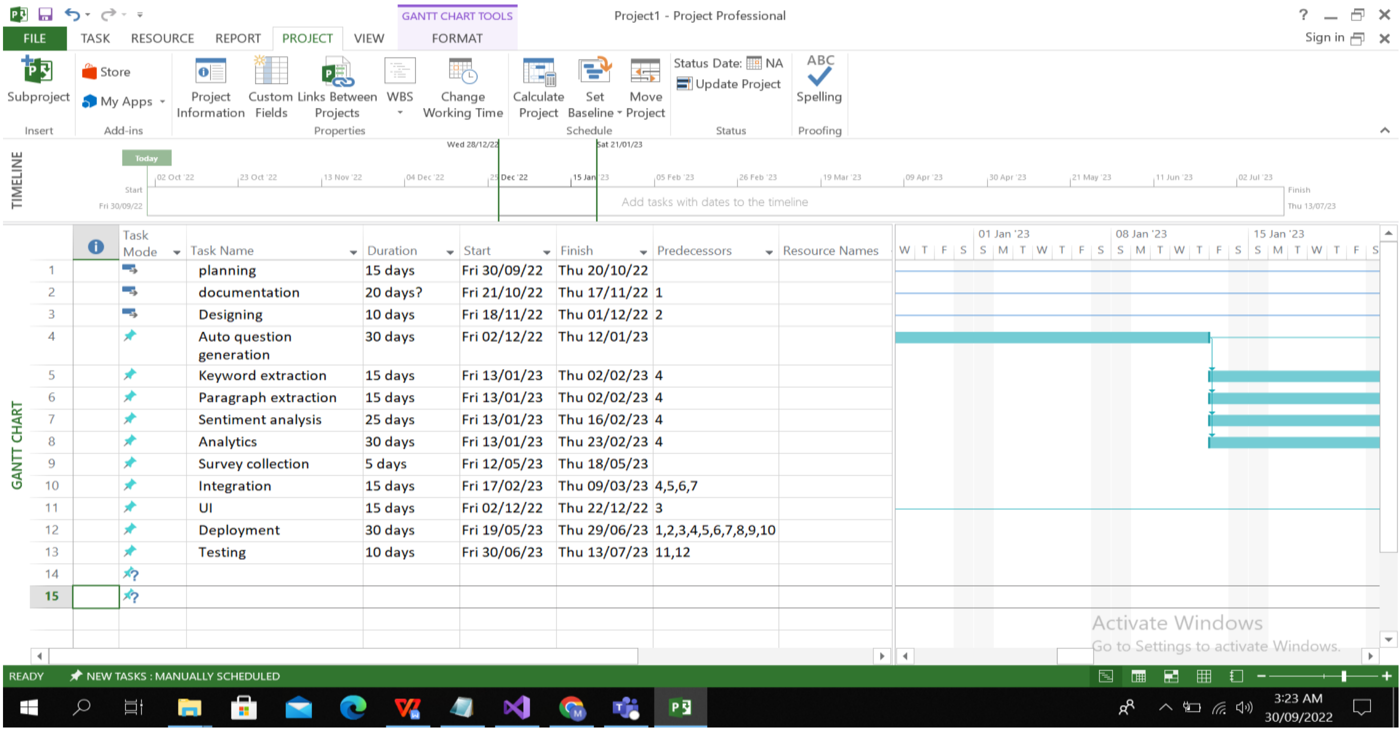


Figure 4.6: Project Milestone

How To Organize Tasks Into Phases?

1. Select task you want to divide in phases
2. Then click on task tab
3. Click on indent option

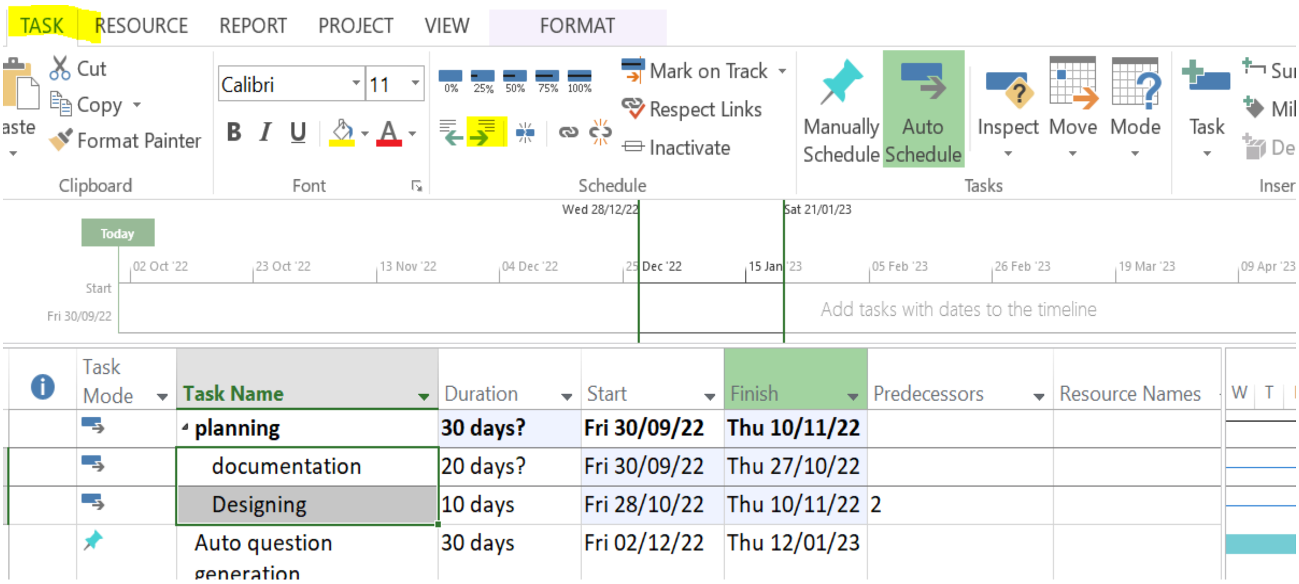


Figure 4.7: Phases

3) Graded Lab Tasks

Note: The instructor can design graded lab activities according to the level of difficult and complexity of the solved lab activities. The lab tasks assigned by the instructor should be evaluated in the same lab.

Lab Task 1

You are tasked with creating a project plan for organizing a team training workshop. Follow the steps to create a project plan using Microsoft Project Professional. Apply appropriate filters to focus on specific task groups, and then generate a customized tasks list for presentation.

Lab Task 2

You have been given the responsibility to manage a software development project. Create a project plan for developing a new software application. Apply filters to view tasks with specific characteristics, and then generate a filtered tasks list for presentation.

Lab # 05

Develop Work Break Down structure using MS project professional and Trello

Objective:

The objective of this lab is to familiarize participants with two popular project management tools, Microsoft Project Professional and Trello. During the lab, participants will learn how to create a structured WBS for a project using Microsoft Project Professional, breaking down the project into manageable tasks and subtasks. Additionally, participants will explore how to use Trello to organize and visualize the WBS in a collaborative and agile manner, enabling effective task tracking, assignment, and progress monitoring.

Activity Outcomes:

After successful completion of this lab, the students will be able to create a structured WBS for a project using Microsoft Project Professional, breaking down the project into manageable tasks and subtasks. Additionally, students will be able to organize and visualize the WBS in a collaborative and agile manner, enabling effective task tracking, assignment, and progress monitoring.

Instructor Note:

As pre-lab activity, read Chapter 16 from the book "Microsoft Project 2019 Step by Step, Cindy Lewis, Carl Chatfield, Timothy Johnson., Microsoft Press, 2019".

1) Useful Concepts:

Even some of the most experienced construction project managers and schedulers face challenges with scheduling and scheduling using *MS Project*. However, creating a construction project schedule in *MS Project* can be complex, but it doesn't have to be difficult. Learning how to create an MS Project schedule can then be helpful to both seasoned professionals as well as beginners.

Below, we are going to walk you through step-by-step how to build and create your own construction schedule using *MS Project* that's specifically customized for construction. This lab will teach you how to create a new schedule, how to design your columns (or what data points to capture unique to construction), what fields to include, how to create a baseline so you can track *planned vs. actual*.

In Trello, a work breakdown structure (WBS) is a visual framework that allows users to organize and structure tasks within a project or initiative. While Trello primarily employs a card-based system for task management, users can create a virtual WBS by leveraging lists and cards. Lists represent different stages or phases of a project, while cards represent individual tasks or activities within those stages. By using Trello's intuitive drag-and-drop interface, users can easily arrange and rearrange cards to reflect the hierarchical relationship between tasks. This approach offers a clear overview of the project's scope, tasks, and their dependencies, making it simpler to manage and monitor progress. Additionally, labels, due dates, attachments, and comments can be added to cards to provide more context and information, enhancing collaboration and communication among team members.

2) Solved Lab Activities

Sr.No	Allocated Time	Level of Complexity	CLO Mapping
1	30	Low	CLO-6
2	30	Low	CLO-6
3	30	Low	CLO-6

ACTIVITY 1

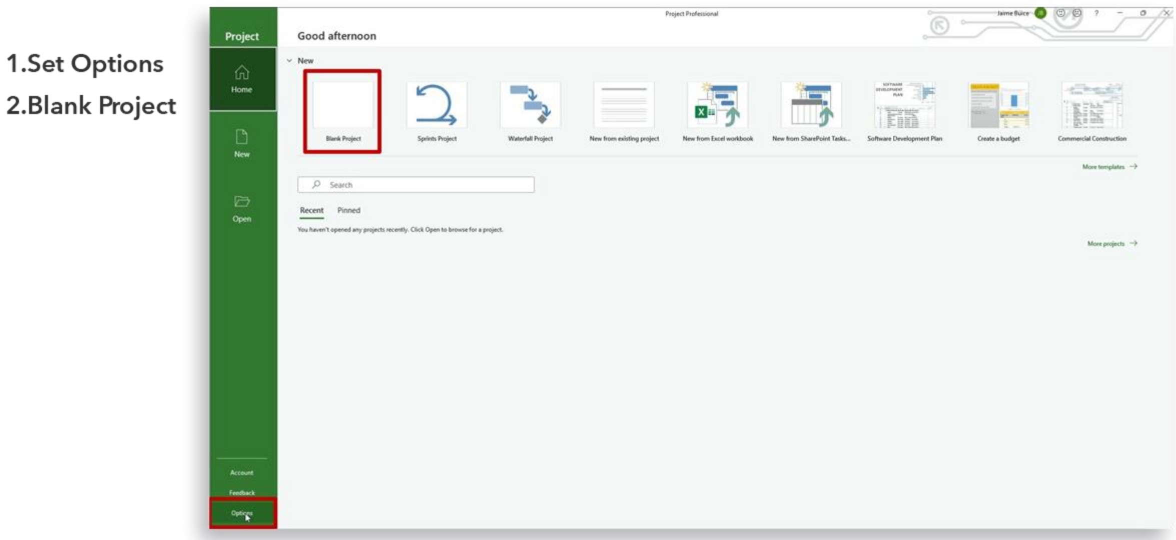


Figure 5.1: Opening Project

Create a New Project

1. First, you need create a project. You can do so in several ways:

- From an industry template
- Brand new from scratch (blank slate that you populate)
- Based on an existing project
- Open from a previous vs. of MSP

2. Second, set your options by clicking '*Set Options*' in the lower-left corner.

   1. Navigate to '*Schedule*.'
   2. Then, check '*Auto-Schedule,'* this is very important.important.

3. Under '*Default Pass Type'*, make sure also to select the following as well:
    1. Split in-progress tasks.
    2. Update manually scheduled tasks when updating links.
4. Select Blank Project (SmartPM) and assign a name.

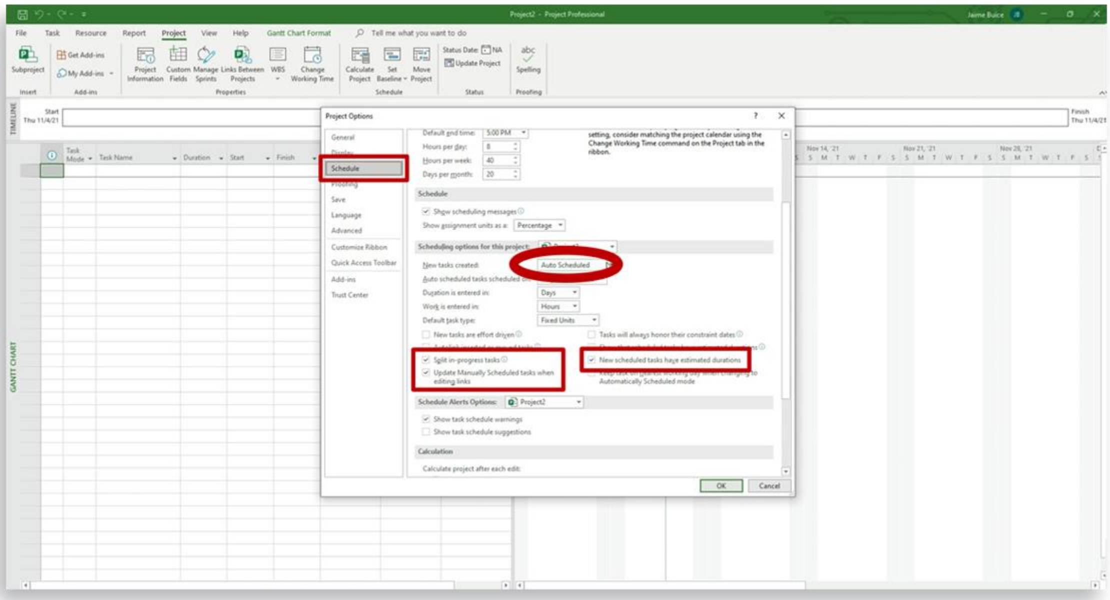


Figure 5.2: Project Properties

Adjust Construction Schedule Columns

Next, you want to adjust the auto-populated columns.

1. Remove or "hide" the '*Resources'* column temporarily.
2. Then, add* (+) the following new columns:
    1. *% complete*
    2. *Successors*
    3. *Total Slack* (float)
3. Last, rearrange the columns so that percent *complete* is before the *predecessor*.

* Remember, there are over 500 auto-populated choices to choose from. Simply begin typing in the column header, and *MS Project* will fill it in for you.

Add Schedule Start Date

The last thing you need to do is to add a 'start date' for your project. Then, you're ready to begin filling in the schedule.

1. Go to the project tab at the top.
2. Select '*Project Information*.'
3. Go to the '*Start Date*' field, and from the pull-down calendar, select your start date.

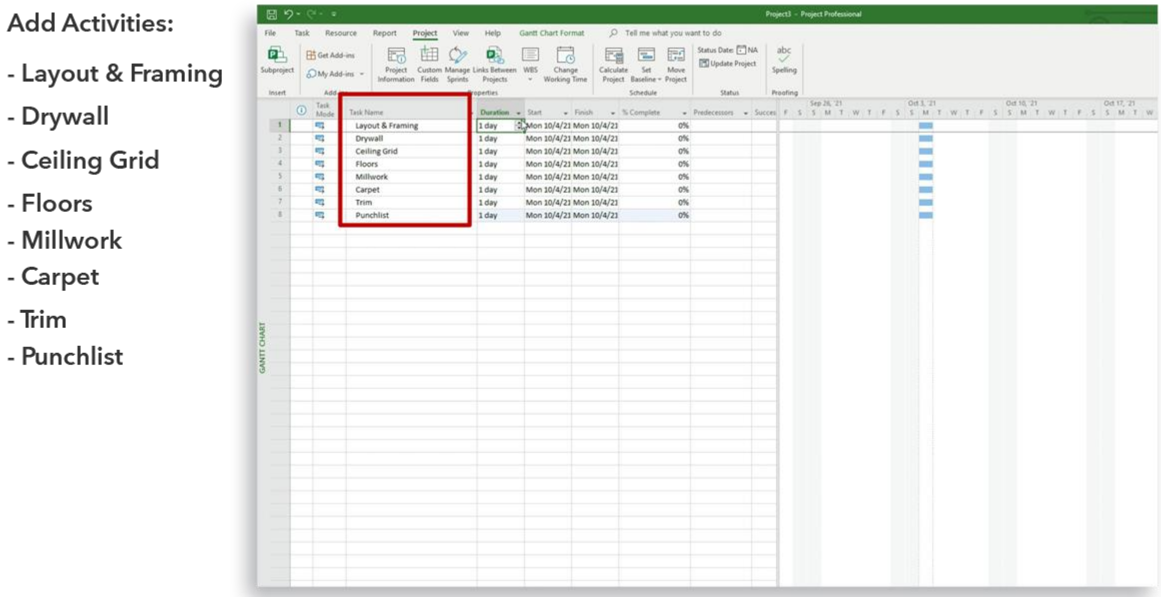


Figure 5.3: Project Information

Add Schedule Tasks and Activities

Once your project is created, add tasks and activities to your project:

1. Go to the column titled, '*Task Name'* and begin populating each field with the key work required to complete your project. Some key tasks here for construction might include:
1. Layout & Framing
2. Drywall
3. Ceilings
4. Floors
2. Next, go into the *'Durations"* column. Change the default of '*1 day*' to the appropriate length of time that each activity will take for your unique project.

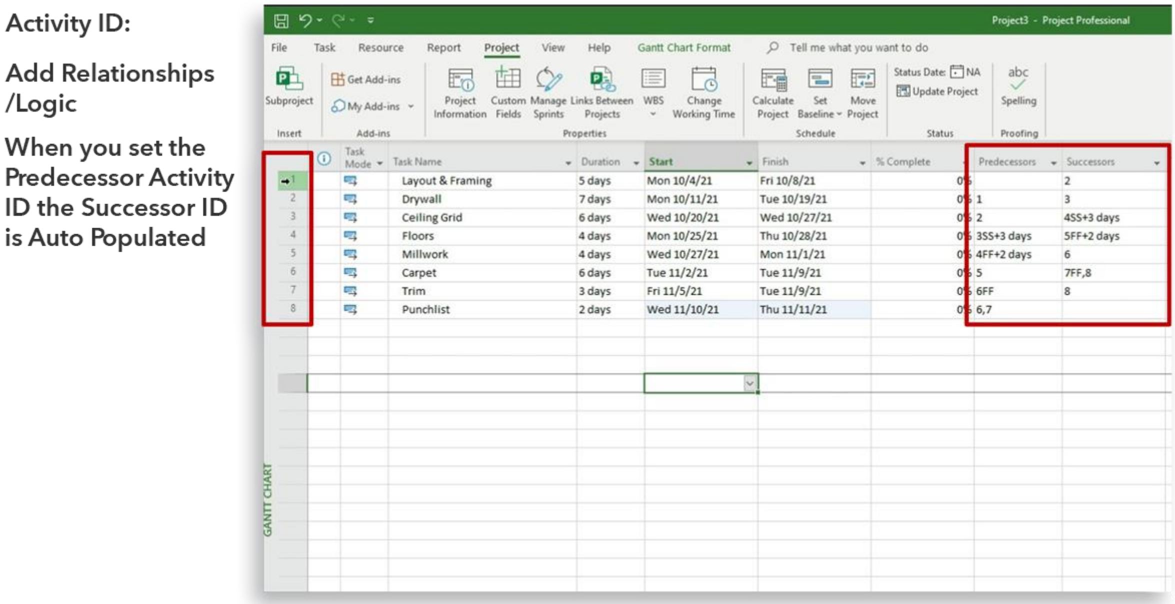


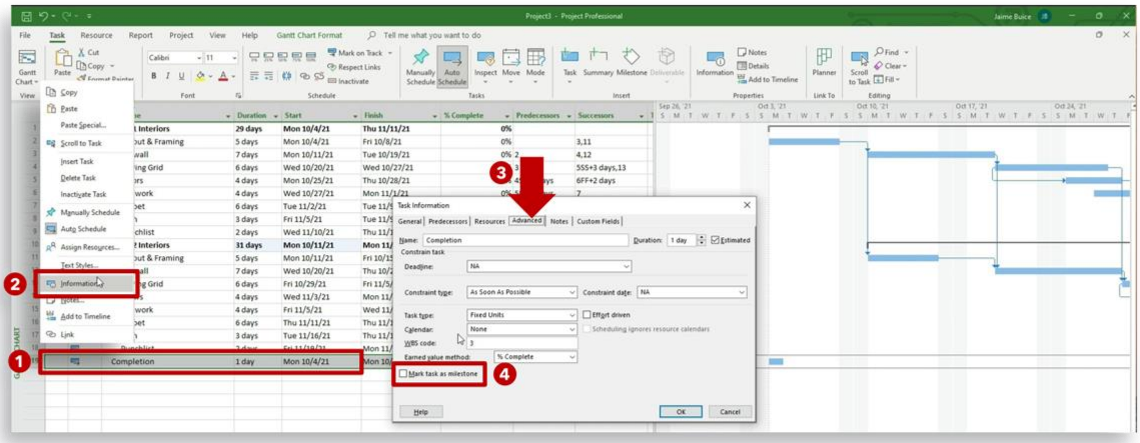
Figure5.4: Tasks and Activities

Order Schedule Tasks and Define Relationships

After you have created your construction work activities and added their durations, you'll want to establish the relationships between each task. Tell MS Project what order they should occur in and which one should follow the next. We do this by adding tasks dependencies:

1. Go to the predecessor and successor columns.
2. Take the *'Line* Item' number of the task and add that number to either the predecessor column or the successor columns
1. For example, for task #2, Drywall, you would put the number 1 in its predecessor column. For Layout & Framing (task #1), you would put the number 2 in its successor column
3. After that, you then want to include any start-to-start logic or finish-to-finish logic in these two columns, including any lag time if necessary*
4. If you've missed a task or the plan has changed, simply select the row where you want the new task and select 'insert a new task' from the pop-up menu. Type in the name of the new activity.

*Never adjust the Start and Finish Dates while doing this. MS Project will perform those functions and calculations for you.



1. Right Click Completion Line
2. Select Information
3. Advance Tab
4. Check Milestone Box

Figure 5.5: Tasks Ordering

Create Construction Schedule Wbs, Summary Bars & Milestones

You could continue creating a long list of job tasks. However, at this stage, it might be important to create "*summary bars*" to refine task relationships. Summary bars or WBS (work breakdown structures) are a good way to "bucket" work activities. All tasks that complete a larger activity or milestone can be grouped together. To do this, you insert a new task (or modify a previous one), give it a name, and then "indent" all the sub-tasks below it. Those changes and new relationships will update on the Gantt chart view.

To make any *summary bar* a *milestone*, you have to indicate to *MS Project* that it should now be considered a milestone:

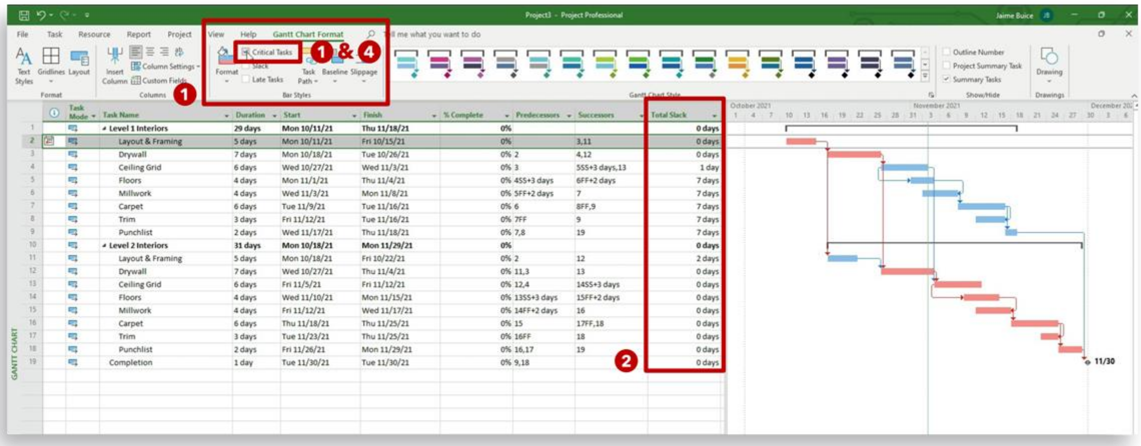
1. Double-click on the task.
2. Select "Advanced" from the pop-up menu.
3. Mark as "Milestone" by clicking the check box.

Add Schedule Constraints

As we all know, even with the best-laid plans, things often go awry. Although not recommended, this is where "constraints" come in. When something that is scheduled for a certain date can no longer start on that date, as an example, you would need to go back into that task and put a constraint on it:

1. Click on the task.
2. Select Advanced from the pop-up menu.
3. Go to the '*Constraint Type'* pull-down menu and select the appropriate constraint.
4. Change the date.

For our example above, you would select the '*Must Start On'* constraint and then select the new date it should start on.



1. Under Gantt Chart Format Click Critical Tasks
2. Total Slack determines the criticality of each activity- Total Slack is a Program Calculation
3. Activities on the Critical Path are now Red
4. Uncheck Critical Tasks to close Critical Path View

Figure 5.6: Constraints

ACTIVITY 2

4 Key Power-Ups Used For Trello Gantt Charts

Power-Ups predominantly have helped Trello become popular among the other tools. With Power-Ups, additional features can be integrated to give additional functionalities in a Trello Board. To build a Trello Gantt Chart, below are a few Power-Ups available in Trello:

Planyway

Planyway helps in glancing at the entire work of an organization. It visualizes plans and breaks big tasks into simple steps by building a team schedule. This software also assists in balancing workloads mapping all tasks and subtasks allotted on a timeline view. One can also create a roadmap and set dependencies to mark the prior work importance before starting a new work. Planyway also improves productivity by estimating the task completion time, with Time Tracking feature managers deep-dive to find prolificity of team members.

Teamgantt

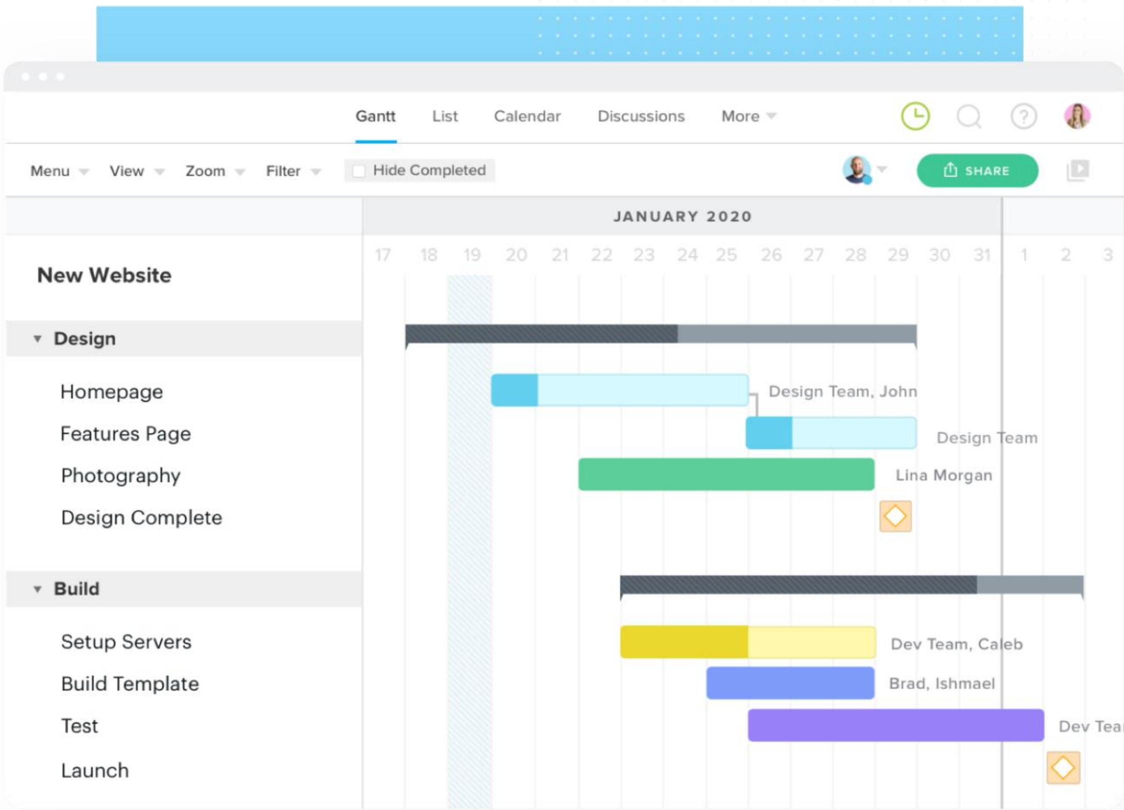


Figure 5.7: Gantt Chart in TeamGantt

TeamGantt is a Project Scheduling software used to make Trello Gantt Charts online for decision-makers to get a pulse of a project. With its collaboration feature, one can invite team members or clients to plan, manage, or schedule a project. As it comes with a drag and drop interface, building Trello Gantt Charts using TeamGantt requires no training. All the necessary files, documents related to tasks are made available in one place, thus avoiding unnecessary emails and discussions. TeamGantt is mobile, and any change in progress or comments gets updated on other devices. However, a user can even import these timelines as CSV and print PDF to pitch meetings' plans.

Bigpicture

BigPicture is a Project Portfolio Management (PPM) Power-Up tool in Trello. It helps in Planning Tasks, Automating Work, and Tracking Project Progress. It comes with a Work Breakdown Structure (WBS) panel while working on Gantt Charts to distribute tasks evenly. As most projects may come with teams with various methodologies, BigGantt features all the Milestones, Auto Schedules, and Baselines with critical paths for project completion.

Projects By Placker

Placker offers flexible Project Management and an extensive set of features for making a Gantt Chart. Placker and Trello Board are bidirectionally connected. Any changes in the former reflect the latter. It also helps users to effectively plan and track Cards on single or multiple Boards.

Steps to Create Trello Gantt Charts Using Teamgantt

Trello Boards consist of Cards arranged in Kanban style. With TeamGantt Power-Up, one can easily synchronize Trello Boards manually or automatically to create Trello Gantt Charts. Below are the steps to create a Trello Gantt Chart using TeamGantt:

- **Step 1:** Search for **TeamGantt** in the Power-Up section of the Trello account, and click on the '**Add Power-Up**' button. One can also add power-up from the 'show menu option' on the Trello board.
- **Step 2:** TeamGantt plugin can prompt users to make an account before it gets added automatically in the Trello Board. To connect a Trello Board, a user needs to create a blank project in TeamGantt and click on '**View in Gantt chart**'.

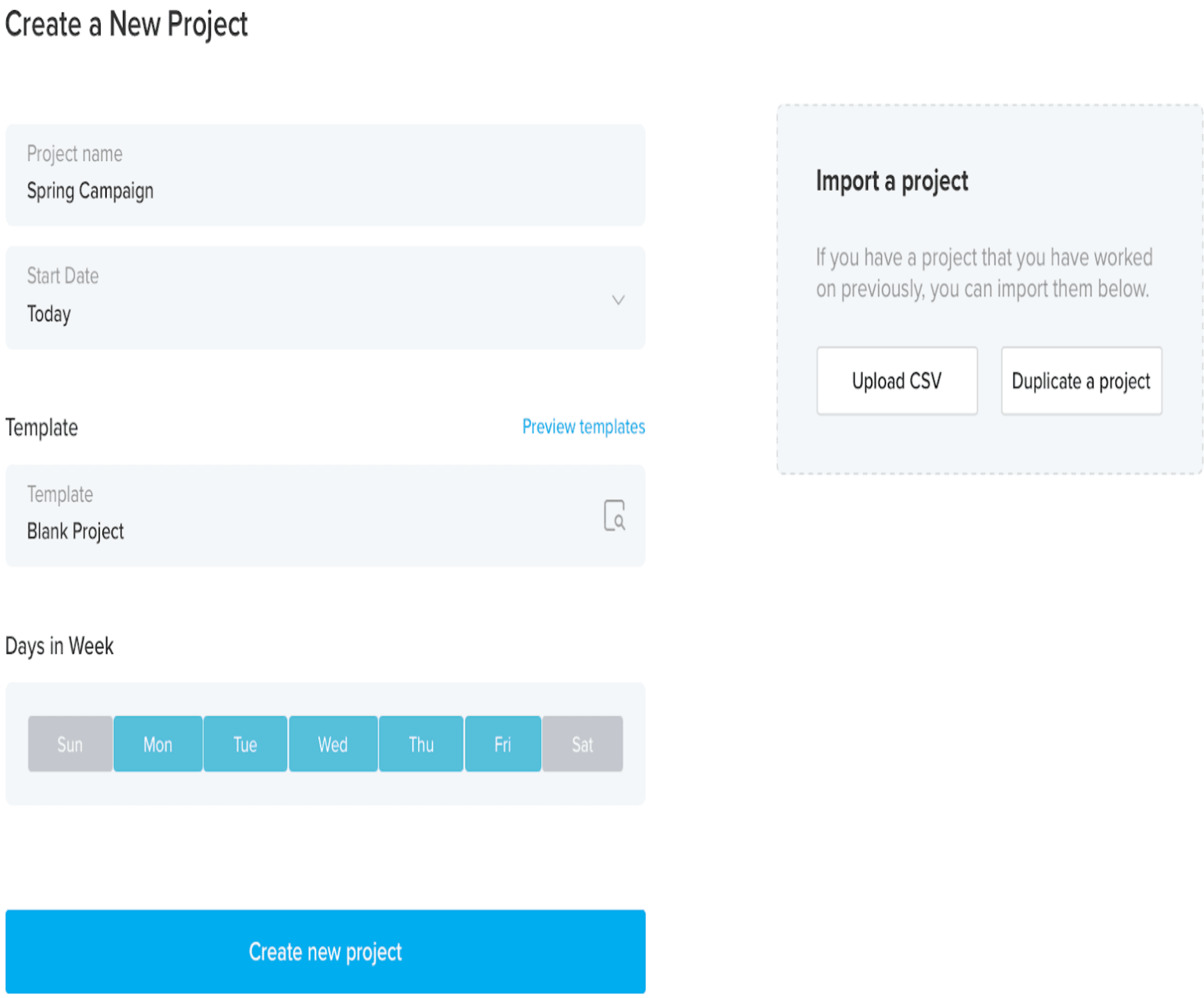


Figure 5.8: Create new project

- **Step 3:** All the project tasks are arranged in a list form, and the schedule of the tasks can be added on selecting a particular task in the Gantt view of the TeamGantt Board (left side).

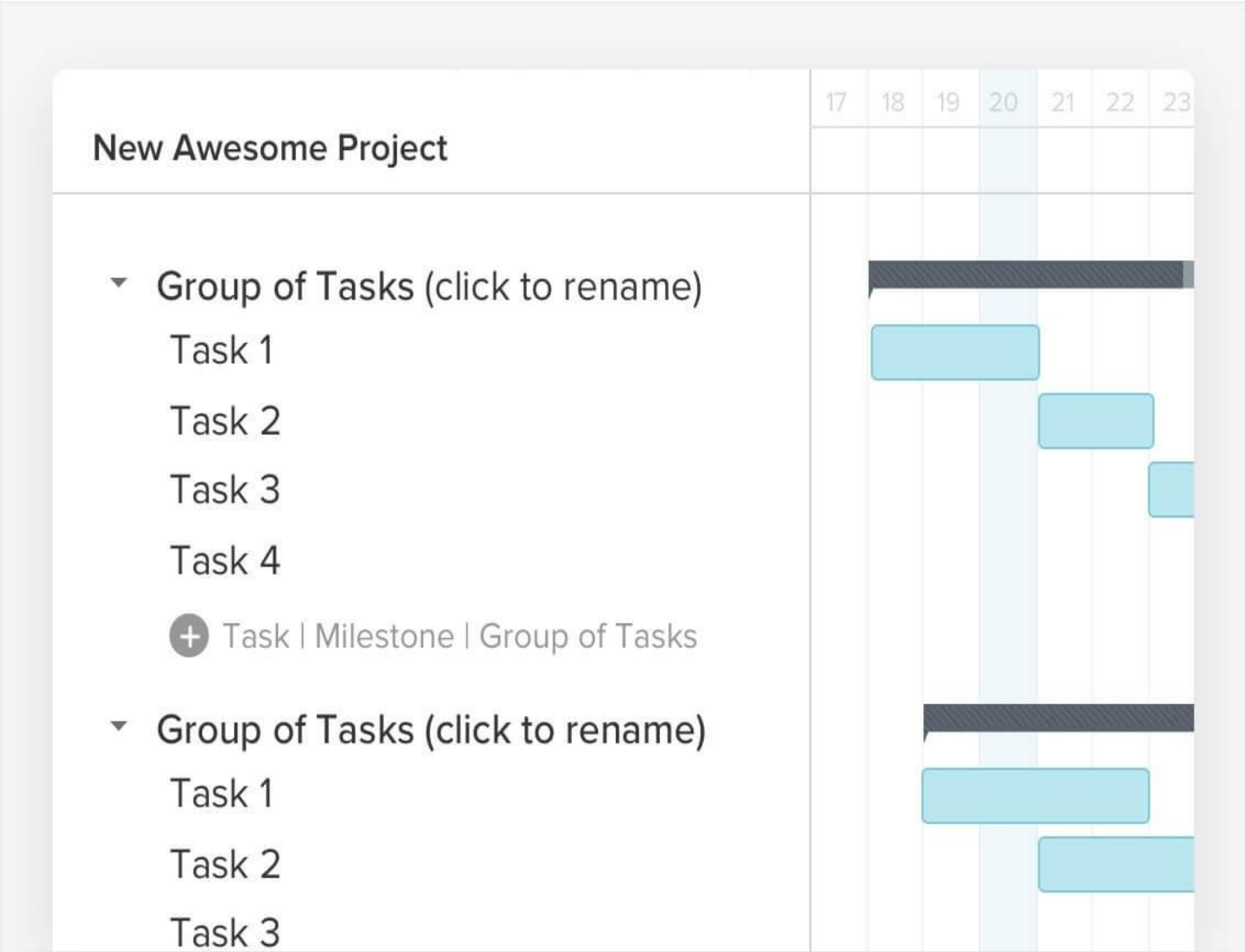


Figure 5.8: Task list and schedule in TeamGantt

- **Step 4:** Hover the mouse on the Board to create a Trello Gantt chart (default of one day), and this can be later extended by dragging either side to suit the requirements of tasks.

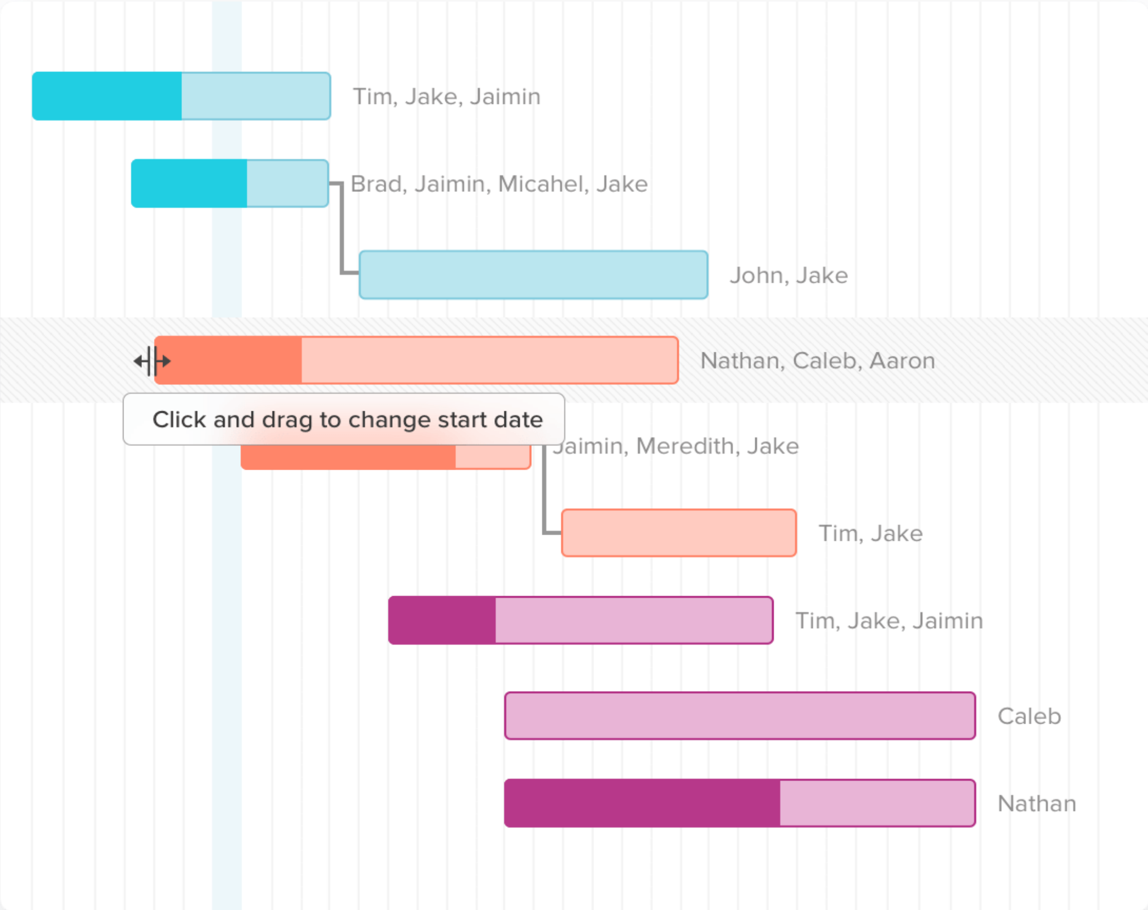


Figure 5.9: Different tasks

- **Step 5:** Upon hovering over a particular task, task progress can be updated by clicking the '**Edit Task**' option. A dependency can also be added by clicking a dot and connecting it to the dependent task.
- **Step 6:** A task in TeamGantt can be assigned a resource by selecting the person logo (last option) and selecting relevant names for a task. The color of a task can also be customized by selecting the blue square box.

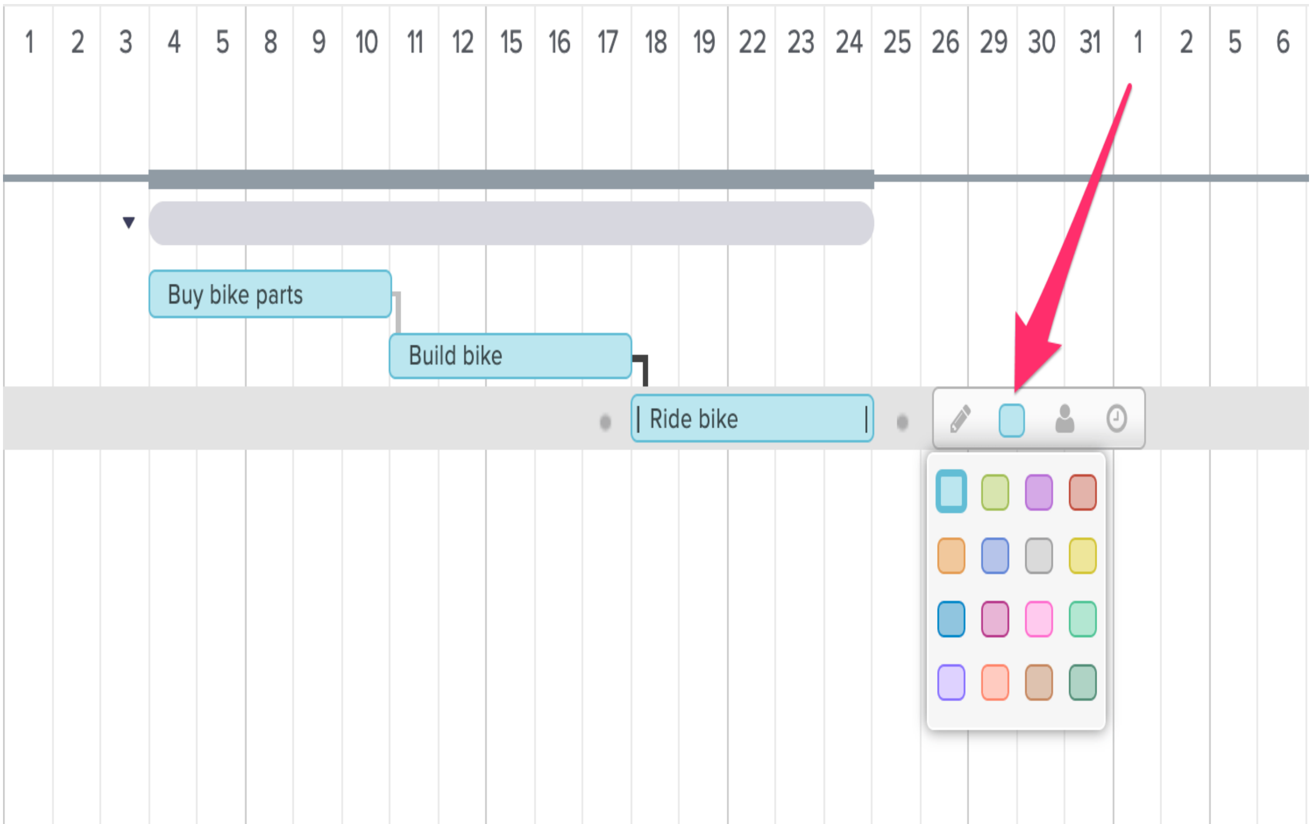


Figure 5.10: Different Task Customizations

Advantages of Trello Gantt Charts

Advancement in Cloud Technology has made this Chart become more flexible, and below are a few benefits exhibited by Trello Gantt Charts:

- **Visualize Project:** A schedule can be prepared using a simple table. However, Trello Gantt charts visualize the entire project to give better insight. It also visualizes some additional features like Assigned Tasks and Added Dependencies.
- **Productive Work:** Certain tasks have high priority, team collaboration becomes necessary to extensively solve them. As the tasks are to be completed within a stipulated time, a Trello Gantt chart provides high-level visibility to increase productivity.
- **Resource Management:** A project, once scheduled, requires discussion among team members. All necessary files and documents during the meeting are shared as email or attachments in other software. Such practice often leads to data duplication or data loss. However, modern Trello Gantt charts have a separate space to store all relevant data like discussions and documents to decrease complexity.

3)GRADED LAB TASKS

Note: The instructor can design graded lab activities according to the level of difficult and complexity of the solved lab activities. The lab tasks assigned by the instructor should be evaluated in the same lab.

LAB TASK 1

Provide a detailed construction project plan with various tasks like excavation, foundation, framing, plumbing, electrical work, and finishing. Students should establish realistic task dependencies and optimize the schedule for efficiency, considering the dependencies between different trades and regulatory requirements.

LAB TASK 2

Create a scenario where a software development project involves tasks like requirements gathering, design, coding, testing, and deployment. Students should establish dependencies based on the software development life cycle and allocate resources effectively, considering that certain tasks might require specialized skills.

Lab # 06

Developing the schedule and Establishing Task Dependencies using MS project professional and Trello

Objective:

This lab allows you to create detailed project plans by breaking down tasks, estimating durations, assigning resources, and defining task dependencies using MS Project and Trello. Establishing task dependencies helps in determining the order in which tasks should be executed. This ensures that tasks are completed in a logical sequence, avoiding unnecessary delays or conflicts.

Activity Outcomes:

After successful completion of this lab, the students will be able to:

- Create a new project
- Set the project start or finish date
- Set the name or title and other file properties
- Add tasks
- Organize tasks
- Set up the project calendars
- Save and publish the schedule

Instructor Note:

As pre-lab activity, read Chapter 4 from the book “Microsoft Project 2019 Step by Step, Cindy Lewis, Carl Chatfield, Timothy Johnson., Microsoft Press, 2019”.

1) Useful Concepts:

Developing schedules and establishing task dependencies are fundamental concepts in project management, critical for ensuring projects are completed efficiently and on time. Both Microsoft Project Professional and Trello offer unique approaches to these concepts, each with its own set of useful features and concepts.

In **Microsoft Project Professional**, the concept of **Work Breakdown Structure (WBS)** plays a pivotal role. A WBS involves breaking down the project into smaller, manageable tasks, creating a hierarchical structure that helps visualize the project's scope. This structure forms the foundation for scheduling and task dependency establishment. Task **dependencies** are relationships that define the sequence in which tasks must be executed. These can include **Finish-to-Start**, where one task must finish before the next starts, or more complex relationships like **Finish-to-Finish** and **Start-to-Start**. These dependencies establish the logical flow of work and help determine the project's **critical path**, the sequence of tasks that directly impact the project's duration.

In contrast, **Trello** adopts a more visual and agile approach. Instead of rigid task dependencies, Trello encourages the concept of **Kanban boards**. These boards visualize tasks as cards that move through different stages of a process, such as "To Do," "In Progress," and "Done." While Trello doesn't inherently offer complex dependency management, it can be augmented with Power-Ups or add-ons to introduce basic dependencies. However, **prioritization** and **visualizing workflows** are key concepts in Trello. By arranging tasks in order of priority and moving cards through different stages, teams maintain a dynamic view of their project's progress.

Both tools emphasize the importance of **resource allocation and management**. In Microsoft Project, you can assign specific resources to tasks, ensuring that workloads are balanced and resources are optimally utilized. In Trello, the concept of collaboration is highlighted, allowing team members to communicate, attach files, and discuss tasks on each card.

Lastly, both tools foster the idea of **continuous improvement and adaptability**. In Microsoft Project, the ability to simulate different scenarios helps in understanding the effects of changes on the project timeline. In Trello, the agile nature of the Kanban board allows teams to adapt quickly to changes, continuously optimizing their workflow.

In conclusion, while Microsoft Project Professional emphasizes detailed planning, complex scheduling, and critical path analysis, Trello focuses on visualizing workflows, collaboration, and flexibility. Understanding these concepts within the context of each tool enables project managers to choose the most suitable approach based on the project's complexity, team dynamics, and management style.

2) Solved Lab Activities:

Sr.No	***Allocated Time***	***Level of Complexity***	***CLO Mapping***
1	***30***	***Low***	***CLO-6***
2	***30***	***Low***	***CLO-6***
3	***30***	***Low***	***CLO-6***
4	***30***	***Low***	***CLO-6***

ACTIVITY 1

Start with an Existing Project

1. Open the project or template you want to use as the basis for the new project.
2. Click **File** > **Info**.
3. Under **Project Information**, change the project’s start date and finish date.
4. Click **Save As** and pick a new name and a new home for the project.

You can start working on your new project right away, but chances are you’ll need to clean up some of the existing project information before you do.

One way around this clean up work, especially if you want to use the old file to create more than one project, is to save the old project as a template. You have the option of stripping out all that progress information as you save it, leaving only tasks and resources behind.

Start With a Template

1. Click **File** > **New**.
2. Search for templates in the box or click the template you want below.
3. In the preview dialog box, click **Create**.

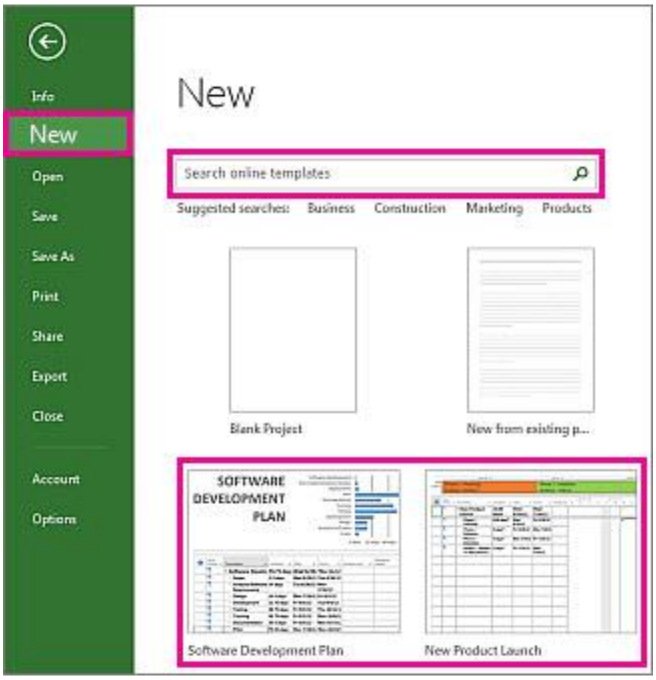


Figure 6.1: Project Template

Make sure that the resources, tasks, and durations in the template are right for your project. Of course, you’re now free to change them as you see fit. More templates are available on Microsoft templates.

Insert A Task Between Existing Tasks

1. Select the row *below* where you want a new task to appear.
2. Select **Task** > **Task**.

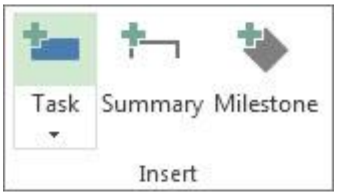


Figure 6.2: Task

3. Type the task name in the inserted row.

The task IDs are automatically renumbered, but the inserted task isn't automatically linked to the surrounding tasks. You can set Project to automatically link inserted tasks to the surrounding tasks.

Add a Task to a Network Diagram

1. Select **View** > **Network Diagram**.

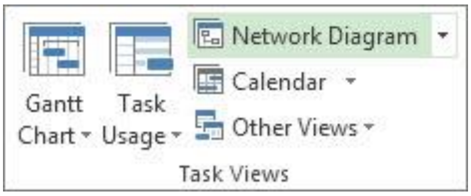


Figure 6.3: Network Diagram

2. Select **Task** > **Task**.

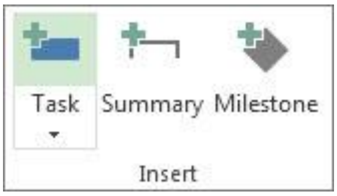


Figure 6.4: Task

3. Type the task name in the new task box.

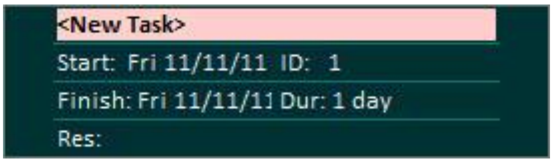


Figure 6.5: Task Name

Add Multiple Tasks at One Time

The Task Form can help you add several tasks at one time, especially if the tasks have resource assignments and task dependencies.

1. Select **View** > **Gantt Chart**.

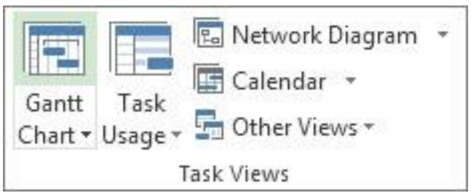


Figure 6.6: Gantt Chart

2. Select the **Details** check box.

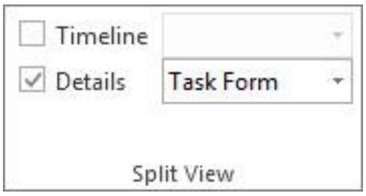


Figure 6.7: Details

The window splits, showing the Gantt Chart on top and the Task Form on the bottom.

3. In the Gantt Chart (top), click the first empty row at the end of the task list.
4. In the Task Form (bottom), type information about the new task:
   - In the **Name** box, type the new task's name.
   - In the **Duration** box, add the task duration.
   - If you want the task duration to stay fixed even if resource assignments are changed, check **Effort driven**.
   - Add details about the task in the form columns (such as the assigned resources and predecessor tasks).
5. Click **OK** to save the new task, and then click **Next** to move to the next row in the Gantt Chart.

Create a Recurring Task

1. Click **View** > **Gantt Chart**.
2. Select the row **below** where you want the recurring task to appear.
3. Click **Task**, click the bottom part of the Task button and then click **Recurring Task**.

Figure 6.8: Recurring Tasks

4. In the **Task Name** box, type the recurring task’s name.
5. In the **Duration** box, add the duration of each occurrence of the task.
6. In the **Recurrence pattern** section, click **Daily**, **Weekly**, **Monthly**, or **Yearly**.

   You can fine-tune how often the task repeats in the area to the right of these options. For example, you can create a task that repeats every Tuesday and Thursday, or one that repeats every three weeks.

7. In the **Start** box, add a start date and then decide when the repeating task will end:
   - Pick **End after**, and then type the number of times the task will repeat.
   - Pick **End by**, and then enter the date you want the recurring task to end.
8. Pick an item from the **Calendar** list, but *only* if you want the recurring task to have a different calendar than the rest of the project.

   For example, the recurring task could happen during the night shift, while the rest of the project happens during daytime business hours.

9. Check **Scheduling ignores resource calendars** if you want Project to schedule the recurring task even if it does not happen when any resources are available to work on it.

Create a Milestone with Zero Duration

The quickest way to create a milestone is to add a task with no duration to your project plan.

1. Click **View**, and then in the **Task Views** group, click **Gantt Chart**.

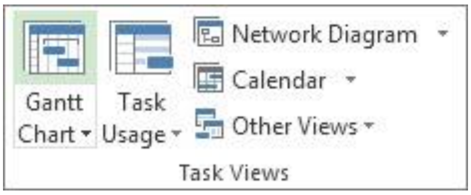


Figure 6.9: Tasks Views

2. Type the milestone name in the first empty row or pick a task you want to turn into a milestone.
3. Type **0** in the **Duration** field, and then press Enter.

The milestone symbol is now part of your Gantt Chart.

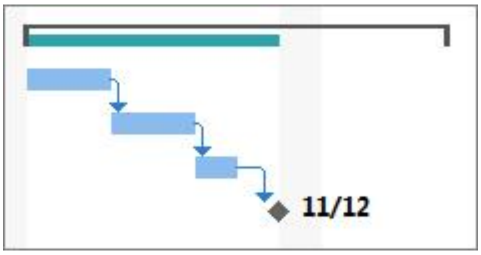


Figure 6.10: Milestone

Add A Milestone With Duration

Sometimes a milestone takes time. For example, the approval process at the end of a phase might take a week, so that milestone would need to take place over time like a normal task.

1. Click **View**, and then in the **Task Views** group, click **Gantt Chart**.

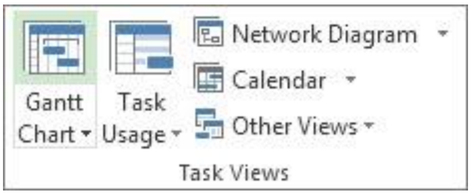


Figure 6.11: Tasks Views

2. Type the milestone name in the first empty row or pick a task you want to turn into a milestone.
3. Select the milestone, and then click **Task**. In the **Properties** group, click **Task Information**.

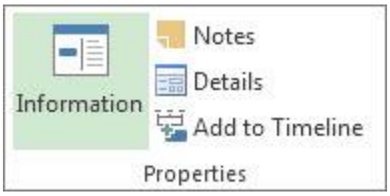


Figure 6.12: Task Information

4. Click the **Advanced** tab, and then type the milestone duration in the **Duration** box.
5. Check **Mark task as milestone**, and then click **OK**.

On the Gantt Chart, the milestone symbol appears on the last day of the task. It doesn't appear as a bar, even though it has duration.

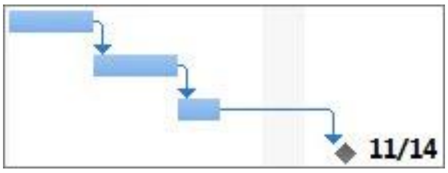


Figure 6.13: Milestone

Add an External Milestone

Sometimes you might need a milestone to track a task that's outside the scope of your project.

- **If the milestone depends on a project that is beyond your control**, such as software that is being developed by another company, create a milestone using the steps in the previous section. You'll have to keep an eye on the external task and update it manually.
- **If the milestone is part of a project in your organization**, you can track it with a cross-project link.

Change a Task Duration

You can change a task's duration at any time to reflect the actual amount of time it needs.

1. Choose **View** > **Gantt Chart**.
2. In the **Duration** column for the task, type the duration in minutes (**m**), hours (**h**), days (**d**), weeks (**w**), or months (**mo**).
3. If the new duration is an estimate, type a question mark (**?**) after it.
4. Press Enter.

Be careful about changing a duration by using the **Start** and **Finish** columns, especially for linked tasks or automatically scheduled tasks. Your changes might conflict with the task dependencies or task constraints Project is tracking, which can throw a carefully constructed schedule off track.

Change Summary Task Duration

You can change the duration of a summary task the same way you change a regular task. But be careful — changing the duration of the summary task doesn't necessarily change the durations of the subtasks.

On the Gantt Chart, a summary task bar always shows the summary task duration (black line) and the sum of the durations of its subtasks (blue bar), like in the picture below. You can see the difference between the two at a glance.

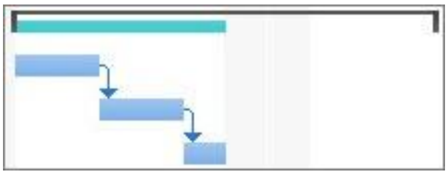


Figure 6.14: Summary Task

Use Estimated Durations

Is the question mark disappearing when you try to add an estimated task duration? Estimated durations are probably not turned on for your project. To turn them on:

1. Chose **File** > **Options**.
2. In the **Project Options** dialog box, choose **Schedule**.
3. Scroll down to the **Scheduling options for this project** area.
4. Select the **Show that scheduled tasks have estimated durations** check box.

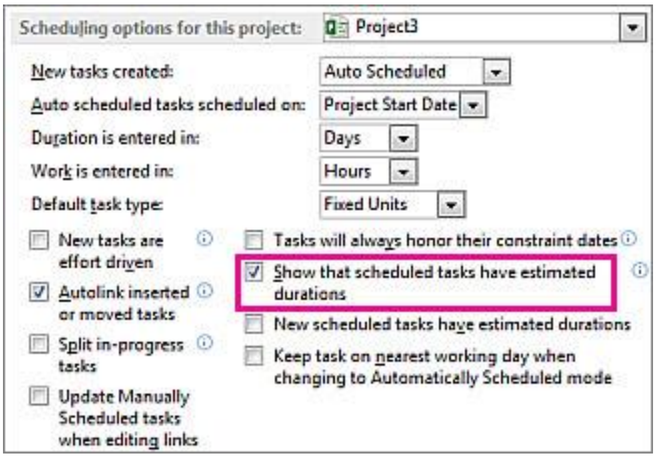


Figure 6.15: Estimated Durations

More about Task Durations

- Change the default task duration unit
- How changing the duration affects linked tasks
- How automatic scheduling can affect task durations

Duration Dependencies

The following two dependencies exist between a task's duration and its start and finish dates:

- A task's duration changes when you change its start and finish dates.
- A task's start and finish dates change when you adjust the duration.

If you enter a task's start and finish dates instead of a duration, Project calculates the duration based on the working time between those dates. If you later change the start date or finish date (but don't change the duration), the duration is recalculated.

Likewise, if you enter a task's duration, Project calculates when the task should start and finish based on the project's start date and the duration, as well as any nonworking time, such as weekends. If you later change the task's duration, Project recalculates the dates to correspond to the duration.

Examples:

- If you enter a start date of May 1 and a finish date of May 4, Project calculates the duration as three days. If you change the finish date to May 5, the duration is recalculated to four days. But if you change the finish date to May 8 instead of editing the duration, Project recalculates the start date to May 4, keeping the duration constant at four days because the last two fields that you changed were for the finish date and duration.

  If you must enter start and finish dates, and if you want the duration to remain constant, make sure the new start and finish dates represent the same number of working days as the previous dates. The duration will remain the same as before.

- If you enter a duration of three days, and the project's start date is March 1, Project calculates the task's start date as March 1 and the finish date as March 3 (unless those dates happen to fall on a weekend or other nonworking time). If you change the duration to four days, the finish date is recalculated to March 4.

  If you want to enter specific start and finish dates for a task, enter the task's start and finish dates and Project recalculates the duration. Be aware that entering the start and finish dates sets a date constraint on the task, which limits the flexibility of your schedule. If you enter a start date and then enter a finish date, a Finish No Earlier Than (FNET) constraint is set. Or if you enter a finish date and then enter a start date, a Start No Earlier Than (SNET) constraint is set. If you want a task to start or finish on a certain date but want to retain the flexibility of the schedule while being alerted if a task will not start or finish on time, you can enter a deadline date for the task. Deadlines do not constrain tasks nor do they affect constraints. As the schedule is updated, an indicator appears if a task is pushed beyond its deadline.

Create And Work With Subtasks And Summary Tasks In Project Desktop

In Project, an indented task becomes a subtask of the task above it, which becomes a summary task. A summary task is made up of subtasks, and it shows their combined information.

To create a subtask or a summary task, indent a task below another one. In the **Gantt Chart** view, select the task you want to turn into a subtask, then click **Task > Indent**.

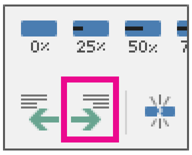


Figure 6.16: Indent

The task you selected is now a subtask, and the task above it, that isn't indented, is now a summary task.

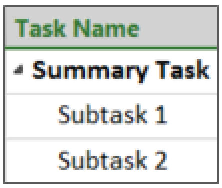


Figure 6.17: Summary Task

Tip: Click **Outdent** to move the task back to the level of the task above it.

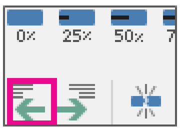


Figure 6.18: Outdent

Show or Hide Subtasks

To show or hide all subtasks for all summary tasks in Project, in the **View** tab, click **Outline** in the **Data** section, and then click **All Subtasks** to show all the subtasks or click one of the **Level** options below it to show all the subtasks up to that level.

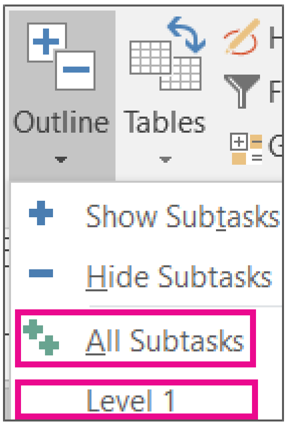


Figure 6.19: All Subtasks

To show and hide all subtasks for a single summary task, simple click the **expand** or **collapse** button to the left of the summary task name to show them or hide them respectively.

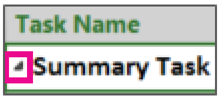


Figure 6.20: Summary tasks

Tip: Outline numbers can be useful for viewing the organization of your tasks at a glance. Learn how to add or hide them.

Tips for organizing and working with outlined tasks

If your tasks aren't indenting or outdenting, there could be a variety of reasons why. Learn more.

When organizing the tasks for a project, you should plan the outline for the project in one of two ways; the top-down method or the bottom-up method.

With the top-down method, you identify the major phases first and then break the phases down into individual tasks. The top-down method gives you a version of the plan as soon as you decide on the major phases.

- With the bottom-up method, you list all the possible tasks first, and then you group them into phases.

Note: Don't forget to link tasks after you finish organizing them. Subtasks and summary tasks create structure, but they don't create task dependencies.

- You can include the project as a summary task.

- When you move or delete a summary task, Project moves or deletes all of its subtasks. Before you delete a summary task, outdent the subtasks you want to keep.
- You can change the duration of a summary task without changing each subtask. But be careful — changing the duration of the summary task does not necessarily change the durations of the subtasks.
- Avoid assigning resources to summary tasks. Assign them to the subtasks instead, or you might not be able to resolve overallocations.
- Summary tasks don't always add up. Some summary task values (cost and work) are the total of the subtask values, others (duration and baseline) aren't. Learn more.

Inactivate a Task

When you **inactivate** a task, it stays in the project plan, but does not affect resource availability, the project schedule, or how other tasks are scheduled.

This feature is only available in Project Professional.

1. Click **View**, and then in the **Task Views** group, click **Gantt Chart**.

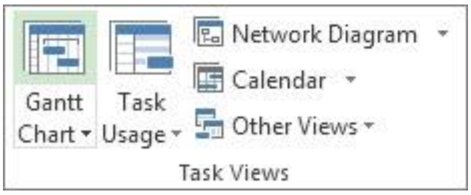


Figure 6.21: Tasks Views

2. In the table, select the number of the task you want to inactivate.
3. Click **Task**, and then in the **Schedule** group, click **Inactivate**.

Figure 6.22: Inactivate tasks

The inactive task stays in the task list, but the text is grayed out and has a line through it.

To reactivate a task, select the inactive task and then click **Inactivate**.

Inactivated Vs. Completed Tasks

Inactivation is not a good way to archive completed tasks, because it could have unexpected effects on the remaining schedule.

Instead, mark the tasks as **completed**. Select the task and click **Task**. In the **Schedule** group, click **100%.**

Figure 6.23: Completed tasks

ACTIVITY 2

Organize Tasks

Link Tasks

1. Choose **View** > **Gantt Chart**.

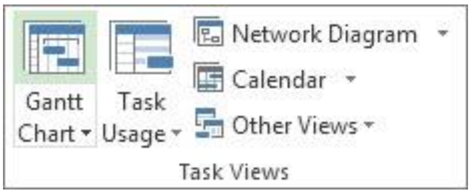


Figure 6.24: Views

2. Hold down Ctrl and select the two tasks you want to link (in the **Task Name** column).
3. Choose **Task** > **Link Tasks**.

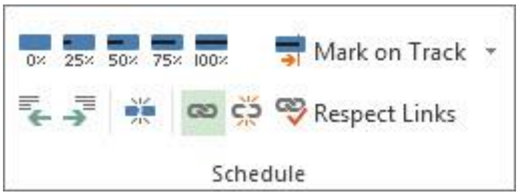


Figure 6.25: Linking Tasks

To remove a link, select the two linked tasks and then choose **Task** > **Unlink Tasks** .

Project creates a simple *finish-to-start* task link by default, which means the first task (the predecessor) needs to finish before the second task (the successor) can start.

If you are a Project Online subscriber, you can also select specific tasks to link to using a drop-down box from the **Predecessors** or **Successors** column.

1. Choose **View** > **Gantt Chart**.
2. Find the **Predecessors** or **Successors** column and select the cell for the task you want to link.

   Note: The **Successor** column is not shown by default. To show it, go to the last column in the row, and select **Add New Column**. Choose **Successors**.

3. In the hierarchical list of all of your tasks, scroll up or down to find the task you want, select the check box next to it, and then click anywhere outside of the drop-down box. The task ID of the task you're linking to will appear in the cell.

Selecting links from the drop-down is especially useful if the task you want to link to is not located close to the task you're linking from.

Insert a Task Between Linked Tasks

You can set up Project so that when you insert a new task between linked tasks, the new task is automatically linked to the surrounding tasks.

1. Choose **File** > **Options**.
2. In the **Project Options** dialog box, choose **Schedule** and scroll to the **Scheduling options in this project** section.
3. Select the **Autolink inserted or moved tasks** check box.
4. Insert the new task.

Link Tasks in a Network Diagram

1. Choose **View** > **Network Diagram**.

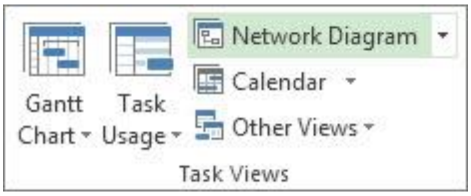


Figure 6.26: Network Diagram

2. Point to the center of the predecessor task box.
3. Drag the line to the successor task box.

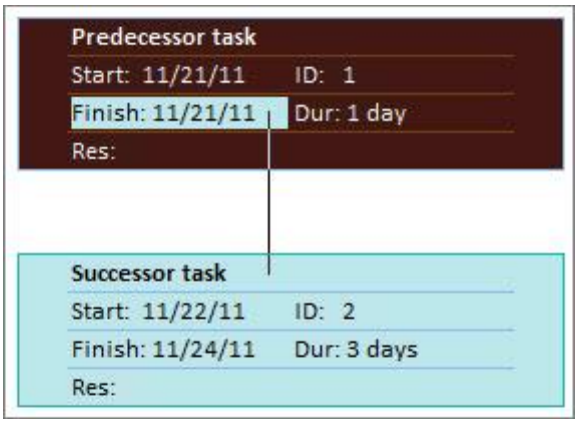


Figure 6.27: Predecssor

Link Tasks in Calendar

1. Choose **View** > **Calendar**.

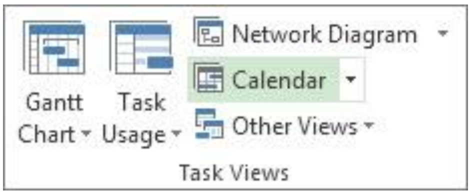


Figure 6.28: Calender

2. Point to the predecessor task's calendar bar.
3. Drag the line to the successor task's calendar bar.

Link Manually Scheduled Tasks

When you link a manually scheduled task to another task, Project places the manually scheduled task relative to the other task.

You can configure Project so that a manually scheduled task does not move when you link it to another task:

1. Choose **File** > **Options**.
2. In the **Project Options** dialog box, choose **Schedule** and scroll to the **Scheduling options in this project** section.
3. Clear the **Update manually scheduled tasks when editing links** check box.

Change or Remove Task Dependencies

After tasks are linked to create task dependencies, you can easily change or remove the dependencies, if needed, by doing any of the following:

Change the link type of a task dependency

By default, Project links tasks in a finish-to-start dependency. You can, however, easily change the type of task dependency on the **Advanced** tab of the **Task Information** dialog box. To open the **Task Information** dialog box, double-click the name of the task whose link type you want to change.

Note: Double-clicking a link to an external task opens the project containing the task, if the project is available. Externally linked tasks appear dimmed in the task list. If you want to modify the link type for an external task, double-click the externally linked task to open the project that contains the task, and then do the following to modify the linked task. For example, if you linked Task A in Project Z to Task 1 in Project 5, you can modify the link type in the task information for Task 1.

1. Choose **View** > **Gantt Chart**.
2. Double-click the link line that you want to change.

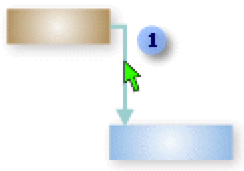


Figure 6.29: View Task

1 To change a link type, double-click here.

3. To change the type of task dependency, select a different type from the **Type** list.

Link to a different task

If you have a task that is currently linked to the wrong task, you can easily modify the link so that it reflects the correct dependency.

1. Choose **View** > **Gantt Chart**.
2. In the task list, select the task for which you are changing the link.
3. On the **Task** tab, in the **Properties** group, choose **Information**.
4. On the **Predecessors** tab, in the **Task Name** column, select the task that should not be linked, and then select a task from the list to identify the correct dependency.
5. Modify the link type, lag time, or lead time in the **Type** and **Lag** columns.

   Tip: To enter lead time, type a negative value in the **Lag** column, such as **–2** for two days of lead time.

Remove all dependencies for a task

If your task is no longer dependent on any other tasks, you can remove all of the task's dependencies at once.

1. Choose **View** > **Gantt Chart**.
2. In the **Task Name** field, select the tasks that you want to unlink.

   To select multiple tasks that are not listed consecutively, hold down CTRL and select each task. To select tasks that are listed consecutively, select the first task, hold down SHIFT, and then select the last task that you want to select in the list.

3. On the **Task** tab, in the **Schedule** group, choose **Unlink Tasks**.

   The tasks are rescheduled, based on any links to other tasks or constraints.

Note: Removing a task link removes the dependency between two tasks, as indicated by the link lines between the tasks. If you want to change the hierarchical structure of a task or subtask as part of an outline structure for your project, then you need to outdent the task rather than remove the task link.

Remove specific task dependencies

If your task is linked to several tasks and you need to remove specific links while leaving some intact, you can choose which links to remove in the **Task Information** dialog box.

1. Choose **View** > **Gantt Chart**.
2. In the task list, select the task for which you are removing dependencies.
3. On the **Task** tab, in the **Properties** group, choose **Information**.
4. On the **Predecessors** tab, select the dependency that you are removing, and then press DELETE.

Turn Auto linking On or Off

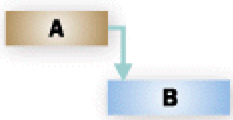
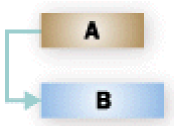
You can configure Project so that when you insert a task among linked tasks, the new task is automatically linked to the surrounding tasks. This is called **autolinking**. For example, if you have three tasks with finish-to-start links and you add a new task between them, the new task will take on a finish-to-start link with the tasks above and below it.

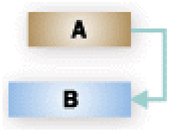
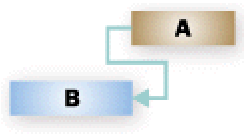
By default, autolinking is disabled. Follow these steps to turn autolinking on.

1. Choose **File** > **Options**.
2. In the **Project Options** dialog box, click **Schedule** and scroll to the **Scheduling options in this project** section.
3. To turn autolinking on, select the **Autolink inserted or moved tasks** check box. To turn autolinking off again, clear the check box.

Types of Tasks

When you link tasks in Project, the default link type is finish-to-start. However, a finish-to-start link does not work in every situation. Project provides the following additional types of task links so that you can model your project realistically:

Link type	Example	Description
Finish-to-start (FS)	 Figure 6.30: Finish to start	The dependent task (B) cannot begin until the task that it depends on (A) is complete. For example, if you have two tasks, "Dig foundation" and "Pour concrete," the "Pour concrete" task cannot begin until the "Dig foundation" task is completed.
Start-to-start (SS)	 Figure 6.31: Start to start	The dependent task (B) cannot begin until the task that it depends on (A) begins. The dependent task can start at any time after the task that it depends on begins. The SS link type does not require that both tasks begin at the same time.

Link type	Example	Description
		For example, if you have two tasks, "Pour concrete" and "Level concrete," the "Level concrete" task cannot begin until the "Pour concrete" task begins.
Finish-to-finish (FF)	 Figure 6.32: Finish to finish	The dependent task (B) cannot be completed until the task that it depends on (A) is completed. The dependent task can be completed at any time after the task that it depends on is completed. The FF link type does not require that both tasks be completed at the same time. For example, if you have two tasks, "Add wiring" and "Inspect electrical," the "Inspect electrical" task cannot be completed until the "Add wiring" task is completed.
Start-to-finish (SF)	 Figure 6.33: Start to finish	The dependent task (B) cannot be completed until the task that it depends on (A) begins. The dependent task can be completed at any time after the task that it depends on begins. The SF link type does not require that the dependent task be completed concurrent with the beginning of the task on which it depends. For example, the roof trusses for your construction project are built off-site. Two of the tasks in your project are "Truss delivery" and "Assemble roof." The "Assemble roof" task cannot be completed until the "Truss delivery" task begins.

Troubleshooting

If you linked your tasks but the successor task doesn't move, there may be several causes:

- If a task has any actuals applied, such as an actual start date or a percentage of work completed, the task can't be rescheduled any earlier than the date when the task actually began. If no progress is entered and the task has an inflexible constraint, the constraint can take precedence over task dependencies.

  For example, if a task has a Start No Earlier Than (SNET) constraint set to July 1, the task is tied to that date and won't be rescheduled to an earlier date, even if its predecessor finishes on June 28 and the successor task could start earlier than July 1.

- In Project, if you create a task by dragging the pointer on the chart portion of a Gantt Chart view, a Start No Earlier Than (SNET) constraint is set on the task for projects that are scheduled from the start date. For projects that are scheduled from a finish date, a Finish No Later Than (FNLT) constraint is set on the task.
- If the successor task is completed, it will not move to reflect the link.

Here are some possible resolutions:

- To reset the task constraint to be more flexible, select the task, choose **Information**, and then select the **Advanced** tab. In the **Constraint type** list, select **As Soon As Possible**. Project then schedules the task's start date according to the task dependency.
- To make task dependencies override inflexible constraints for all tasks, choose **File** > **Options** > **Schedule**. In the **Scheduling options** section, clear the **Tasks will always honor their constraint dates** check box

Split a Task

If you need to interrupt work on a task, you can split the task so that part of it starts later in the schedule. You can split a task into as many sections as you need.

1. On the **View** tab, in the **Task Views** group, choose **Gantt Chart**.

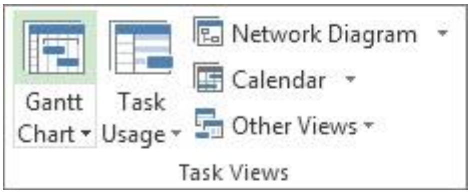


Figure 6.34: Task View

2. On the **Task** tab, in the **Schedule** group, choose **Split Task**.

Figure 6.35: Spliting of task

3. On the task's Gantt bar, click the bar on the date where you want the split to occur, and drag the second part of the bar to the date that you want work to begin again.

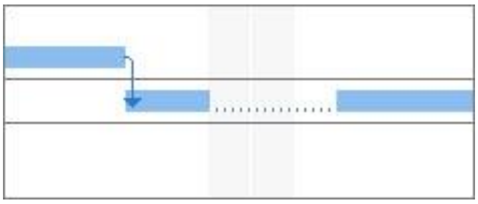


Figure 6.36: Task's Gantt bar

To remove a split in a task (join), drag a section of the task until it touches another section.

Note: If your project has tasks that were already split, but you aren't sure why, your project may have been leveled while allowing splits to occur. On the **Resource** tab, in the **Level** group, choose **Leveling Options**. In the **Resource Leveling** dialog box, select the **Leveling can create splits in**

remaining work check box to allow splits as a result of leveling, or clear this check box to prevent splits during resource leveling.

Move Parts of a Split Task

1. On the **View** tab, in the **Task Views** group, choose **Gantt Chart**.

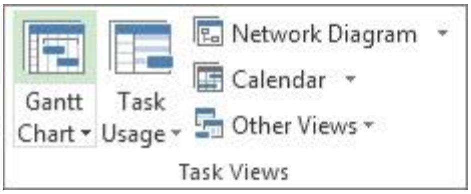


Figure 6.37: Task Views

2. **To move one part:** Hover over any part of the split task *other than the first one*.

   To move the whole thing: Hover over the first part of the split task.

3. The cursor changes to a four-way arrow.
4. Drag the bar left to start earlier or drag it right to start later.

Note: If your project has tasks that were already split, but you aren't sure why, your project may have been leveled while allowing splits to occur. On the **Resource** tab, in the **Level** group, choose **Leveling Options**. In the **Resource Leveling** dialog box, select the **Leveling can create splits in remaining work** check box to allow splits as a result of leveling, or clear this check box to prevent splits during resource leveling.

Change The Duration of a Split Task

To change the duration of the entire split task (including all of its sections), type a new duration in the **Duration** field.

You can also adjust the duration of each section of a split task:

1. On the **View** tab, in the **Task Views** group, choose **Gantt Chart**.

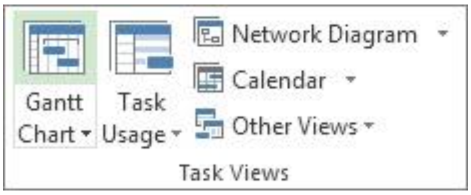


Figure 6.38: Task Views

2. Hover over the right end of any part of the split task until the cursor changes to an expansion arrow.
3. Drag to the left to shorten the duration of the part, or drag to the right to lengthen the duration.

Modify the Gantt Bar Style for Split Tasks

By default, split tasks are represented in the Gantt Chart view with dotted lines connecting each split portion. You can change the appearance of the split by modifying the split bar style.

1. On the **View** tab, in the **Task Views** group, choose **Gantt Chart**.
2. On the **Format** tab, select the **Format Bar Styles** button in the lower right corner of the **Gantt Chart Style** group.
3. In the **Bar Styles** dialog box, select **Split** in the **Name** column of the table.
4. On the **Bars** tab, under **Middle**, select options for the split bar in the **Shape**, **Pattern**, and **Color** lists.

ACTIVITY 3

Set Up The Project Calendars

How Do All Of These Calendars Work Together?

There are four types of calendars in Project: **base calendars**, **project calendars**, **task calendars**, and **resource calendars**.

Base calendars are used almost like a template for project, task, and resource calendars. They define the standard working and non-working times for all projects in your organization. They specify the work hours for each day, the work days for each week, and any exceptions, like company holidays. Three default base calendars are already set up in Project: Standard, 24 Hours, and Night Shift.

Project, resource, and task calendars use a base calendar as a template, and then are modified to reflect the unique working days and hours for individual projects, resources, or tasks. These unique calendars are particularly helpful when accounting for things like a leave of absence, shift work, or tasks that are completed by equipment that runs throughout nights and weekends.

These calendars all stack together to frame how work is scheduled on a project.

An example...

Let's say your organization is located in Seattle, and has a typical work week of 8-hour work days, Monday through Friday. The base calendar reflects these working days and hours, as well as company holidays, like July 4 for Independence Day, and December 25 for Christmas.

A project manager starts a new project that runs from July 1 to July 15, and requires a Tuesday-Saturday work week.

The **base calendar** shows the Monday through Friday work week, with July 4 identified as a non-working holiday.

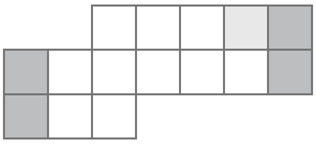


Figure 6.39: Base Calender

The **project calendar** uses the base calendar as a starting point, but adds that Saturdays are now working days, and Mondays are now non-working days.

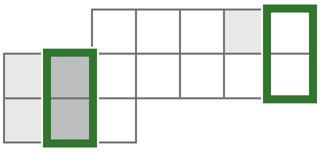


Figure 6.40: Project Calender

One task in the project takes place in another country, so the July 4 holiday won't apply. The **task calendar** adds that July 4 is a working day.

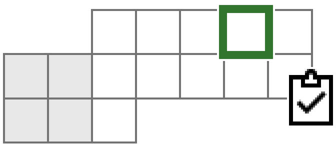


Figure 6.41: Task Calender

Finally, the resource that will be working on the task has a different working schedule than the rest of your organization. The **resource calendar** adds that Tuesday is a non-working day, and Wednesday through Saturday are 10-hour days.

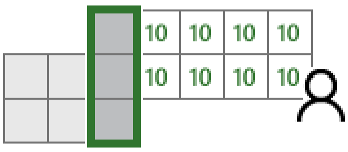


Figure 6.42: Resource Calender

When Project schedules the task, it accounts for the assigned resource's calendar, the task calendar, the project calendar, and the base calendar.

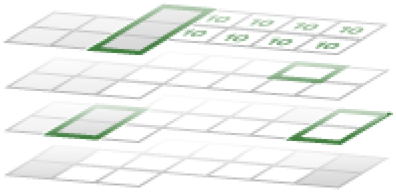


Figure 6.43: All Calenders

The resulting availability contributes to how Project calculates the start and finish dates for the task and the project.

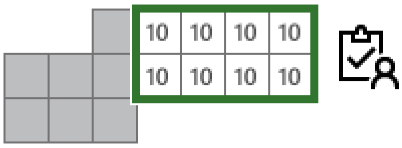


Figure 6.44: Result of Different Calenders

Save a Project File In Project Desktop

Saving a project file to your computer is similar to saving a file in any other Office program.

1. Click **File** > **Save As**.
2. Locate the folder where you want to save it.

   If you are using the subscription version of Project (Project Online Desktop Client), you will first be shown the folders in which you most recently saved a project file. You can also select another location.

3. Type a name in the **File name** box, and then click **Save**.

Of course, you're not limited to saving to just your computer or to just the Project file format.

Other Save Options

- Sync a project with a tasks list
- Save a project to a different file format
- More ways to save a project file

Save a Project To A Different File Format

1. Click **File** > **Save As**.
2. Under **Choose a Location**, choose **Computer** or a web location to save the file.
3. Under **Choose a Folder**, click a folder at the location.
4. In the **File name** box, type the project name.
5. Click the **Save as type** box and choose a file format.
6. Click **Save**.

You can save a project file to these formats:

- Project 2007
- Project 2000 – 2003 (not available in Project 2016)
- Project template
- PDF
- XPS
- Excel workbook
- Tab delimited text
- CSV (comma delimited)
- XML

ACTIVITY 4

Calendar View of Trello

Plan faster and better with the Calendar view. From tracking team progress to staying on top of your work, give a time dimension to your projects. Calendar helps you anticipate any roadblocks, events or deadlines, to take your team smoothly to the finish line.

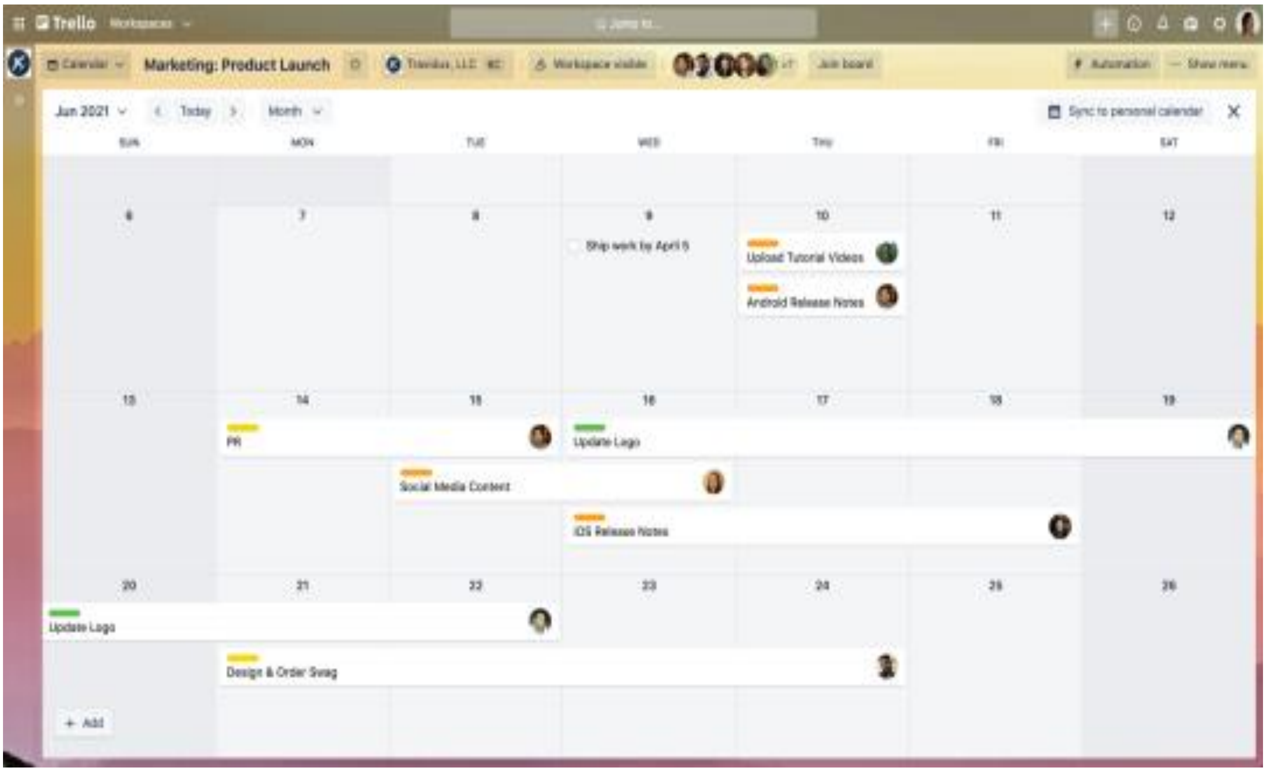


Figure 6.45: Trello Calender

Feature focus:

- **Calendar:** Cards can be viewed in the Calendar in a monthly view, to get the big picture, or weekly view, to focus on the work ahead of you. At a glance see all the relevant dates of your project—add start and due dates to cards and your cards will appear in Calendar view.
- **Today and time arrows:** Easily navigate through time with header controls for finding the right day. You can always click “today” just in case you get lost.

- **Add start/due dates:** Get the granularity you need to get your work done. With full-day events and multi-date events, you and your team can more easily grasp the big picture.
- **Drag and drop cards:** The Trello way to organize and move things around is now in Calendar, too. Drag your cards to new dates with the calendar and it will auto-magically update the dates on your cards, conveniently syncing within Calendar view as well.
- **Stretch cards:** Easily pick a side of a card and stretch it to set up date ranges, which can span multiple days or weeks.
- **Advanced checklists:** Items with due dates are supported , where you can view, drag, and mark as an advanced checklist item is completed.
- **Create new cards:** If you think that a card is missing for your project to have all the pieces it needs, quickly add cards from the Calendar view by double clicking on a date.

The calendar view is great for:

- **Tracking team progress:** from keeping track of sales meetings with leads and close dates to managing interviews in a hiring pipeline, quickly discover the status of every task.
- **Planning work that lies ahead:** Whether you are building a strong marketing editorial calendar or planning your next event or conference, you can easily map out any time-based event.
- **On the go due date tracking:** Check out the Calendar view on the go from the Trello app. Using this view on mobile, you can visualize critical due dates and tasks from anywhere.
- **Staying on top of tasks:** For you or your team, easily be reminded of any work that needs to be done.

Switch to The Calendar View

When you have Planyway set up, your scheduled Trello cards are automatically visualized in one of the calendar views: daily, 3-day, weekly or monthly.

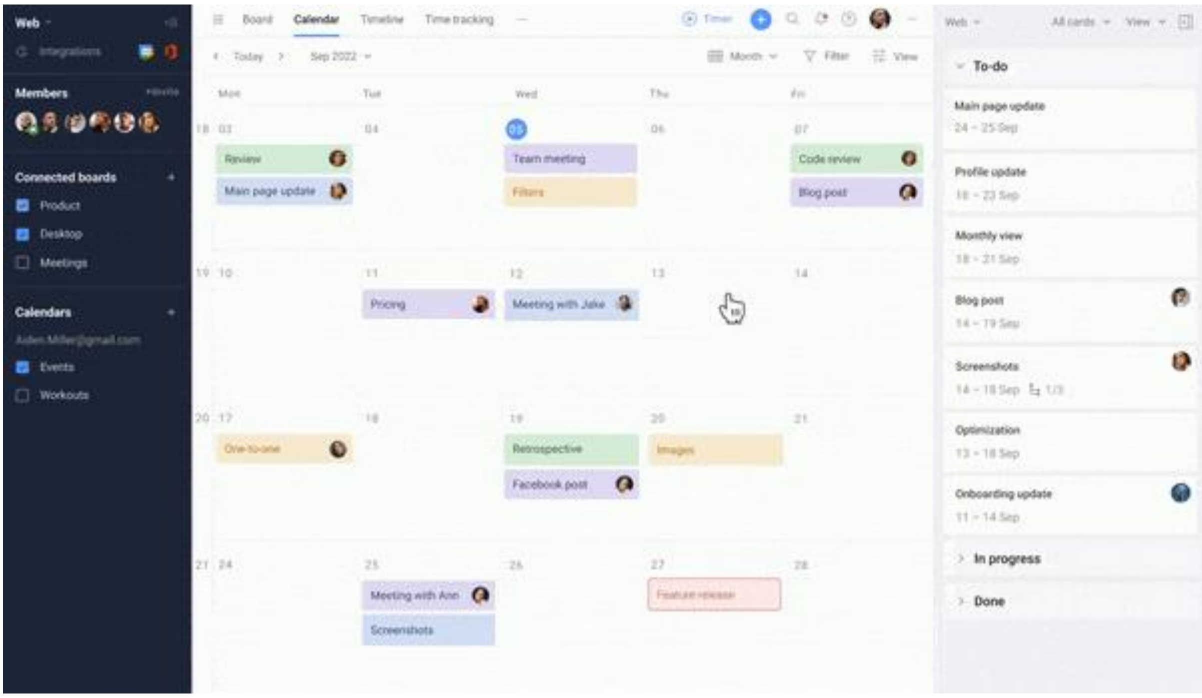


Figure 6.46: Calender View

You can keep creating cards on the board and then drag them to the calendar, or simply create cards right on the calendar at a click.

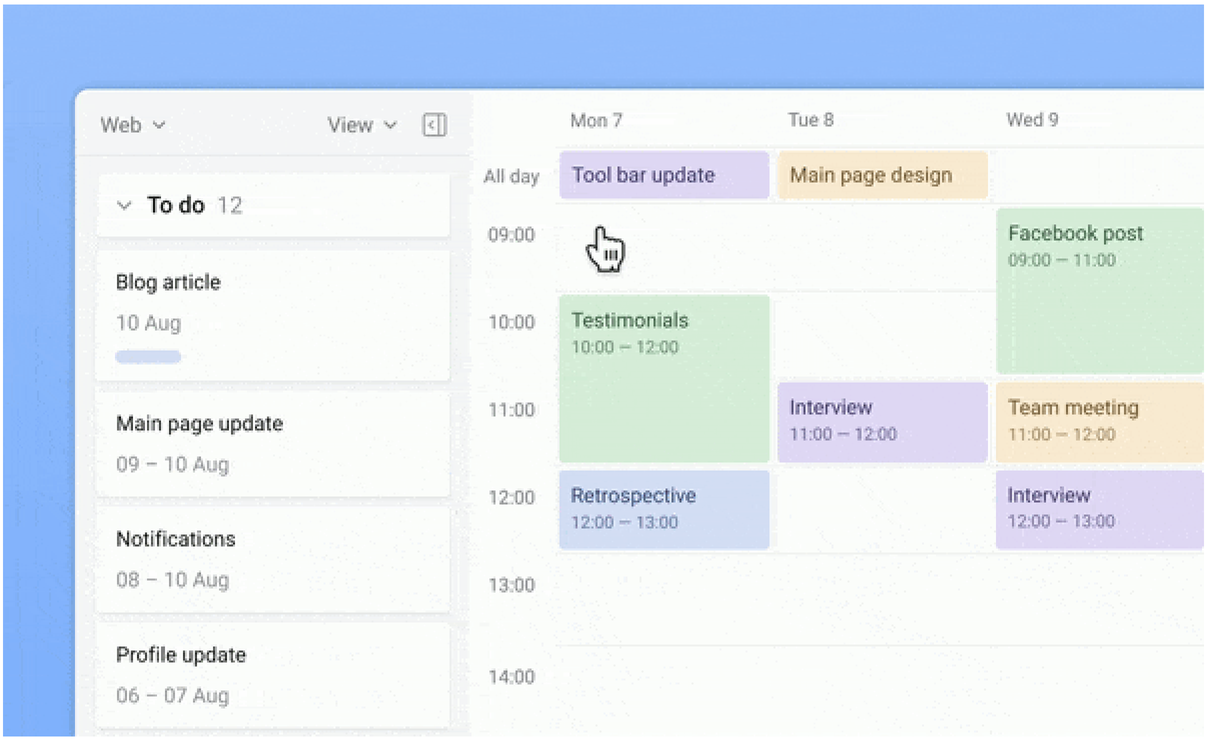


Figure 6.47: Cards in Trello

CREATE CHECKLISTS

If your cards seem complex and include multiple stages, don’t hesitate to split them with checklists. And in case you want to see checklist items in your planner, just drag them to the calendar the same way as cards.

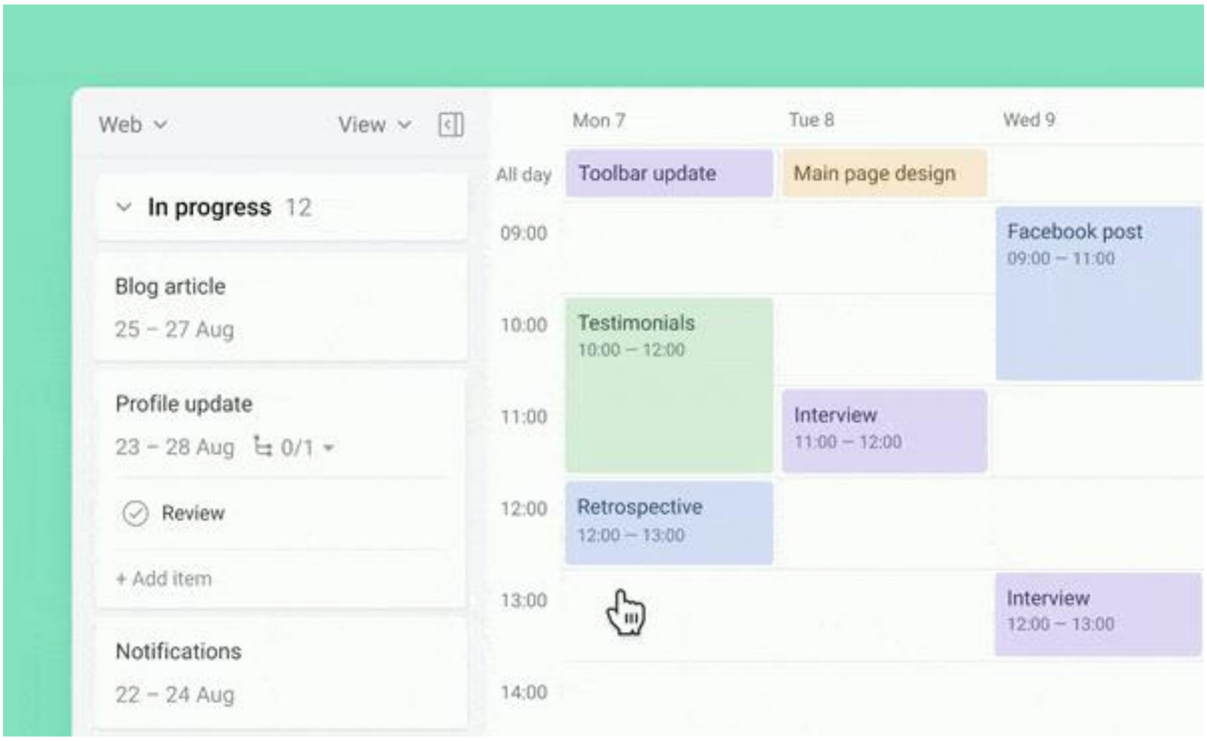


Figure 6.48: Checklists in Trello

Put Tasks On Repeat

Create the recurrence rule for those cards that have to repeated in your planner on certain dates. Once set, recurrent cards will show up automatically.

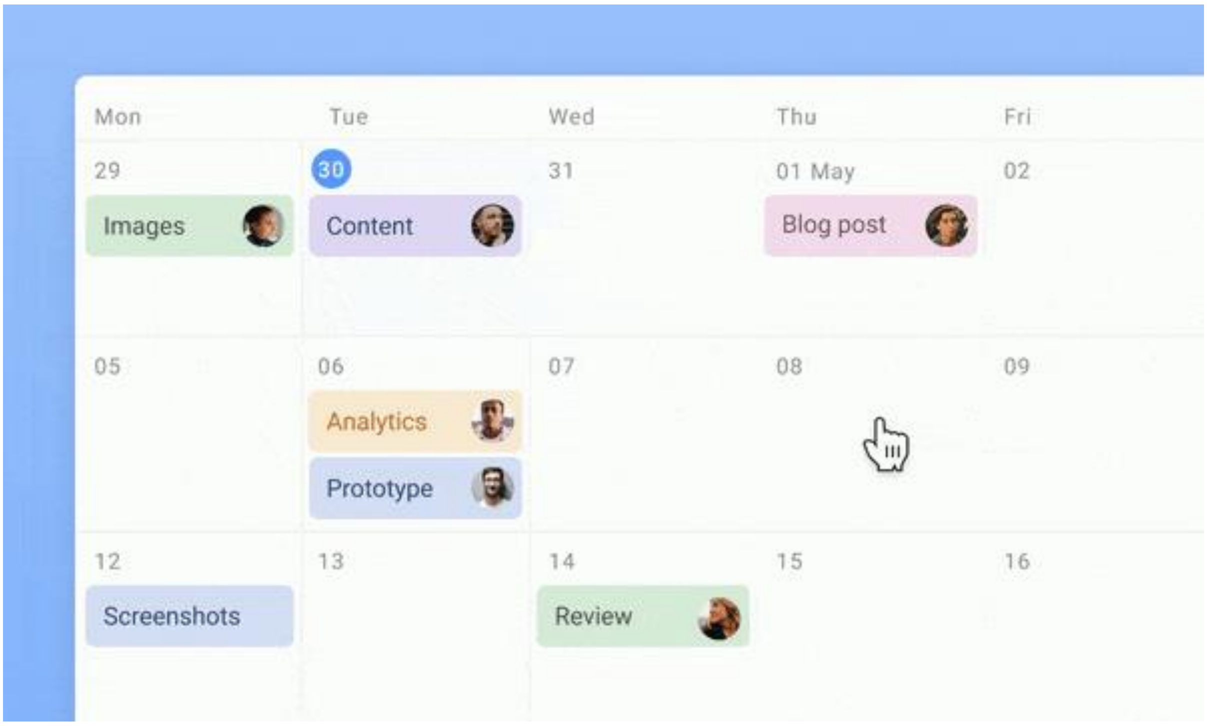


Figure 6.48: Recurrence Tasks in Trello

Set Priorities

Priority labels will help you prioritize work and distinguish more important tasks from the less important.

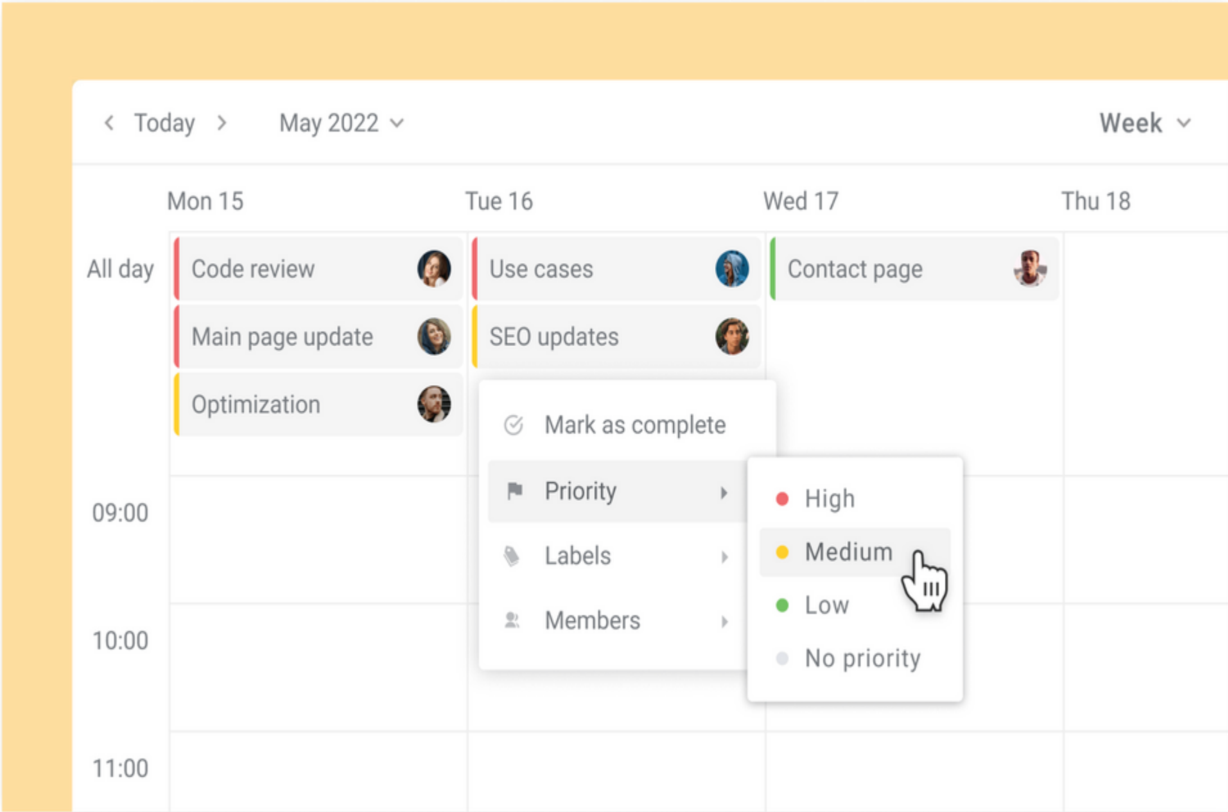


Figure 6.49: Setting Priorities in Trello

Manage Tasks Across Boards

When working with several Trello boards, you might find it easier to keep track of all of them together. Connect any boards you need and see cards in one planner.

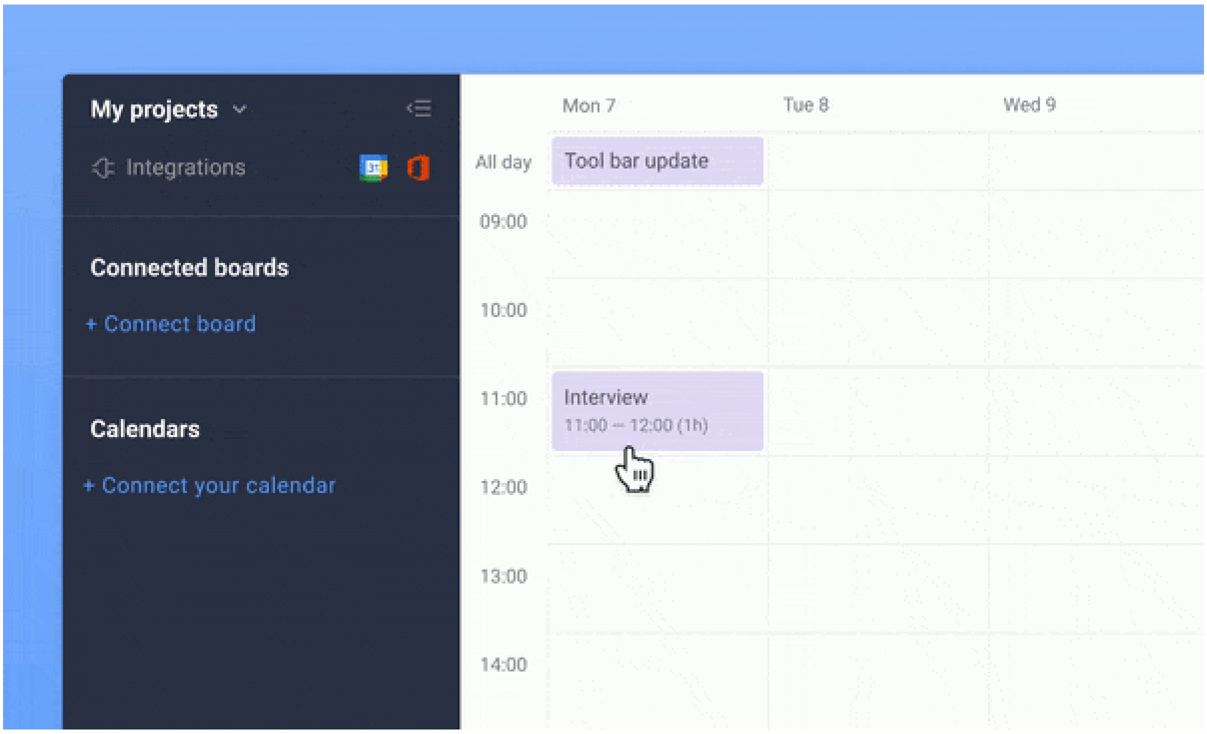


Figure 6.50: Managing Tasks

Add External Calendars

Planyway allows you to see Trello cards with events from Google Calendar, Outlook, Apple, etc. This way you get a single source of your availability across multiple apps.

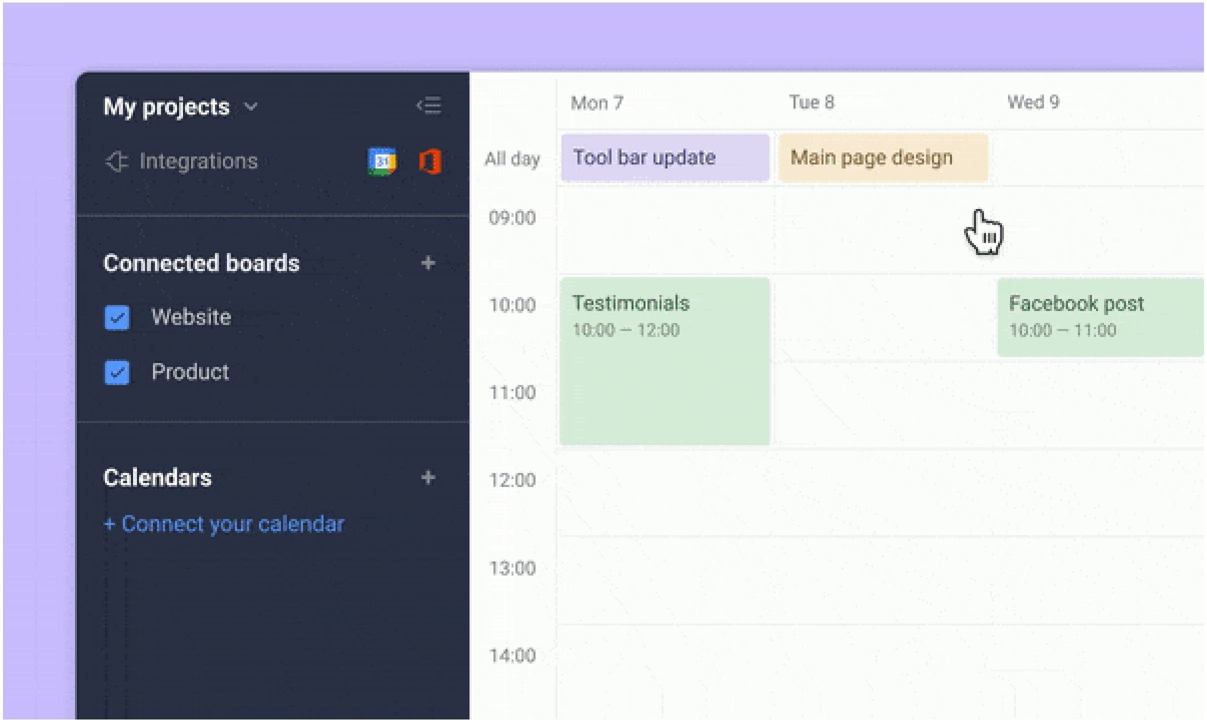


Figure 6.51: External Calenders

Set Reminders

Add reminders to your tasks and events not to forget about what’s important. You’ll receive them in-app, as browser notifications or, in case you use our mobile app, as mobile notifications.

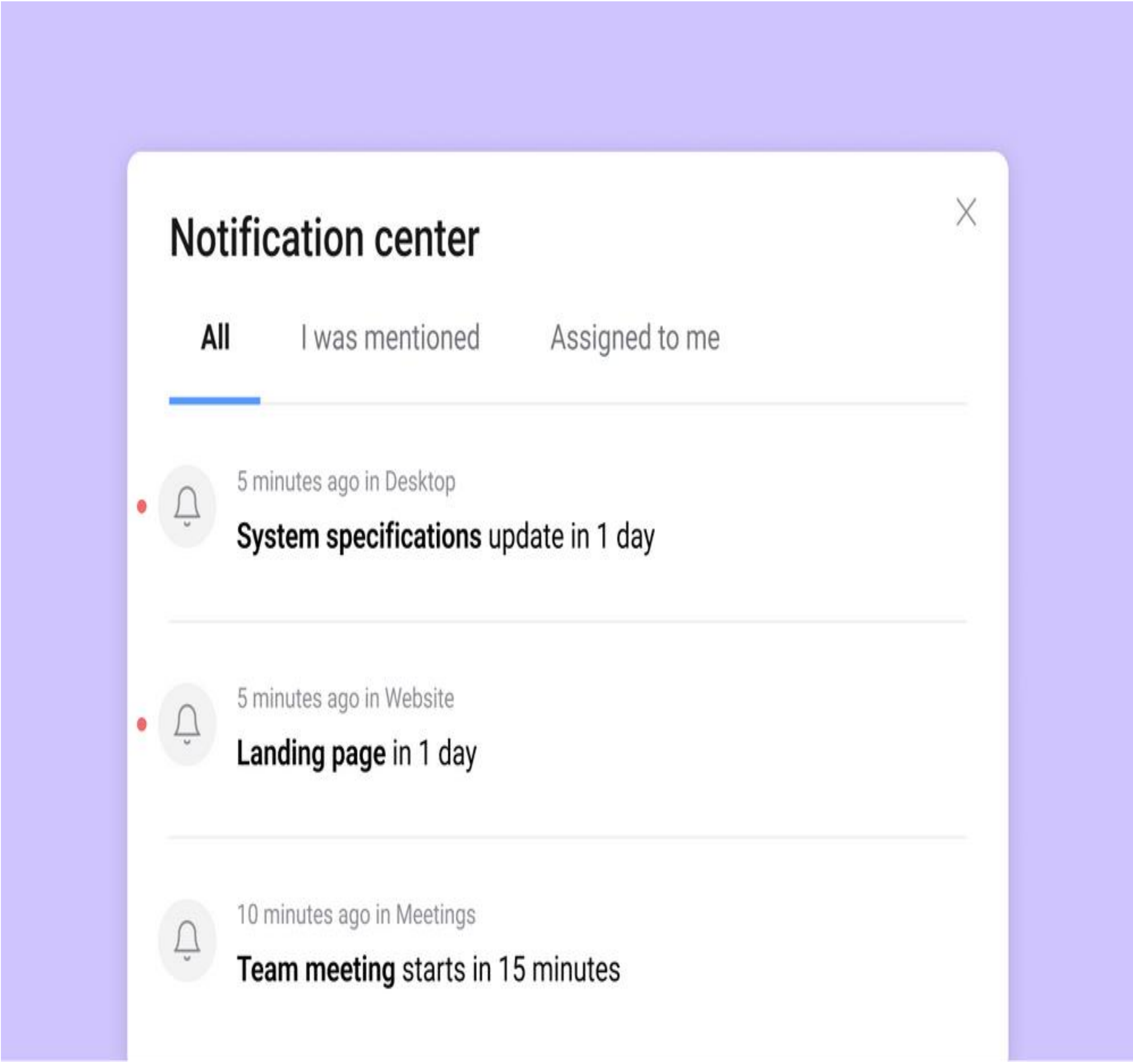


Figure 6.52: Reminders and Notifications

3) Graded Lab Tasks

Note: The instructor can design graded lab activities according to the level of difficult and complexity of the solved lab activities. The lab tasks assigned by the instructor should be evaluated in the same lab.

LAB TASK 1

Sequential Tasks:

Create a project plan for organizing a workshop. Define tasks like "Select Venue," "Prepare Materials," and "Send Invitations." Set up dependencies so that "Prepare Materials" cannot start until "Select Venue" is complete.

LAB TASK 2

Project Planning:

Create a Trello board for planning a marketing campaign. List tasks such as "Content Creation," "Design Development," and "Social Media Promotion." Add due dates to each task to create a schedule.

Lab # 07

Developing Gantt chart, Network Diagrams, and perform Critical Path Analysis using MS Project Professional

Objective:

The objective of this lab is to make students proficient in Microsoft Project Professional to create Gantt charts and network diagrams, establish task dependencies accurately to represent project workflows. Identify and analyze the critical path using both graphical representations for effective project management. It also helps students to learn and optimize project schedules and communicate them visually to stakeholders.

Activity Outcomes:

After successful completion of this lab, the students will be able to:

1. **Visual Representation:** Students will be able to create clear and informative Gantt charts and network diagrams to represent project schedules and dependencies.
2. **Accurate Task Dependencies:** Students will establish precise task dependencies, showcasing an understanding of how tasks relate to each other in project workflows.
3. **Critical Path Identification:** Students will learn to identify the critical path and recognize tasks that impact project duration the most.
4. **Strategic Planning:** Students will be equipped to optimize project schedules by managing tasks on the critical path and adjusting task sequences

Instructor Note:

As pre-lab activity, read Chapter 7 from the book "Microsoft Project 2019 Step by Step, Cindy Lewis, Carl Chatfield, Timothy Johnson., Microsoft Press, 2019".

1) Useful Concepts

Developing Gantt charts, Network Diagrams, and performing Critical Path Analysis are essential components of project management, and Microsoft Project Professional offers a robust set of tools to facilitate these tasks.

Gantt Chart: A Gantt chart is a visual representation of a project schedule that shows tasks, their durations, and their interdependencies over time. In MS Project Professional, you can create a Gantt chart by listing tasks, setting start and end dates, and defining task relationships (dependencies). The chart provides a clear overview of the project timeline, helps identify task overlaps, and allows for easy tracking of task progress. Gantt charts are particularly useful for stakeholders to understand project timelines at a glance.

Network Diagrams: Network diagrams, also known as PERT (Program Evaluation and Review Technique) charts, display the logical flow and interdependencies among project activities. MS Project Professional enables you to create these diagrams by defining tasks, their dependencies, and their estimated durations. The resulting network diagram helps in identifying the critical path (the longest sequence of tasks that determines the project's minimum duration) and non-critical paths, aiding in resource allocation and schedule optimization.

Critical Path Analysis: Critical Path Analysis is a method used to identify the sequence of tasks that determine the shortest possible duration for a project. In MS Project Professional, this analysis is automated through the identification of the critical path based on task dependencies and durations. By determining which tasks are critical (any delay in these tasks will delay the entire project) and non-critical, project managers can focus their attention on tasks that have the most impact on project completion. Critical Path Analysis assists in resource allocation, risk assessment, and overall project time management.

In conclusion, MS Project Professional offers an integrated platform for developing Gantt charts, creating Network Diagrams, and performing Critical Path Analysis. These tools empower project managers to visualize project timelines, understand task interdependencies, and identify critical tasks for efficient project execution. Utilizing these concepts and tools helps enhance project planning, monitoring, and decision-making, ultimately leading to successful project outcomes.

2) Solved Lab Activities

Sr.No	***Allocated Time***	***Level of Complexity***	***CLO Mapping***
1	***30***	***Low***	***CLO-6***
2	***30***	***Low***	***CLO-6***
3	***30***	***Low***	***CLO-6***

ACTIVITY 1

How to Make A Gantt Chart in Microsoft Project

Open Your Project File

Enable Microsoft Project and open your project file (.mpp). As you can see, the interface consists of 2 separate panes. The left pane is a worksheet in which you enter the data of task names, duration days, start dates, finish dates, predecessors, and resources names. The right pane is the Gantt Chart, which shows the timeline of the project.

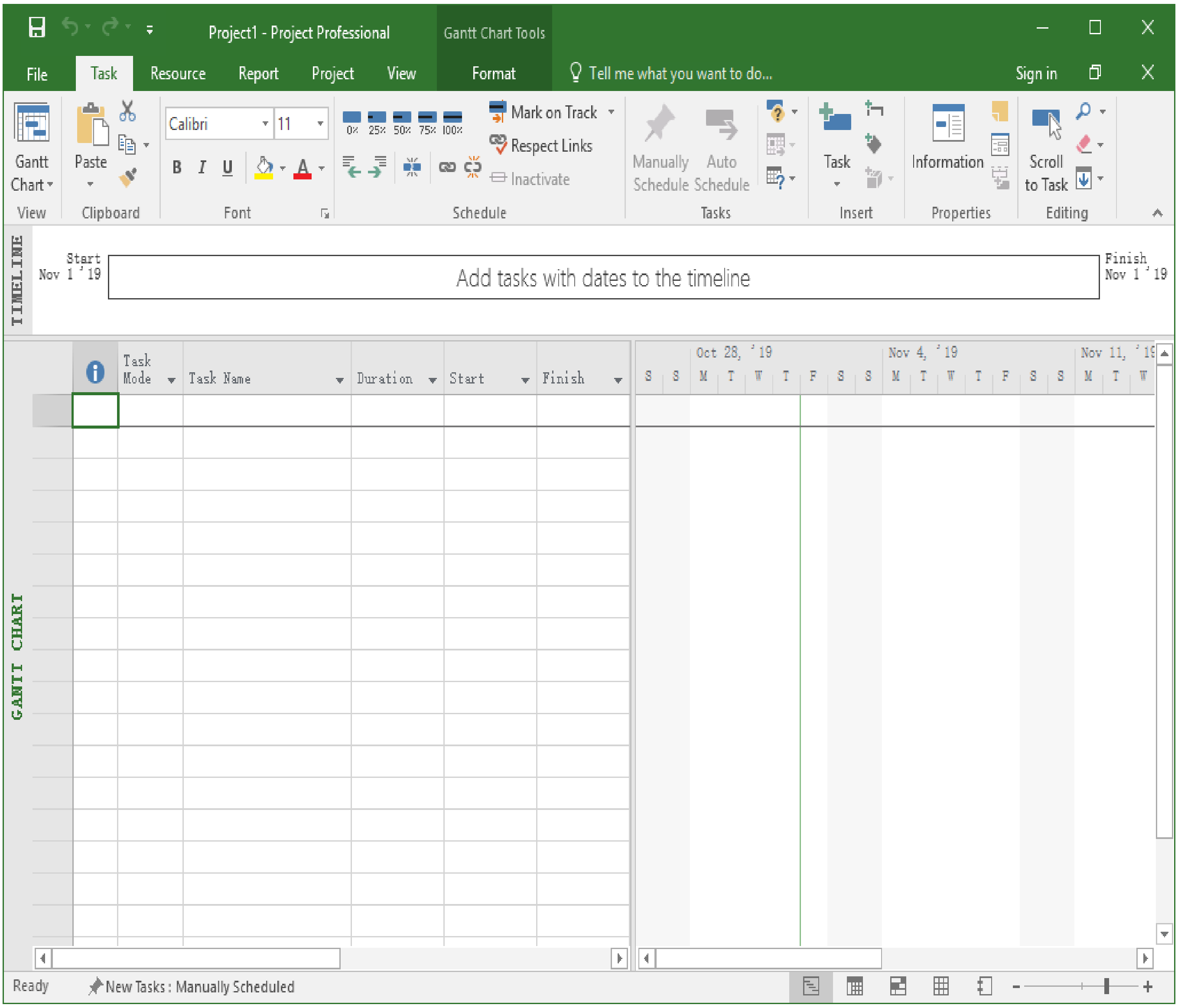
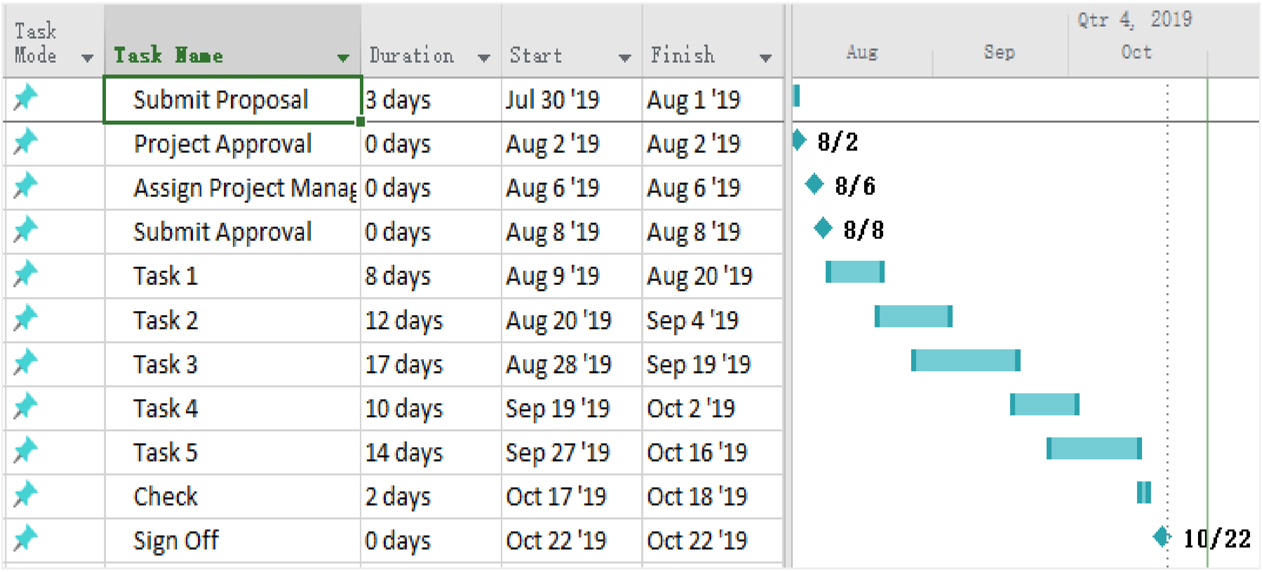


Figure 7.1: Sample Project Opened

Enter the Project Data

Enter the data of your project on the left pane and the software will automatically create a Gantt chart on the right pane. In this example, I just input the task names, duration days, start dates, and finish dates.



Task Mode	Task Name	Duration	Start	Finish
	Submit Proposal	3 days	Jul 30 '19	Aug 1 '19
	Project Approval	0 days	Aug 2 '19	Aug 2 '19
	Assign Project Manag	0 days	Aug 6 '19	Aug 6 '19
	Submit Approval	0 days	Aug 8 '19	Aug 8 '19
	Task 1	8 days	Aug 9 '19	Aug 20 '19
	Task 2	12 days	Aug 20 '19	Sep 4 '19
	Task 3	17 days	Aug 28 '19	Sep 19 '19
	Task 4	10 days	Sep 19 '19	Oct 2 '19
	Task 5	14 days	Sep 27 '19	Oct 16 '19
	Check	2 days	Oct 17 '19	Oct 18 '19
	Sign Off	0 days	Oct 22 '19	Oct 22 '19

Figure 7.2: Sample Data

Step 3: Add Gantt Chart Wizard on the Ribbon

To add the **Gantt Chart Wizard** on the ribbon, you need to:

1. Click on **File** > **Options** to open the **Project Option** dialog box;
2. Click on **Customize Ribbon**;
3. In the right column under **Main Tabs**, right-click on the tab where the Wizard will be inserted and choose **Add New Group** in the context menu;
4. The newly-added group will be listed as **New Group (Custom)**, right-click on it and rename it. Here I called it as **Wizard**.

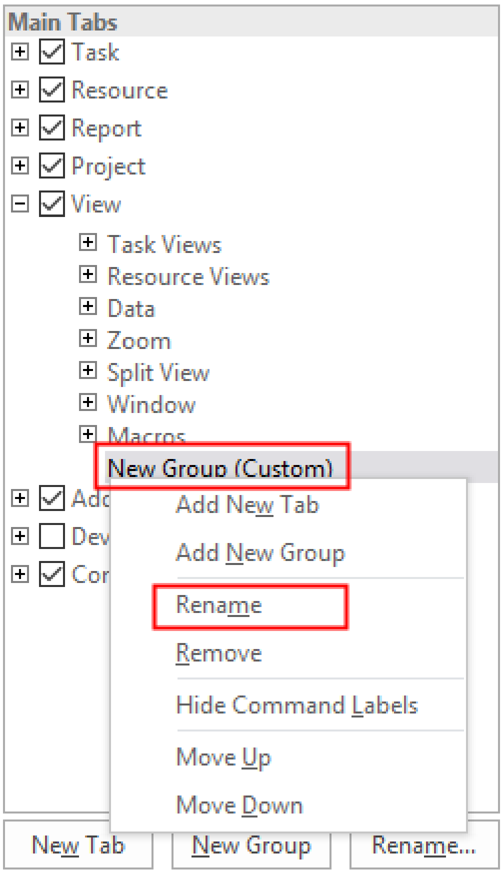


Figure 7.3: Wizard

1. In the left column of the pane, select **Commands Not in the Ribbon** under **Choose commands from**. After that, you need to find **Gantt Chart Wizard** on the scroll list and click **Add** button between the columns.

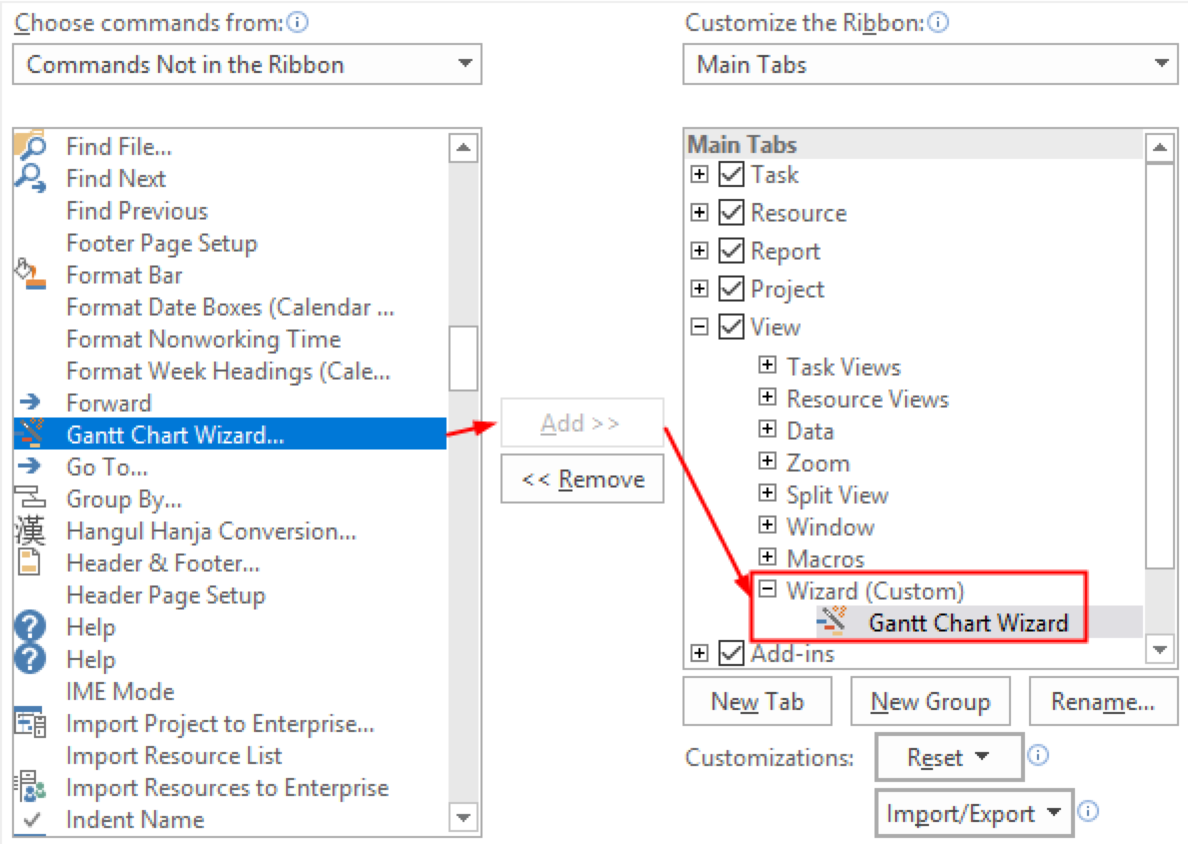


Figure 7.4: Gantt Chard Wizard

Note: Before you add **Gantt Chart Wizard** to the **Main Tabs**, you have to make sure that you select **Wizard (Custom)**. Otherwise, it will be added on another tab.

1. Click **OK** on the dialog box. Here is the **Gantt Chart Wizard** button on the **View** tab.

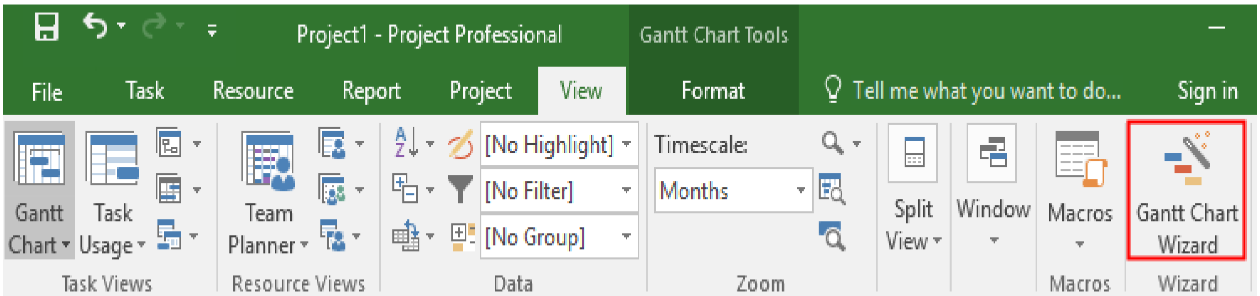


Figure 7.5: Gantt Chart Tools

Customize the Gantt Chart

After you add the **Gantt Chart Wizard** on the ribbon, you can use it to format your Gantt bars.

The first step is to decide what kind of information will be displayed on the chart.

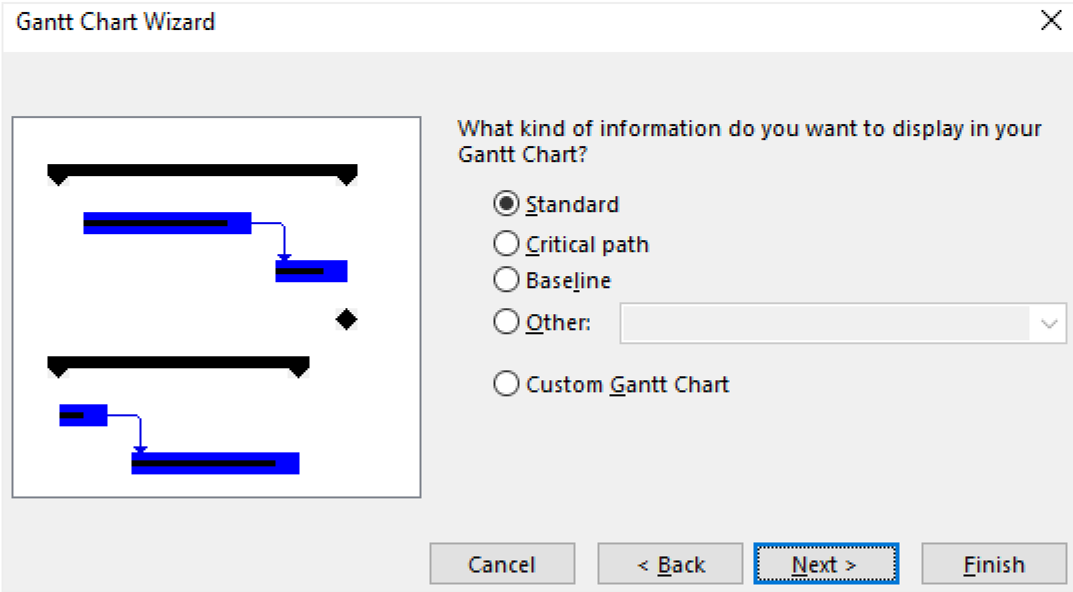
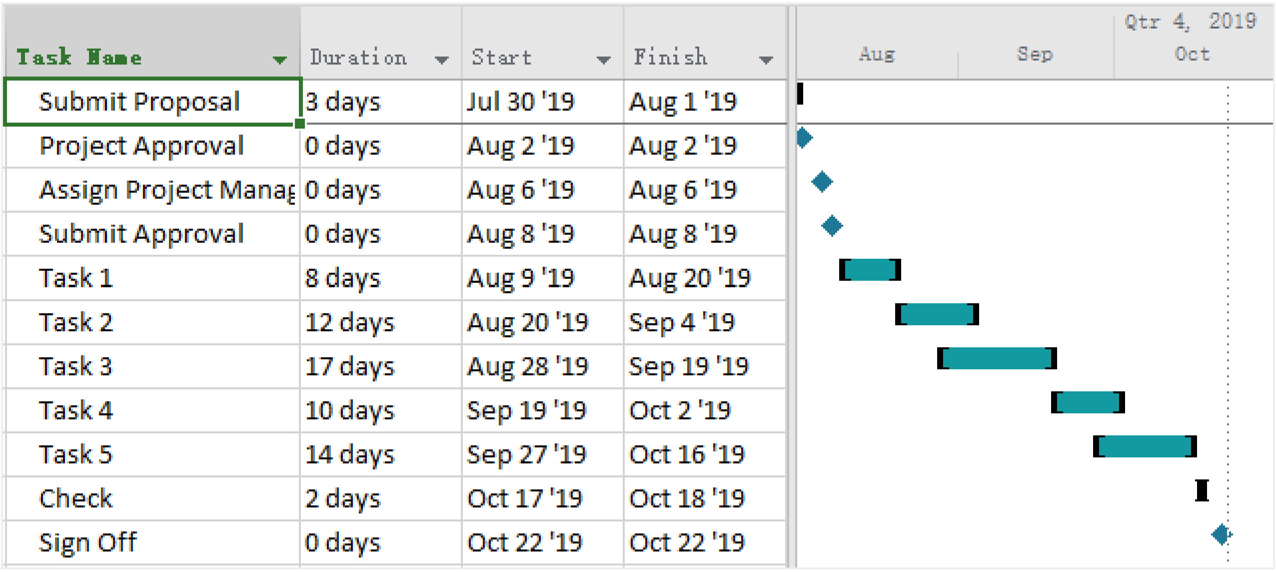


Figure 7.6: Gantt Chart Wizard Display Choice

The second step is to decide what task information will be displayed on the Gantt bars. The third step is to decide whether to show the links between dependent tasks or not. Here is what the Gantt chart looks like now:



Task Name	Duration	Start	Finish
Submit Proposal	3 days	Jul 30 '19	Aug 1 '19
Project Approval	0 days	Aug 2 '19	Aug 2 '19
Assign Project Manag	0 days	Aug 6 '19	Aug 6 '19
Submit Approval	0 days	Aug 8 '19	Aug 8 '19
Task 1	8 days	Aug 9 '19	Aug 20 '19
Task 2	12 days	Aug 20 '19	Sep 4 '19
Task 3	17 days	Aug 28 '19	Sep 19 '19
Task 4	10 days	Sep 19 '19	Oct 2 '19
Task 5	14 days	Sep 27 '19	Oct 16 '19
Check	2 days	Oct 17 '19	Oct 18 '19
Sign Off	0 days	Oct 22 '19	Oct 22 '19

Figure 7.7: Tasks Dependency

ACTIVITY 2

Create a Network Diagram in Project Desktop

A Network Diagram is a graphical way to view tasks, dependencies, and the critical path of your project. Boxes (or nodes) represent tasks, and dependencies show up as lines that connect those boxes. After you've switched views, you can add a legend, customize how your boxes appear, and print your Network Diagram.

- To find the Network diagram view, choose **View** > **Network Diagram**.

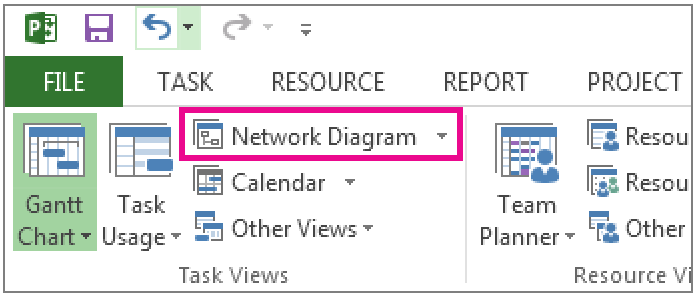


Figure 7.8: Network Diagram

- Add a legend
- Automatically change the way the boxes are laid out
- Manually change the way boxes are laid out
- Change the line style between boxes
- Choose what kind of task information to show

Add A Legend

1. Choose **File** > **Print** > **Page Setup**.
2. On the **Legend** tab, decide how you want your legend to look, which pages it should show up on, and then labels you want.
3. Choose **OK**.

Automatically Change the Way the Boxes Are Laid Out

1. Choose **View** > **Network Diagram**.
2. Choose **Format** > **Layout**.

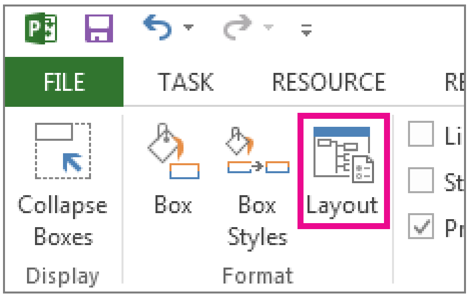


Figure 7.9: Layout

3. Under **Box Layout**, choose the box arrangement, alignment, spacing, height, and width that work best for you. To space boxes evenly, select **Fixed** in the **Height** and **Width** boxes.

Keep in mind that grouped tasks are positioned automatically. You'll need to undo grouping if you want to change them.

Manually Change the Way Boxes Are Laid Out

If you've gotten this far and still don't like how your boxes are positioned, click **Format** > **Layout**, select **Allow manual box positioning**, choose **OK**, and then drag the boxes to the spot you want.

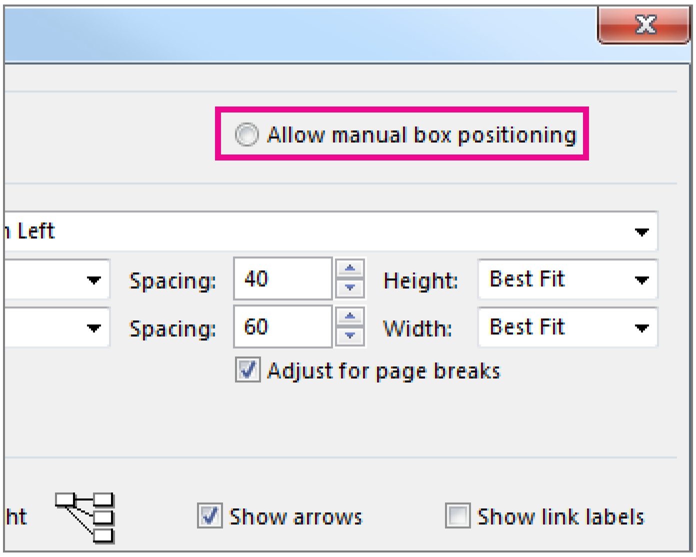
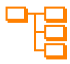
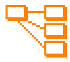


Figure 7.9: Positioning

If you manually reposition a task, you can change the layout of any linked tasks or subtasks associated with it by right-clicking on the task and choosing **Layout Related Tasks Now**.

Change the Line Style Between Boxes

If you have a lot of tasks that you’ve linked to predecessor or successor tasks, the links between boxes can be really hard to follow. Try changing the line style, and then arranging them in way that’s easier to see.

1. Choose **View** > **Network Diagram**.
2. Choose **Format** > **Layout**.
3. Under **Link style**, select **Rectilinear** or **Straight**. Rectilinear links look like this , and straight links look like this .
4. Select **Show arrows** to add arrows that point to predecessor and successor tasks. Select **Show link labels** to add dependency and lead or lag time to the link line.

CHOOSE WHAT KIND OF TASK INFORMATION TO SHOW

If things are looking cluttered (or you start to experience information overload), try changing the task information in each box so you only see what’s most important.

1. Choose **View** > **Network Diagram**.
2. Choose **Format** > **Box Styles**.
3. In the **Style settings for** list, select the task that you want to change.

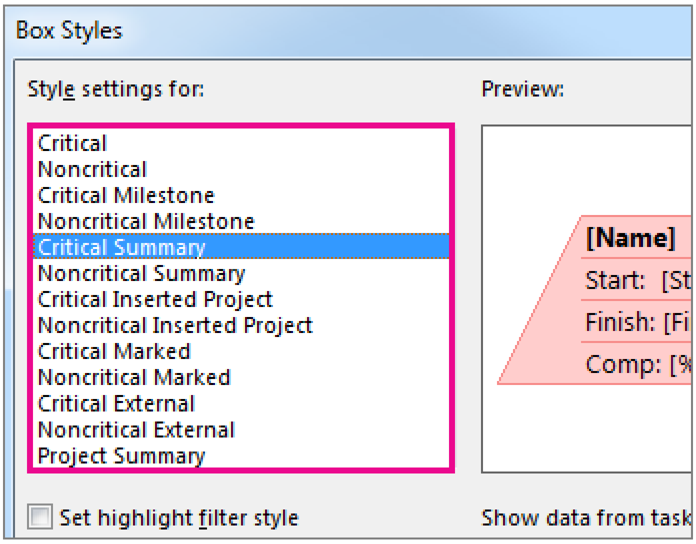


Figure 7.10: Style

4. Under **Border**, choose the shape, color, width, and gridline options to create the look you want.
5. Select a name under **Data template** to apply your changes to an existing template. To create a new template that will use your changes, choose **More Templates**, and then choose **New** (to create a new template), **Copy** (to base the new template on an existing one), **Edit** (to change a template), or **Import** (to import a template from another project).
6. Choose **OK**.

ACTIVITY 3

Show the Critical Path of Your Project

Every task is important, but only some of them are *critical*. The critical path is a chain of linked tasks that directly affects the project finish date. If any task on the critical path is late, the whole project is late.

The critical path is a series of tasks (or sometimes only a single task) that controls the calculated start or finish date of the project. The tasks that make up the critical path are typically interrelated by task dependencies. There are likely to be many such networks of tasks throughout your project plan. When the last task in the critical path is complete, the project is also complete.

Show Critical Path in The Gantt Chart View

The Gantt Chart view will likely be your most used view for showing the critical path.

1. Choose **View** > **Gantt Chart**.
2. Choose **Format**, and then select the **Critical Tasks** check box.

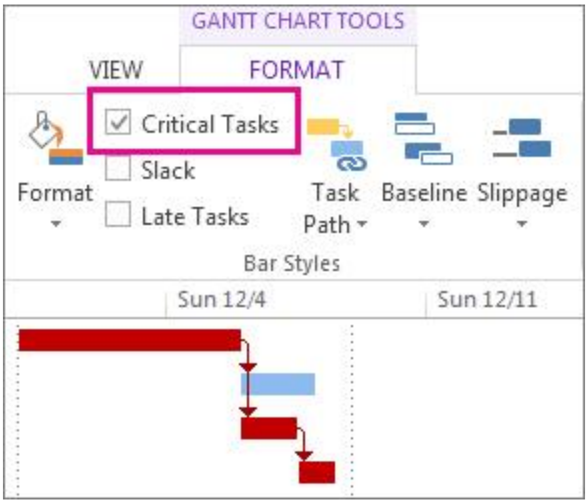


Figure 7.11: Critical Tasks

Tasks on the critical path now have red Gantt bars.

Show the Critical Path in Other Task Views

You can see the critical path in any task view by highlighting it.

1. On the **View** tab, pick a view from the **Task Views** group.
2. Staying on the **View** tab, select **Critical** from the **Highlight** list. The critical path shows up in yellow.
3. To see *only* the tasks on the critical path, choose the **Filter** arrow, then pick **Critical**.

Figure 7.12: Critical Path in Other Tasks.

Tip: In a Network Diagram, tasks on the critical path automatically show up in red. No highlighting is needed.

View the Critical Path in a Master Project

When you're managing a master project, whole subprojects can be on the critical path. You can see if this is true by telling Project to treat the subprojects like they are summary tasks.

1. Choose **File** > **Options**.
2. Choose **Schedule**, and then scroll down to the **Calculation options for this project** area.
3. Make sure the **Inserted projects are calculated like summary tasks** box is selected.

Tip: This setting does not affect other projects. That is, it only applies to the master project you're working on.

Change What Tasks Show Up On The Critical Path

Typically, critical tasks have no slack. But you can tell Project to include tasks with one or more days of slack on the critical path so you can see potential problems coming from farther away.

Choose **File** > **Options**.

Choose **Advanced**, and then scroll down to the **Calculation options for this project** area.

Add a number to the **Tasks are critical if slack is less than or equal to** box.

Show Multiple Critical Paths

You can set up your project schedule to display as many critical paths as you need to keep tabs on your project.

1. Choose **File** > **Options**.
2. Choose **Advanced**, scroll down to the bottom, and then select **Calculate multiple critical paths**.
3. Choose **View** > **Gantt Chart**.
4. Choose **Format**, and then select **Critical tasks**.

By default, Project shows only one critical path, the one that affects the project's finish date. But you might need to see more than one for a couple reasons:

- To make sure each subproject of a master project is on time.

- To track the progress of different phases or milestones.
- To keep an eye on any series of tasks for any reason.

Project management tip: When viewing multiple critical paths, don't forget that there's still only one main critical path. If it falls behind schedule, the whole project falls behind schedule. Try these tips to make better use of multiple critical paths in a project once you've set them up.

Try this	**Here's how**
Work with multiple projects in one schedule	1. Choose **File** > **Options**. 2. Choose **Schedule**, scroll to the bottom, and then select **Inserted projects are calculated like summary tasks**.
Show only critical tasks	On the Gantt Chart, choose **View** > **Filter** > **Critical**. To display all tasks again, select **No filter** in the filter list.
Group critical tasks	On the Gantt Chart, choose **View** > **Group By** > **Critical**. To display all tasks again, select **No Group** in the group list.

3) Graded Lab Tasks

Note: The instructor can design graded lab activities according to the level of difficult and complexity of the solved lab activities. The lab tasks assigned by the instructor should be evaluated in the same lab.

LAB TASK 1

Imagine you're managing the construction of a new office building. Tasks include site preparation, foundation pouring, structural framing, electrical wiring, plumbing installation, interior finishing, and landscaping. You can use MS Project to create a Gantt chart, listing tasks with start and end dates, and dependencies between tasks. The Network Diagram will visually display the sequence of tasks and their interdependencies. By performing Critical Path Analysis, you'll identify the tasks that are crucial for completing the project on time, helping you allocate resources effectively and manage potential delays.

LAB TASK 2

Suppose you're overseeing the development of a new software application. Tasks involve requirement gathering, design, coding, testing, debugging, and documentation. Use MS Project to create a Gantt chart displaying task timelines and their order. Generate a Network Diagram to illustrate how tasks rely on each other. Through Critical Path Analysis, determine which tasks must be closely monitored to prevent project delays.

LAB TASK 3

Imagine you're organizing a large-scale conference. Tasks include venue selection, speaker invitations, marketing campaigns, attendee registration, logistics planning, and post-event surveys. Develop a Gantt chart to visualize task durations and milestones. Create a Network Diagram to understand the flow of activities. Use Critical Path Analysis to identify tasks that are most critical to the event's success, allowing you to allocate resources effectively.

Lab # 08

12/11/2024 Lab

Applying different Cost Estimation Techniques

Objective:

The objective of applying different cost estimation techniques in Microsoft Project Management is to enhance the accuracy and reliability of project cost predictions and budgeting. By utilizing a variety of estimation methods such as bottom-up, top-down, analogous, and parametric estimating, project managers can better understand the potential financial implications of their projects. These techniques enable the identification of potential risks and uncertainties associated with cost factors, allowing for more informed decision-making and resource allocation. The use of diverse estimation approaches also helps in accommodating different project complexities and scope variations, leading to a more comprehensive and realistic cost projection, ultimately contributing to successful project execution and control.

Activity Outcomes:

After successful completion of this lab, the students will be able to apply the following concepts:

1. Accuracy and Reliability: Employing various estimation methods enhances the accuracy and reliability of cost predictions by considering multiple factors and perspectives, leading to more realistic budgeting and financial planning.

2. **Risk Assessment:** Different techniques allow for the identification and analysis of potential cost-related risks and uncertainties, helping project managers develop contingency plans and mitigate financial surprises.

3. **Informed Decision-making:** By comparing results from different estimation approaches, project managers can make more informed decisions about resource allocation, procurement strategies, and project scope adjustments.

4. **Adaptability:** Different projects have varying levels of complexity and data availability. Utilizing a range of estimation techniques accommodates these differences and ensures a tailored approach to cost estimation.

5. **Stakeholder Communication:** Using diverse methods facilitates effective communication with stakeholders, as project managers can present cost estimates from multiple angles, fostering a deeper understanding of the project's financial aspects.

Instructor Note:

As pre-lab activity, read Chapter 5 from the book “Microsoft Project 2019 Step by Step, Cindy Lewis, Carl Chatfield, Timothy Johnson., Microsoft Press, 2019”.

1) Useful Concepts

Cost estimation techniques in project management involve a range of approaches to predict the financial investment required for completing a project. Different techniques suit various project types, sizes, and complexities. Here are some useful concepts for different cost estimation techniques:

1. **Analogous Estimation (Expert Judgment)**:
   - **Concept**: This technique relies on comparing the current project with similar completed projects to estimate costs.
   - **Useful Concepts**: Historical data, past project experience, domain expertise, similarity assessment, adjusting for differences.
2. **Parametric Estimation**:
   - **Concept**: This technique uses statistical relationships between project parameters and costs to estimate costs for new projects.
   - **Useful Concepts**: Metrics correlation, regression analysis, cost drivers, functional relationships, mathematical models.
3. **Bottom-Up Estimation**:
   - **Concept**: Cost is estimated for each task or work package, and then aggregated to determine the overall project cost.
   - **Useful Concepts**: Work breakdown structure (WBS), task-level estimation, resource rates, detailed task analysis.
4. **Top-Down Estimation**:
   - **Concept**: An overall cost figure is determined, and then it is distributed across project components.
   - **Useful Concepts**: High-level estimation, analogy, percentage allocation, expert judgment, top-level assumptions.
5. **Three-Point Estimation (PERT)**:
   - **Concept**: A weighted average of optimistic, most likely, and pessimistic estimates is used to provide a more accurate range.
   - **Useful Concepts**: Triangular distribution, beta distribution, risk assessment, uncertainty, expected value.
6. **Reserve Analysis**:
   - **Concept**: Contingency reserves are added to the cost estimate to cover uncertainties and risks.
   - **Useful Concepts**: Risk identification, risk assessment, risk quantification, risk response planning, risk management.
7. **Vendor Bid Analysis**:
   - **Concept**: Costs are estimated based on bids from potential vendors or suppliers.
   - **Useful Concepts**: Request for proposal (RFP), vendor selection, bid evaluation, competitive bidding, negotiation.
8. **Expert Judgment**:
   - **Concept**: Cost estimates are derived from inputs provided by subject matter experts.
   - **Useful Concepts**: Industry expertise, domain knowledge, expert panels, consensus building, experience-based estimation.
9. **Cost of Quality Estimation**:
   - **Concept**: Estimating costs associated with ensuring project quality and avoiding rework or defects.

- **Useful Concepts**: Prevention costs, appraisal costs, failure costs, cost-benefit analysis, quality assurance.

10. **Learning Curve Estimation**:
    - **Concept**: As more units of work are performed, efficiency and productivity improve, leading to reduced costs.
    - **Useful Concepts**: Learning curve theory, experience curve, cumulative average time, progress ratio.

Choosing the appropriate cost estimation technique or combination of techniques depends on the project's nature, available data, project stage, and the level of accuracy required. It's often beneficial to combine multiple techniques to enhance the reliability of cost estimates and account for uncertainties.

2) Solved Lab Activities

Sr.No	*Allocated Time*	*Level of Complexity*	*CLO Mapping*
1	***30***	***Low***	***CLO-6***
2	***30***	***Low***	***CLO-6***

ACTIVITY 1

PERT (Program Evaluation and Review Technique) is a method to provide accurate estimates of project tasks. The PERT estimate techniques provides three different estimates for each task, including the most Pessimistic (P) estimate considering everything goes wrong. It includes the most Optimistic (O) estimate, assuming everything goes right. Finally, it includes the Most Likely (ML) estimate given standard work processes and effort.

Leveraging all three potential estimates, we derive a weighted final estimate for the project work effort. The resulting PERT estimate is calculated as (P + (4xML) + O) / 6. Notice the Most Likely (ML) estimate is weighted 4 times as much as the other two estimates.

Remember the PERT formula and use it to provide estimates when you have a high level of uncertainty.

Note: To be fair, there are other ways to derive the PERT estimate calculation, but this is the one that PMI references.

Scenario: We have an uncertain task assigned to a resource. Let's say the most likely estimate on the task is 16 hours. The best estimate, assuming everything goes right is 10 hours and the worst estimate, assuming everything goes wrong is 28 hours. Leveraging the PERT formula to provide an accurate estimate, the final estimate is 17 hours.

(28 + 16 (4) + 10) / 6 = 17 hours

Notice the final estimate was pulled slightly towards the Pessimistic estimate, but not by much, considering the final estimate to more closely aligned to the given most likely estimate for the task.

Incorporating PERT into Project Professional

Let’s begin configuring PERT within Project Professional to provide a more accurate estimate of our uncertain tasks. Here are the detailed configurations within Project Professional,

Create Task Level Field

1. Open Project Professional. Within Project Professional, select the Project tab. Within the Project tab, select the Custom Fields button.

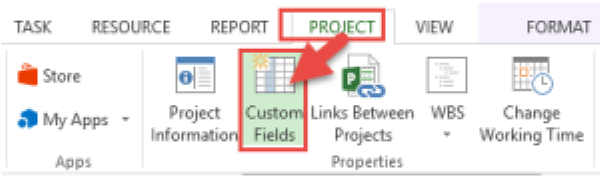


Figure 8.1: Custom Field Button

2. Within the Custom Fields window, be sure to select the Tasks type and select the Number format.

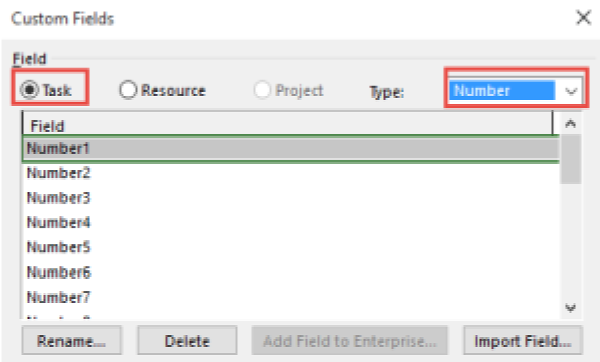


Figure 8.2: Tasks Type in Custom Fields

3. Within the Field list, select one of the unused fields. Click the

[Rename] button. Rename the task field “Optimistic”.

4. Rename three custom task fields within a Number format for PERT estimates:

Field	Custom Name
Number1	Optimistic
Number2	Most Likely
Number3	Pessimistic

Figure 8.3: Renaming Custom Fields

* Other than the field name, you may leave all other field format options set by default.

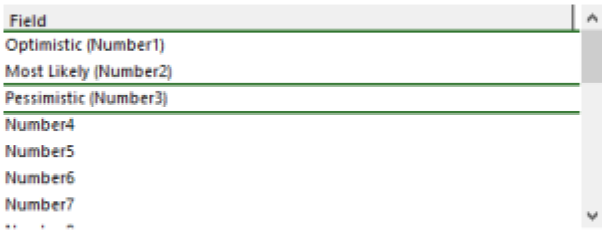


Figure 8.4: Default Names

5. When the PERT estimate fields have been entered, let's establish the PERT formula.

a. Select a fourth task field and rename the field "PERT Estimate".

b. Within the task options, select the [Formula] button within the Custom Attributes section.

c. Within the formula prompt, enter the following formula:

([Optimistic] + ([Most Likely] * 4) + [Pessimistic]) / 6

d. Within the Calculation for task and group summary rows, select Rollup: Sum

e. Click [OK] to complete the custom field entry.

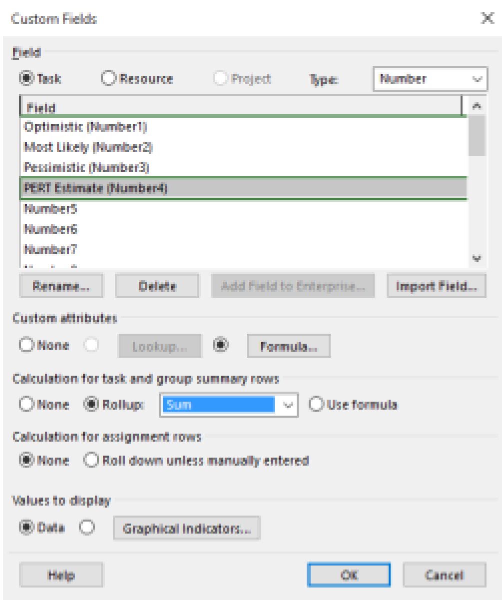


Figure 8.5: Custom Fields Entry

Create PERT View

Let's establish your PERT view for purposed of providing estimates.

1. Within Project Professional, select the View tab

2. Within the View tab, select the Tables button. Within the Tables, select [More Tables].

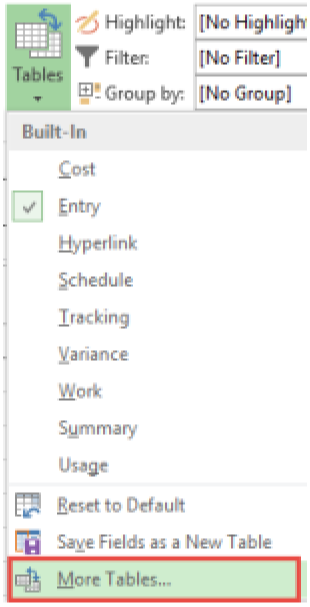


Figure 8.6: Tables View

3. Within the More Tables window, select the [New] button to establish a new table.

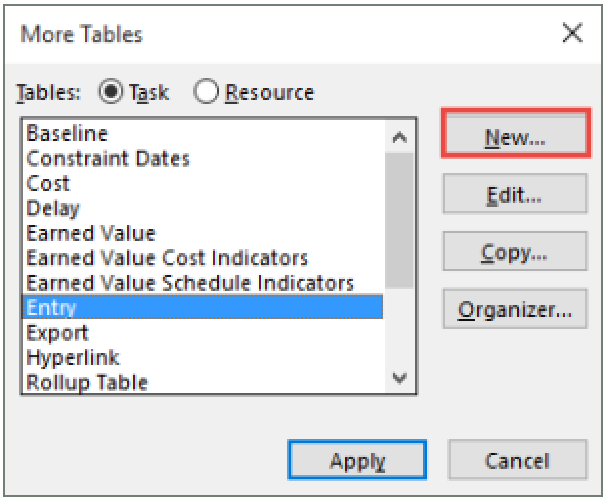


Figure 8.7: Establishing a new table

4. Give the table name "PERT Estimate".

5. Within the Field Name, select the following fields and apply valid Widths.

- ID Width: 6
- Name Width: 15
- Optimistic Width: 15
- Most Likely Width: 15
- Pessimistic Width: 15
- PERT Estimate Width: 15
- Work Width: 10

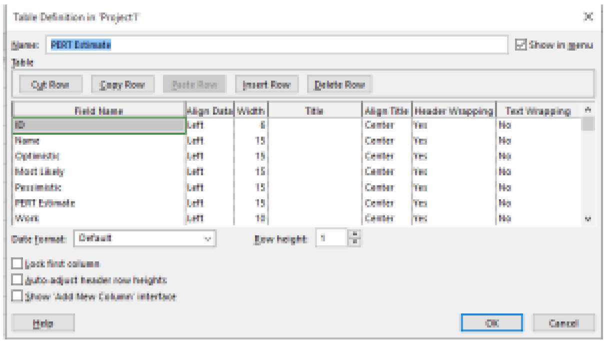


Figure 8.8: Table Definition

6. Click [OK] to save the new table.

7. Within the More Tables window, click [Apply]. Ensure the new table is currently displayed.

8. Within the Views tab, select the More Views option.

9. Within the More Views window, select [New] button.

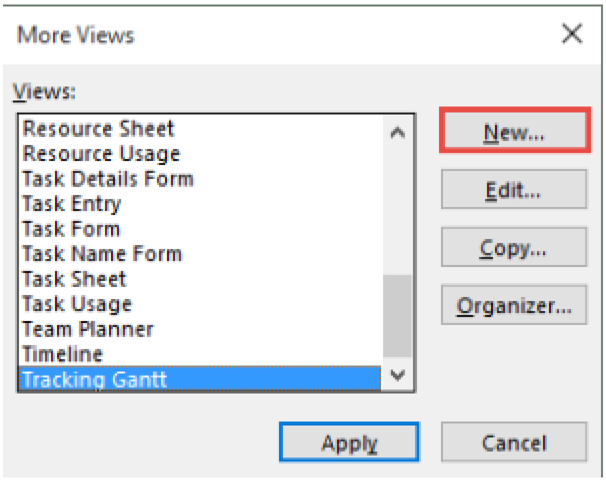


Figure 8.9: Views

10. Within the Define New View window, select Single view and click [OK].

11. Within the View Definition window, enter the following items:

- Name: PERT Estimate
- Screen: Gantt Chart
- Table: PERT Estimate
- Group: No Group
- Filter: All Tasks
- Be sure to Show in Menu

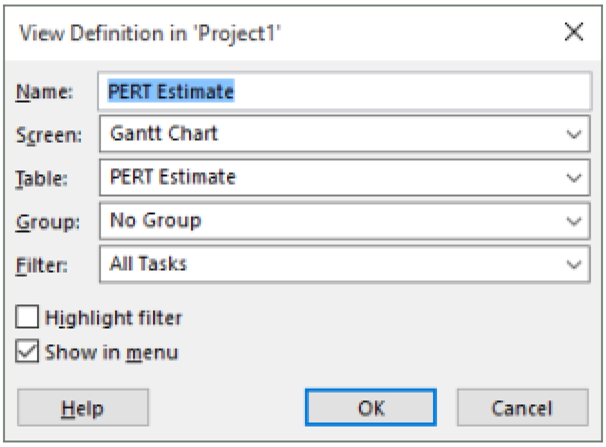


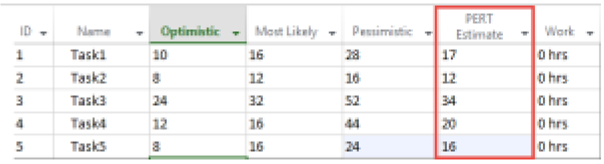
Figure 8.10: View Definition

12. Click [OK] to save your new view.

13. Within the More Views window, click Apply.

14. Confirm the display of the PERT Estimates view.

15. Begin entering your tasks and PERT Estimates to validate the calculation within the PERT Estimate field.



ID	Name	Optimistic	Most Likely	Pessimistic	PERT Estimate	Work
1	Task1	10	16	28	17	0 hrs
2	Task2	8	12	16	12	0 hrs
3	Task3	24	32	52	34	0 hrs
4	Task4	12	16	44	20	0 hrs
5	Task5	8	16	24	16	0 hrs

Figure 8.11: Estimates

16. At the conclusion, you may also enter the PERT Estimate task field into the default Entry table by clicking the standard [Add a new column] option.

ACTIVITY 2

In this scenario where you're tasked with estimating and managing the construction of a simple project: building a garden shed. We'll use PERT (Program Evaluation and Review Technique) along with Microsoft Project Professional 2019 to plan and manage this project. PERT involves estimating three time estimates for each task: optimistic, most likely, and pessimistic. The formula for calculating the expected duration (TE) is (Optimistic + 4 * Most Likely + Pessimistic) / 6.

Scenario: Building a Garden Shed

Project Tasks:

1. Design shed layout
2. Purchase materials
3. Build foundation

4. Assemble walls and roof
5. Paint and finish

Task Duration Estimates (in days):

- Design shed layout: Optimistic: 2, Most Likely: 5, Pessimistic: 8
- Purchase materials: Optimistic: 1, Most Likely: 3, Pessimistic: 5
- Build foundation: Optimistic: 3, Most Likely: 7, Pessimistic: 10
- Assemble walls and roof: Optimistic: 4, Most Likely: 6, Pessimistic: 9
- Paint and finish: Optimistic: 2, Most Likely: 4, Pessimistic: 6

Now, let's set this up in Microsoft Project Professional 2019:

1. **Create a New Project:**
   - Open Microsoft Project Professional 2019.
   - Click on "File" and select "New."
   - Choose "Blank Project."
2. **Define Tasks:**
   - Click on the "Task Name" column and enter the task names listed above.
3. **Enter Task Durations:**
   - Click on the "Duration" column for each task and enter the PERT estimates.
4. **Calculate Expected Durations:**
   - For each task, calculate the expected duration using the formula mentioned earlier: (Optimistic + 4 * Most Likely + Pessimistic) / 6.
   - Enter these expected durations in a new column, e.g., "Expected Duration."
5. **Create Dependencies:**
   - Link the tasks in the logical order they need to be performed. For example, "Build foundation" should come after "Purchase materials."
6. **View the PERT Chart:**
   - Go to the "View" tab and select "Pert Chart."
7. **View the Gantt Chart:**
   - You can also switch to the Gantt Chart view to see the project timeline with task dependencies.
8. **Analyze Critical Path:**
   - Microsoft Project will automatically calculate the critical path, which is the sequence of tasks that determines the project's overall duration.
9. **Resource Allocation (Optional):**
   - You can assign resources to tasks and manage resource allocation in the "Resource Sheet" view.
10. **Save the Project:**
    - Click on "File" and choose "Save" to save your project file.

This scenario demonstrates how PERT can be applied using Microsoft Project Professional 2019 for a simple project. It helps in estimating the expected duration of the project tasks and identifying the critical path, which helps in effective project planning and management. Remember that this is a simplified scenario, and in real-world projects, you would need to consider more factors, such as resource availability, costs, and potential risks.

3) Graded Lab Tasks

Note: The instructor can design graded lab activities according to the level of difficult and complexity of the solved lab activities. The lab tasks assigned by the instructor should be evaluated in the same lab.

LAB TASK 1

In this scenario apply PERT (Program Evaluation and Review Technique) using Microsoft Project Professional 2019. In this scenario, you'll be managing a small event planning project: organizing a company picnic.

Scenario: Organizing a Company Picnic

Project Tasks:

1. *Choose picnic location*
2. *Plan activities*
3. *Arrange catering*
4. *Send invitations*
5. *Set up decorations*
6. *Conduct the picnic*
7. *Clean up after the picnic*

Task Duration Estimates (in days):

- *Choose picnic location: Optimistic: 1, Most Likely: 2, Pessimistic: 4*
- *Plan activities: Optimistic: 2, Most Likely: 3, Pessimistic: 6*
- *Arrange catering: Optimistic: 1, Most Likely: 3, Pessimistic: 5*
- *Send invitations: Optimistic: 1, Most Likely: 2, Pessimistic: 3*
- *Set up decorations: Optimistic: 2, Most Likely: 3, Pessimistic: 4*
- *Conduct the picnic: Optimistic: 1, Most Likely: 1, Pessimistic: 2*
- *Clean up after the picnic: Optimistic: 1, Most Likely: 2, Pessimistic: 3*

LAB TASK 2

In this scenario, we'll focus on a basic software development project: Creating a simple website.

Scenario: Creating a Simple Website

Project Tasks:

1. *Gather requirements*
2. *Design website layout*
3. *Develop HTML/CSS*
4. *Implement functionality*
5. *Review and testing*
6. *Deploy the website*

You need to assign ***Task Duration Estimates (in days) by yourself like for example***

Optimistic: 1, Most Likely: 3, Pessimistic: 5.

Lab #10

Applying Quality Management Techniques

Objective:

The objective of this lab is to make students understand the Importance of Quality Management and to Explore how quality management ensures software project success. They will also learn MS Project's Role in Quality Management which will help them understand how MS Project can be leveraged for implementing quality management techniques. Students will also implement Quality Control Measures: and monitor Quality Metrics. This lab will also help them learn to make informed project decisions based on quality data.

Activity Outcomes:

After successful completion of this lab, the students will be able to apply the following concepts:

1. Practical Application of Theoretical Concepts: Apply theoretical knowledge of quality management in a practical scenario using MS Project.
2. Skill Enhancement in MS Project: Develop advanced proficiency in using MS Project for quality management.
3. Project Planning and Execution Skills: Enhance abilities in planning and executing projects with a focus on quality.
4. Critical Thinking and Problem-Solving: Strengthen critical thinking and problem-solving skills in the context of project quality.
5. Data Analysis and Interpretation: Improve skills in analyzing and interpreting quality-related data in project management.

Instructor Note:

As pre-lab activity, read Chapter 8 from the book “PMBOK”.

Useful Concepts:

Software Quality Management is a comprehensive discipline focused on ensuring that software products meet specified requirements and customer expectations, delivering high quality and reliable performance. It encompasses a variety of practices, methodologies, and standards designed to systematically manage the quality of software throughout its lifecycle. Key components include quality assurance (QA), which involves the establishment of processes to prevent defects; quality control (QC), which involves the detection and correction of defects through testing; and continuous improvement, which focuses on refining processes and products based on feedback and performance metrics. Effective SQM ensures that software is developed efficiently, within budget, and is robust, scalable, and secure, thereby enhancing user satisfaction and trust.

The concepts covered in this lab are:

- Quality Management Fundamentals: Understanding the basics of quality management in the context of software projects.
- MS Project Tools for Quality Management: Detailed exploration of MS Project features pertinent to quality management.
- Setting Quality Objectives: Techniques for defining clear, measurable quality objectives within MS Project.
- Quality Control and Assurance Processes: Implementing processes for quality control and assurance in a project.
- Quality Metrics and KPIs: Identifying and tracking Key Performance Indicators (KPIs) for quality in MS Project.
- Risk Management in Quality: Addressing potential risks to project quality and mitigation strategies.

2) Solved Lab Activities

Sr.No	*Allocated Time*	*Level of Complexity*	*CLO Mapping*
1	***30***	***Medium***	***CLO-6***
2	***30***	***Medium***	***CLO-6***
3	***15***	***Medium***	***CLO-5***
4	***15***	***Medium***	***CLO-5***
5	***15***	***Medium***	***CLO-5***

<u>Activity 1:</u>

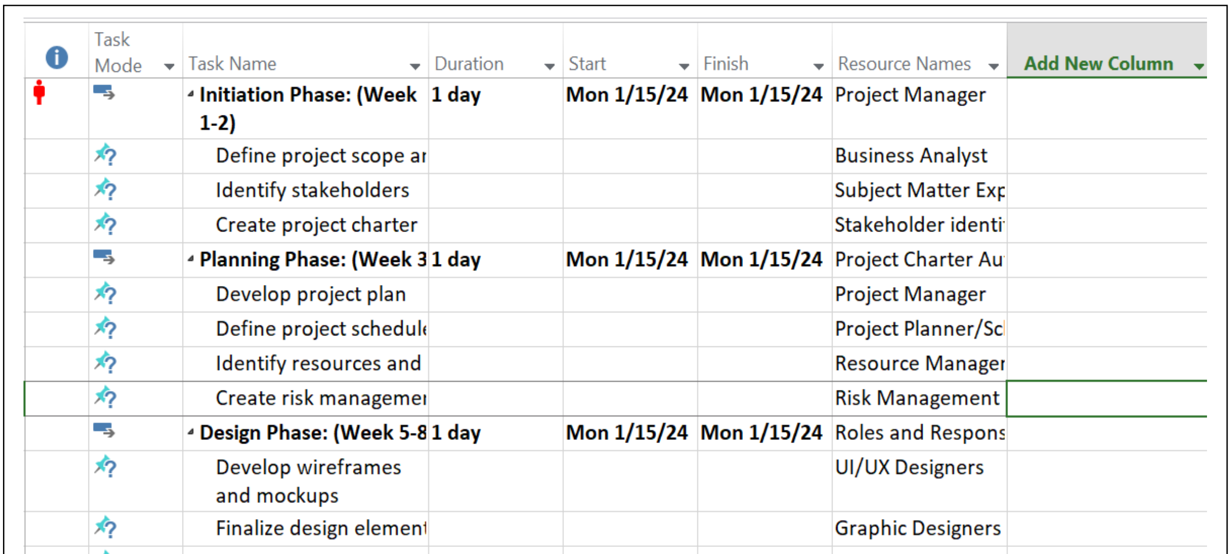
Setting Up a Quality Management Plan in MS Project

1) Open MS Project: Launch MS Project and select 'New Project'.

2) Define Project Information: Enter project name, start date, and other basic information.

3) Set Up Tasks: Create a list of tasks and subtasks necessary for the project.

4) Assign Resources: Allocate resources to each task, including team members and materials.



Task Mode	Task Name	Duration	Start	Finish	Resource Names	Add New Column
	Initiation Phase: (Week 1-2)	1 day	Mon 1/15/24	Mon 1/15/24	Project Manager	
	Define project scope ar				Business Analyst	
	Identify stakeholders				Subject Matter Exp	
	Create project charter				Stakeholder identi	
	Planning Phase: (Week 3	1 day	Mon 1/15/24	Mon 1/15/24	Project Charter Au	
	Develop project plan				Project Manager	
	Define project schedule				Project Planner/Sc	
	Identify resources and				Resource Manager	
	Create risk managemer				Risk Management	
	Design Phase: (Week 5-8	1 day	Mon 1/15/24	Mon 1/15/24	Roles and Respons	
	Develop wireframes and mockups				UI/UX Designers	
	Finalize design element				Graphic Designers	

	Task Mode	Task Name	Duration	Start	Finish	Resource Names	Add New Column
		Initiation Phase: (Week 1-2)	1 day	Mon 1/15/24	Mon 1/15/24	Project Manager	
		Define project scope ar				Business Analyst	
		Identify stakeholders				Subject Matter Exp	
		Create project charter				Stakeholder identi	
		Planning Phase: (Week 3	1 day	Mon 1/15/24	Mon 1/15/24	Project Charter Au	
		Develop project plan				Project Manager	
		Define project schedul				Project Planner/Sc	
		Identify resources and				Resource Manager	
		Create risk managemer				Risk Management	
		Design Phase: (Week 5-8	1 day	Mon 1/15/24	Mon 1/15/24	Roles and Respons	
		Develop wireframes and mockups				UI/UX Designers	
		Finalize design element				Graphic Designers	
		Review and obtain client approval				Client Representative	
		Development Phase:	1 day	Mon 1/15/24	Mon 1/15/24	Design Review	

8) Set Baselines: Set a baseline to compare actual progress with the planned progress.

9) Document Quality Objectives: In the notes section, detail the quality objectives for each task.

10) Save the Plan: Save the project plan for future reference and tracking.

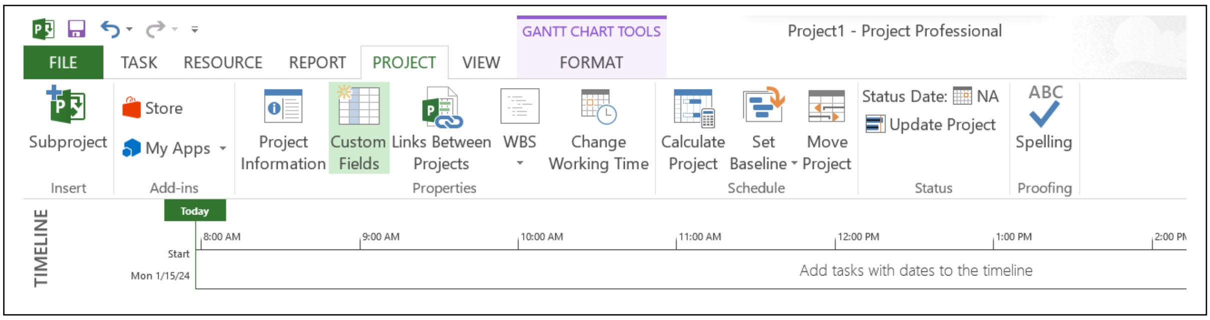
Activity 2:

Defining Quality Metrics and KPIs in MS Project

1) Identify Key Quality Metrics: Determine what quality aspects are critical for your project (e.g., defect rates, client satisfaction scores).

	Task Mode	Task Name	Quality Metrics	Text3	Add New Column
		Create project charter	Code Review Effective		
		Planning Phase: (Week 3	Test Coverage		
		Develop project plan	On-Time Delivery		
		Define project schedul	UAT Success Rate		
		Identify resources and	Performance Metrics		
		Create risk managemer	Documentation Comp		
		Design Phase: (Week 5-8	Change Request Rate		
		Develop wireframes and mockups	Post-Launch Defect Rate		
		Finalize design element	Resource Utilization		
		Review and obtain client approval	Risk Mitigation Success		

2) Custom Fields for Metrics: In MS Project, go to 'Project' tab, select 'Custom Fields' to create new fields for each metric.



3) Setting Baseline for Metrics: Establish a baseline for these metrics to measure performance against.

4) Regular Review Setup: Schedule regular intervals to review these metrics.

5) Data Input Mechanism: Establish how data for these metrics will be collected and updated in MS Project.

6) Analysis Reports: Use MS Project’s reporting features to create periodic reports for these metrics.

7) Reviewing Metric Trends: Analyze the reports to observe trends over time.

8) Save and Update Project Plan: Regularly save the project plan with updated metrics.

Activity 3:

Implementing Quality Control Checkpoints

1) Planning Quality Checkpoints: Identify stages in the project where quality checks are essential.
2) Adding Checkpoints to Tasks: In the Gantt Chart view, add checkpoints as specific tasks.

Task Mode	Task ID	Task Name	Quality Metrics
	1	**Initiation Phase: (Week 1-2)**	Defect Density
	1.1	Define project scope ar	Client Satisfaction Sco
	1.2	Identify stakeholders	Requirements Traceal
	1.3	Create project charter	Code Review Effective
	2	**Planning Phase: (Week 3**	Test Coverage
	2.1	Develop project plan	On-Time Delivery
	2.2	Define project schedul	UAT Success Rate
	2.3	Identify resources and	Performance Metrics
	2.4	Create risk managemei	Documentation Comp
	3	**Design Phase: (Week 5-8**	Change Request Rate
	3.1	Develop wireframes and mockups	Post-Launch Defect Rate

3) Assigning Responsibility: Assign team members to each checkpoint task.

Task Mode	Task ID	Task Name	Quality Metrics	Resource Names
	1	**Initiation Phase: (Week 1-2)**	Defect Density	Project Manager
	1.1	Define project scope ar	Client Satisfaction Sco	Business Analyst
	1.2	Identify stakeholders	Requirements Traceal	Subject Matter Expert
	1.3	Create project charter	Code Review Effective	Stakeholder identifica
	2	**Planning Phase: (Week 3**	Test Coverage	Project Charter Autho
	2.1	Develop project plan	On-Time Delivery	Project Manager
	2.2	Define project schedul	UAT Success Rate	Project Planner/Sched
	2.3	Identify resources and	Performance Metrics	Resource Manager
	2.4	Create risk manageme	Documentation Comp	Risk Management Spe
	3	**Design Phase: (Week 5-8**	Change Request Rate	Roles and Responsibil
	3.1	Develop wireframes and mockups	Post-Launch Defect Rate	UI/UX Designers
	3.2	Finalize design element	Resource Utilization	Graphic Designers
	3.3	Review and obtain client approval	Risk Mitigation Success	Client Representative

4) Setting Deadlines: Define deadlines for each quality checkpoint.

	Task Name	Quality Metrics	Resource Names	Deadline	Add New Column
1	**Initiation Phase: (Week 1-2)**	Defect Density	Project Manager	**Wed 1/31/24**	
2	Define project scope ar	Client Satisfaction Sco	Business Analyst	Tue 2/13/24	
3	Identify stakeholders	Requirements Traceal	Subject Matter Expert	Thu 2/15/24	
4	Create project charter	Code Review Effective	Stakeholder identifica	Thu 2/29/24	
5	**Planning Phase: (Week 3**	Test Coverage	Project Charter Autho	**Thu 3/14/24**	
6	Develop project plan	On-Time Delivery	Project Manager	Fri 3/1/24	

5) Reminder Alarms: Set up reminders for approaching checkpoints.

6) Documentation Requirements: Specify what documents or results need to be produced at each checkpoint.

7) Status Update Routine: Establish a routine for updating the status of these checkpoints.

8) Record Keeping: Maintain detailed records of each checkpoint in the project notes or documents section.

Activity 4:

Monitoring and Analyzing Quality Data

1) Open Project File: Open your project file where quality metrics are tracked.

2) Review Updated Metrics: Check the latest updates on the quality metrics.

3) Variance Analysis: Analyze any variances from the planned quality metrics.

4) Adjust Plans: Make necessary adjustments to tasks based on quality data.

5) Generate Quality Reports: Use MS Project's reporting feature to generate quality reports.

6) Stakeholder Communication: Update stakeholders with the quality status reports.

7) Feedback Integration: Incorporate feedback into the project plan for continuous improvement.

8) Save and Document Changes: Save all changes and document actions taken for future reference.

Activity 5:

Conducting a Quality Review and Feedback Session

1) Schedule Review Meeting: Utilize MS Project to plan a meeting for reviewing project quality. Set a date, time, and list attendees in the calendar.

2) Prepare Quality Report: Before the meeting, generate a detailed report on project quality metrics using MS Project's reporting feature.

3) Gather Team Feedback: During the meeting, discuss the quality report with the team and solicit their feedback on various quality aspects of the project.

4) Analyze Feedback for Insights: Post-meeting, analyze the collected feedback to identify key areas for improvement or concern.

5) Develop an Improvement Plan: Based on the feedback, formulate a plan targeting quality improvement in specific areas of the project.

6) Update Project Plan: Incorporate the improvement strategies into the project plan using MS Project, adjusting tasks and timelines as necessary.

7) Communicate Changes: Clearly communicate any changes or updates in the project plan to the team and stakeholders.

8) Document Meeting Minutes: Record the main points discussed, feedback received, and decisions made during the meeting. Include these minutes in the project documentation for reference.

Graded Lab Tasks:

Note: The instructor can design graded lab activities according to the level of difficulty and complexity of the solved lab activities. The lab tasks assigned by the instructor should be evaluated in the same lab.

- **Task 1:**
  Improving an Existing Project: Take an existing project plan and identify areas where quality management was insufficient. Use MS Project to enhance the plan, focusing on integrating more robust quality control measures.

- **Task 2:**
  FYP Quality Management Application: Students are to apply quality management techniques to their ongoing Final Year Project. They should develop a comprehensive quality management plan in MS Project, tailoring the metrics and checkpoints specifically to the needs of their FYP. This task requires them to integrate theoretical knowledge with practical application in a real-world scenario.

Lab #11

Establishing Communication Management Techniques in MS Project

Objective:

The objective of this lab is to make students Understanding Communication Management and to Grasp the essentials of effective communication in project settings. This lab will make them able to use MS Project for Communication Planning i-e Utilize MS Project to plan and monitor communication. This lab focuses on different Communication Methods and Tools: Explore various communication methods and tools within MS Project.

Activity Outcome:

After successful completion of this lab, the students will be able to:

1. Apply Communication Theories: Implement communication management theories using MS Project.
2. Enhanced Collaboration Skills: Improve skills in facilitating team and stakeholder communication.
3. Make Effective Communication Plans: Develop proficiency in creating communication plans.
4. Provide Stakeholder Management: Gain expertise in managing stakeholder expectations and engagement.
5. Apply these skills in FYPs and future projects.

Instructor Note:

As pre lab activity, read chapter 10 “Project Communication Management” from PMBOK.

Useful Concepts:

Software Communication Management is a critical aspect of software project management that focuses on the effective exchange of information among all stakeholders involved in a project. This includes team members, project managers, clients, and other relevant parties. It involves establishing clear communication channels, defining protocols for information sharing, and ensuring that all stakeholders are kept informed about project progress, changes, and issues. Effective Communication Management facilitates collaboration, reduces misunderstandings, and helps in making informed decisions, which are crucial for the successful execution of a project. By prioritizing robust communication management, projects can achieve better coordination, improved problem-solving, and enhanced overall efficiency. This lab covers the following concepts in more detail:

1. Communication Management Principles: Explore principles of effective communication in projects.
2. MS Project’s Communication Tools: Delve into MS Project features for communication planning and tracking.

3. Stakeholder Analysis: Techniques for identifying and categorizing stakeholders.
4. Developing Communication Plans: Crafting communication plans tailored to project needs.
5. Managing Information Distribution: Strategies for efficient information sharing and distribution.
6. Monitoring Communication: Techniques for assessing the effectiveness of communication strategies.

2) Solved Lab Activities:

Sr.No	*Allocated Time*	*Level of Complexity*	*CLO Mapping*
1	***30***	***Medium***	***CLO-6***
2	***30***	***Medium***	***CLO-6***
3	***15***	***Medium***	***CLO-5***
4	***15***	***Medium***	***CLO-5***
5	***15***	***Medium***	***CLO-5***

<u>Activity 1:</u>

Creating a Communication Plan in MS Project

1) Launch MS Project and Open New Project: Start MS Project and create a new project file.

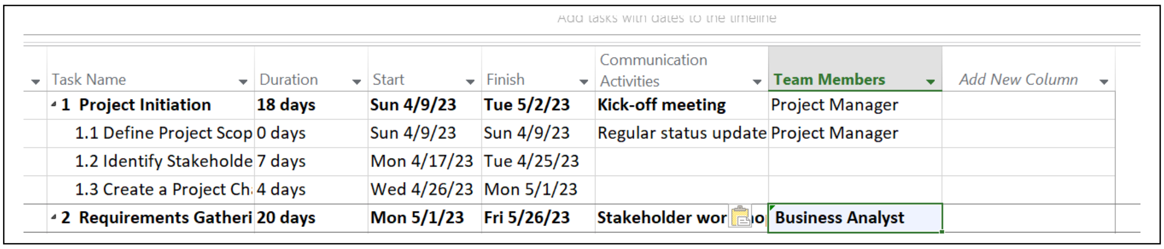
2) Define Project Scope and Objectives: Enter basic project information like name, start date, and objectives in the 'Project Information' dialog.

3) Insert Communication Plan Section: Add a new section in your project plan dedicated to communication. This can be done by inserting new tasks labeled as 'Communication Activities'.

4) Identify Communication Activities: List key communication activities such as meetings, reports, and updates.

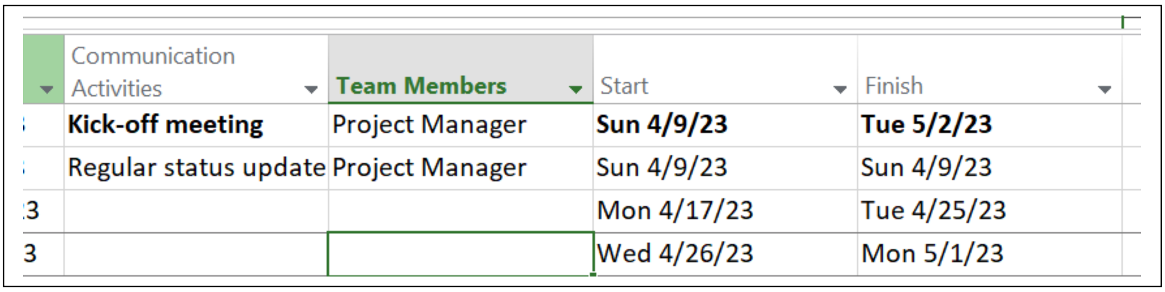
Task Mode	Task Name	Duration	Start	Finish	Communication Activities	Add New Column
	1 Project Initiation	**18 days**	**Sun 4/9/23**	**Tue 5/2/23**	**Kick-off meeting**	
	1.1 Define Project Scop	0 days	Sun 4/9/23	Sun 4/9/23	Regular status update	
	1.2 Identify Stakeholde	7 days	Mon 4/17/23	Tue 4/25/23		
	1.3 Create a Project Ch	4 days	Wed 4/26/23	Mon 5/1/23		
	2 Requirements Gatheri	**20 days**	**Mon 5/1/23**	**Fri 5/26/23**	**Stakeholder worksho**	
	2.1 Collect User Requir	7 days	Mon 5/1/23	Tue 5/9/23	Regular progress repo	
	2.2 Prioritize requireme	5 days	Wed 5/10/23	Tue 5/16/23	Updates on competitc	
	2.3 Create user person	5 days	Wed 5/17/23	Tue 5/23/23		
	2.4 Conduct competito	2.6 days	Wed 5/24/23	Fri 5/26/23		
	3 Design and Planning	**23 days**	**Thu 6/1/23**	**Mon 7/3/23**	**Meetings**	
	3.1 Wireframe the app	6 days	Thu 6/1/23	Thu 6/8/23	Project plan presentat	
	3.2 Create a design pro	5 days	Fri 6/9/23	Thu 6/15/23		

5) Assign Roles and Responsibilities: Assign team members to each communication activity, specifying their roles.



Task Name	Duration	Start	Finish	Communication Activities	Team Members	Add New Column
1 Project Initiation	**18 days**	**Sun 4/9/23**	**Tue 5/2/23**	**Kick-off meeting**	Project Manager	
1.1 Define Project Scop	0 days	Sun 4/9/23	Sun 4/9/23	Regular status update	Project Manager	
1.2 Identify Stakeholde	7 days	Mon 4/17/23	Tue 4/25/23			
1.3 Create a Project Ch	4 days	Wed 4/26/23	Mon 5/1/23			
2 Requirements Gatheri	**20 days**	**Mon 5/1/23**	**Fri 5/26/23**	**Stakeholder wor**	**Business Analyst**	

6) Schedule Communication Activities: Set start and end dates for each communication activity.



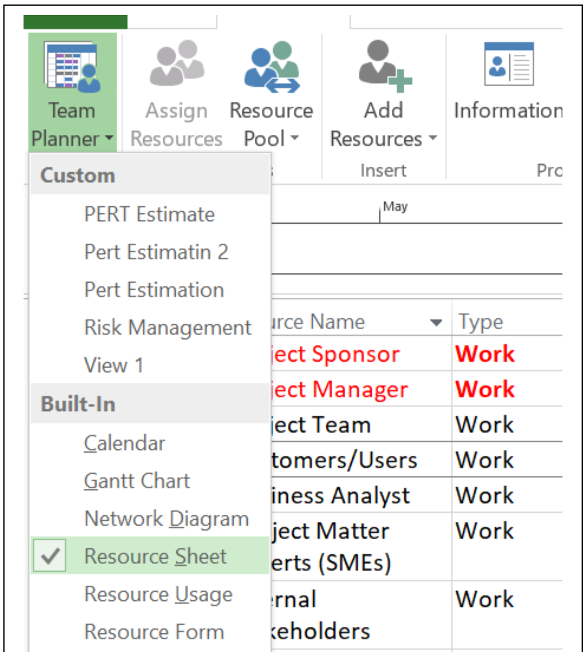
Communication Activities	Team Members	Start	Finish
Kick-off meeting	Project Manager	**Sun 4/9/23**	**Tue 5/2/23**
Regular status update	Project Manager	Sun 4/9/23	Sun 4/9/23
		Mon 4/17/23	Tue 4/25/23
		Wed 4/26/23	Mon 5/1/23

7) Set Up Reminders: Use the 'Task Information' dialog to set reminders for upcoming communication activities.
8) Review and Save Communication Plan: Ensure all communication activities are correctly planned and then save your project.

<u>Activity 2:</u>

Stakeholder Mapping and Analysis

1) Open Existing Project Plan: Load your project in MS Project.



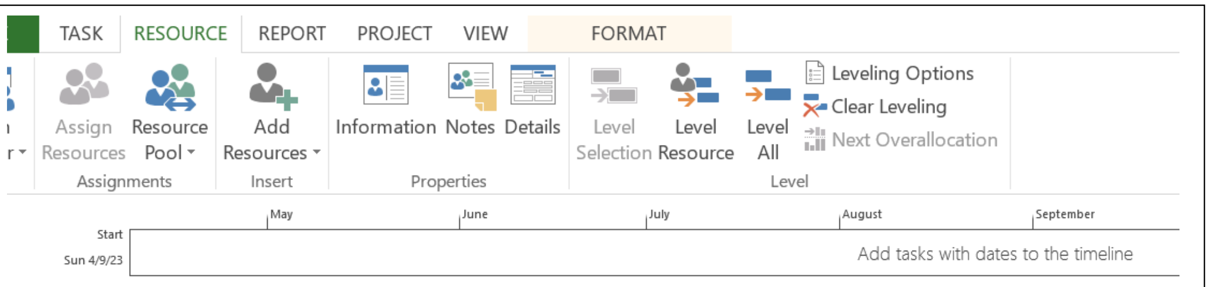
2) Identify Stakeholders: List all project stakeholders in the 'Resource Sheet' view.

		Resource Name	Type	Material	Initials	Group	Max.
1		Project Team	Work		P		100%
2		Customers/Users	Work		C		50%
3		Business Analyst	Work		B		100%
4		Subject Matter Experts (SMEs)	Work		S		100%
5		Internal Stakeholders	Work		I		100%
6		External Stakeholders	Work		E		100%
7		Regulatory Authorities	Work		R		100%
8		Investors/Sharehold	Work		I		100%
9		Project Management Office (PMO)	Work		P		100%
10		Quality Assurance/Testing Team	Work		Q		100%
11		Marketing Team	Work		M		100%
12		Legal Team	Work		L		100%
13		Operations and	Work		O		100%

3) Categorize Stakeholders: Categorize each stakeholder based on their interest and influence in the project.

4) Assign Stakeholders to Tasks: Link stakeholders to specific tasks they are interested in or can influence.

5) Develop Engagement Strategies: Note down engagement strategies for each stakeholder in the 'Notes' section of tasks.



		Resource Name	Type	Material	Initials	Group	Max.	Std. Rate	Ovt. Rate
1		Project Team	Work		P		100%	$60.00/hr	$16.00/hr
2		Customers/Users	Work		C		50%	$62.00/hr	$18.00/hr
3		Business Analyst	Work		B		100%	$0.00/hr	$0.00/hr
4		Subject Matter	Work		S		100%	$0.00/hr	$0.00/hr

6) Regular Updates: Update the stakeholder list and their engagement strategies as the project progresses.

7) Review and Adjust: Regularly review the stakeholder analysis and adjust strategies as needed.

8) Save Updated Project Plan: Save all changes made in your stakeholder analysis.

Activity 3:

Developing a Project Information Distribution Plan

1) Access Communication Section: Go to the communication section in your MS Project plan.

2) List Information Distribution Activities: Add tasks for distributing information like email updates, reports, and meetings.

3) Assign Responsible Team Members: Allocate team members to handle each information distribution task.

Task Name	Duration	Start	Finish	Communication Activities	Te
1 Project Initiation	**18 days**	**Sun 4/9/23**	**Tue 5/2/23**	**Kick-off meeting**	**P**
1.1 Define Project Scop	0 days	Sun 4/9/23	Sun 4/9/23	Regular status update	P
1.2 Identify Stakeholde	7 days	Mon 4/17/23	Tue 4/25/23		
1.3 Create a Project Ch	4 days	Wed 4/26/23	Mon 5/1/23		
2 Requirements Gatheri	**20.4 days**	**Mon 5/1/23**	**Mon 5/29/23**	**Stakeholder worksho**	**B**
2.1 Collect User Requir	7 days	Mon 5/1/23	Tue 5/9/23	Regular progress repo	
2.2 Prioritize requireme	5 days	Wed 5/10/23	Tue 5/16/23	Updates on competito	
2.3 Create user person	5 days	Wed 5/17/23	Tue 5/23/23		
2.4 Conduct competito	3 days	Wed 5/24/23	Mon 5/29/23		
3 Design and Planning	**24 days**	**Thu 6/1/23**	**Tue 7/4/23**	**Meetings**	
3.1 Wireframe the app	6 days	Thu 6/1/23	Thu 6/8/23	Project plan presentat	
3.2 Create a design pro	5 days	Fri 6/9/23	Thu 6/15/23		
3.3 Select the technolo	3 days	Fri 6/16/23	Tue 6/20/23		

4) Define Information Distribution Schedule: Schedule when each type of information will be distributed.

Task Name	Duration	Start	Finish	Communication Activities	Team Members
1 Project Initiation	**18 days**	**Sun 4/9/23**	**Tue 5/2/23**	**Kick-off meeting**	Project Manager
1.1 Define Project Scop	0 days	Sun 4/9/23	Sun 4/9/23	Regular status update	Project Manager
1.2 Identify Stakeholde	7 days	Mon 4/17/23	Tue 4/25/23		
1.3 Create a Project Ch	4 days	Wed 4/26/23	Mon 5/1/23		
2 Requirements Gatheri	**20.4 days**	**Mon 5/1/23**	**Mon 5/29/23**	**Stakeholder worksho**	**Business Analyst**

5) Set Up Distribution Methods: Specify the method of distribution for each type of information (e.g., email, face-to-face meetings).

6) Monitor and Update: Regularly monitor the effectiveness of information distribution and update methods as necessary.

7) Document Changes: Make notes of any changes or updates in information distribution methods.

8) Save Distribution Plan: Save the updated project file with all changes in the distribution plan.

Activity 4:

Implementing Communication Strategies in MS Project

1) Review Communication Plan: Open your project and review the existing communication plan.

2) Add Communication Strategy Tasks: Implement specific communication strategy tasks within the project timeline.

Task Name	Duration	Start	Finish	Communication Activities	Team Members	Communication Strategies	Add Ne
1 Project Initiation	**18 days**	**Sun 4/9/23**	**Tue 5/2/23**	**Kick-off meeting**	Project Manager	Stakeholder Communication Pla	
1.1 Define Project Scop	0 days	Sun 4/9/23	Sun 4/9/23	Regular status update	Project Manager	Project Kickoff Meeting	
1.2 Identify Stakeholde	7 days	Mon 4/17/23	Tue 4/25/23			Regular Status Meetings	
1.3 Create a Project Ch	4 days	Wed 4/26/23	Mon 5/1/23			Progress Reports	
2 Requirements Gatheri	**20.4 days**	**Mon 5/1/23**	**Mon 5/29/23**	**Stakeholder worksho**	**Business Analyst**	Issue Resolution Protocol	
2.1 Collect User Requir	7 days	Mon 5/1/23	Tue 5/9/23	Regular progress repo		Feedback Sessions	
2.2 Prioritize requireme	5 days	Wed 5/10/23	Tue 5/16/23	Updates on competitc		Change Management Communi	
2.3 Create user person	5 days	Wed 5/17/23	Tue 5/23/23			Collaboration Tools Training	
2.4 Conduct competito	3 days	Wed 5/24/23	Mon 5/29/23			Client Demos	
3 Design and Planning	**24 days**	**Thu 6/1/23**	**Tue 7/4/23**	**Meetings**		Post-Launch Communication	

3) Link Strategies to Project Milestones: Associate communication strategies with relevant project milestones.

Communication Activities	Team Members	Communication Strategies	Milestones	Add New (
Kick-off meeting	Project Manager	Stakeholder Communication Pla	**Initiation Phase**	
Regular status update	Project Manager	Project Kickoff Meeting	Planning Phase	
		Regular Status Meetings	Design Phase	
		Progress Reports	Development Phase	

4) Allocate Resources: Assign team members and resources to each communication strategy task.

5) Set Deadlines and Reminders: Define deadlines for each strategy and set reminders.

6) Regular Review and Adjustments: Regularly review the effectiveness of these strategies and make adjustments as needed.

7) Stakeholder Feedback: Incorporate feedback from stakeholders into the plan and update accordingly.

8) Finalize and Save: Finalize the communication strategy implementation and save the project file.

Activity 5:

Evaluating Communication Effectiveness

1) Open Project with Implemented Strategies: Load your project file in MS Project.
2) Review Communication Strategies: Examine each implemented communication strategy within the project timeline.
3) Gather Feedback: Collect feedback on communication effectiveness from team members and stakeholders.
4) Analyze Feedback: Analyze feedback to identify successful and less effective communication practices.
5) Make Adjustments: Based on the analysis, make necessary adjustments to communication strategies.
6) Update Project Plan: Reflect these changes in your MS Project plan.
7) Document Changes and Feedback: Record any changes made and feedback received in the project documentation.
8) Save and Share the Updated Plan: Save the updated project file and share it with relevant stakeholders.

Graded Lab Task:

Note: The instructor can design graded lab activities according to the level of difficulty and complexity of the solved lab activities. The lab tasks assigned by the instructor should be evaluated in the same lab.

1) **Task 1:**
   Create a detailed communication plan for a part of your FYP using MS Project.

2) **Task 2 :**
   Conduct stakeholder analysis for a hypothetical project and outline engagement strategies.

3) **Task 3**
   Develop and implement a comprehensive communication and stakeholder engagement plan in MS Project for a complex project.

4) **Task 4**
   Analyze a real-world case study and suggest improvements in communication management using MS Project.

Lab 12

Project Risk Management Techniques

Objective:

This objective of this lab is to make students Understand Risk Management Fundamentals and gain proficiency in MS Project for Risk Management. They will also learn about Risk Response Planning and Risk Monitoring and Reporting.

Activity Outcomes:

After successful completion of this lab, the students will be able to:

1. Learn application of Risk Management Theories: Ability to apply theoretical risk management concepts in a practical setting using MS Project.
2. Gain Enhanced Decision-Making Skills: Improved skills in making informed decisions regarding risk handling in projects.
3. Plan Effective Risk Response Execution: Mastery in formulating and executing risk response plans in project environments.
4. Apply their knowledge in Risk Monitoring Tools: Competence in utilizing MS Project's risk monitoring and reporting features.
5. Prepare Real-World Project Risk Management plans: Preparedness to manage risks in their FYPs and future professional projects.

Instructor Note:

As pre lab activity, read chapter 11 “Project Risk Management” from PMBOK.

Useful Concepts:

Software Project Risk Management is a systematic approach to identifying, analyzing, and mitigating risks that could potentially impact the successful completion of a software project. This process involves proactive risk assessment at every stage of the project lifecycle, from planning through execution and delivery. Key activities include identifying potential risks, evaluating their likelihood and impact, prioritizing them, and developing strategies to manage or mitigate them. Effective risk management ensures that risks are monitored and controlled, reducing the chances of unexpected issues derailing the project. It also involves continuous communication with stakeholders to keep them informed about risk status and mitigation plans. By integrating risk management into the project management process, teams can enhance resilience, make more informed decisions, and increase the likelihood of delivering a high-quality software product on time and within budget.

2) Solved Lab Activities:

Sr.No	*Allocated Time*	*Level of Complexity*	*CLO Mapping*
1	***30***	***Medium***	***CLO-6***
2	***30***	***Medium***	***CLO-6***
3	***15***	***Medium***	***CLO-5***
4	***15***	***Medium***	***CLO-5***
5	***15***	***Medium***	***CLO-5***

Activity 1:

Risk Register Creation

Step 1: Open MS Project

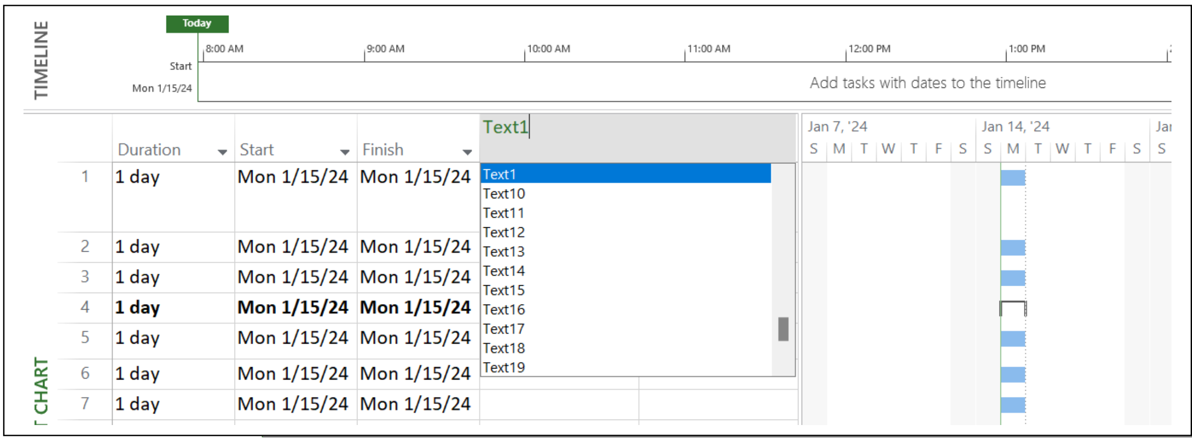
- Click on the search bar on your computer.
- Type "MS Project" and open the application.

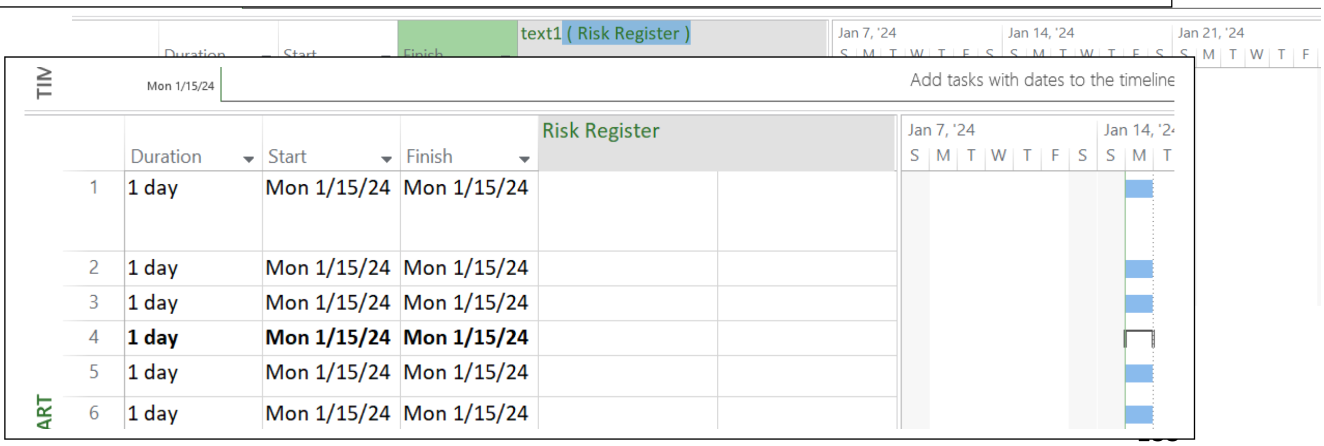
Step 2: Create a New Project File

- In MS Project, click on the 'File' tab in the top-left corner.
- Select 'New' and then choose 'Blank Project.'

Step 3: Insert Risk Register Column

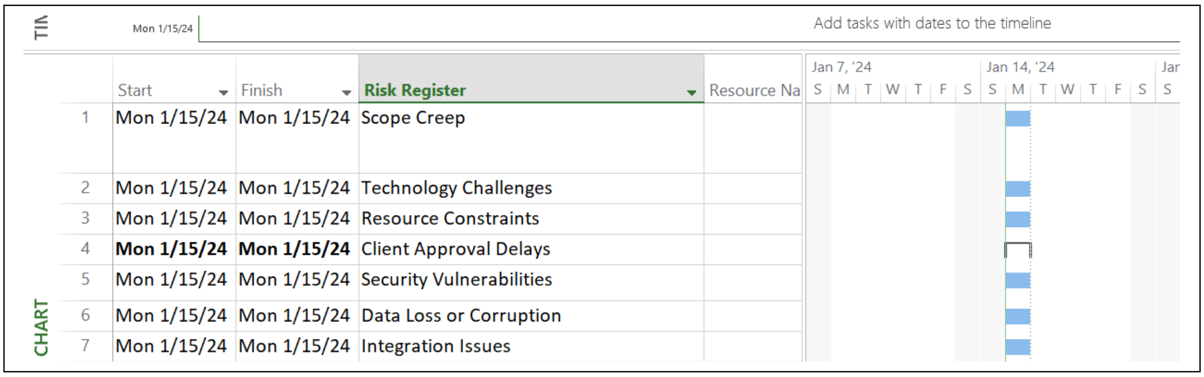
- In the Gantt Chart view, right-click on any column header.
- Select 'Insert Column,' then choose 'Text1' and rename it to 'Risk Register.'





Step 4: List Potential Risks

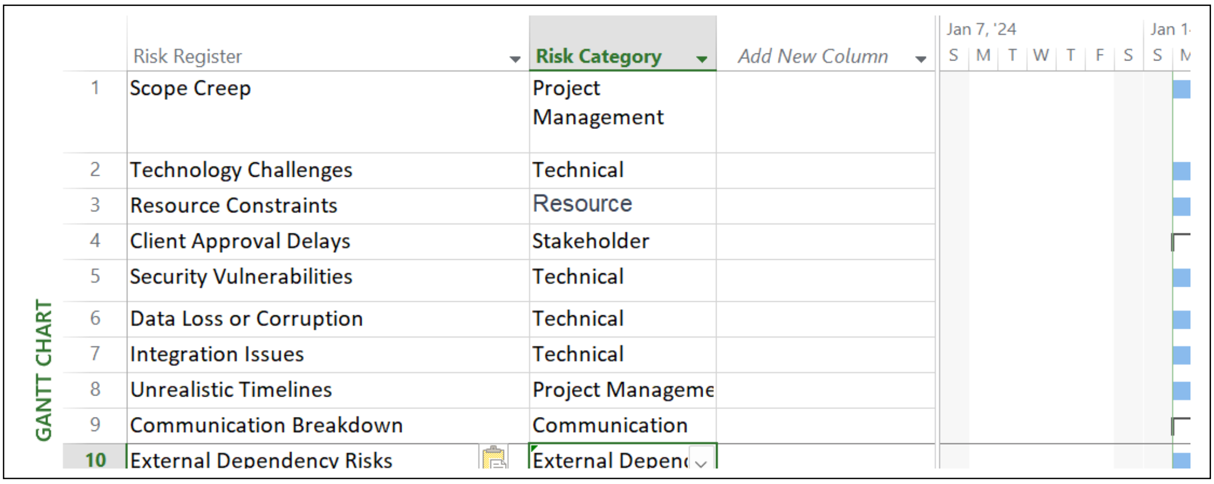
- In the 'Risk Register' column, type in potential risks relevant to your project.



	Start	Finish	Risk Register	Resource Na
1	Mon 1/15/24	Mon 1/15/24	Scope Creep	
2	Mon 1/15/24	Mon 1/15/24	Technology Challenges	
3	Mon 1/15/24	Mon 1/15/24	Resource Constraints	
4	**Mon 1/15/24**	**Mon 1/15/24**	Client Approval Delays	
5	Mon 1/15/24	Mon 1/15/24	Security Vulnerabilities	
6	Mon 1/15/24	Mon 1/15/24	Data Loss or Corruption	
7	Mon 1/15/24	Mon 1/15/24	Integration Issues	

Step 5: Categorize Risks

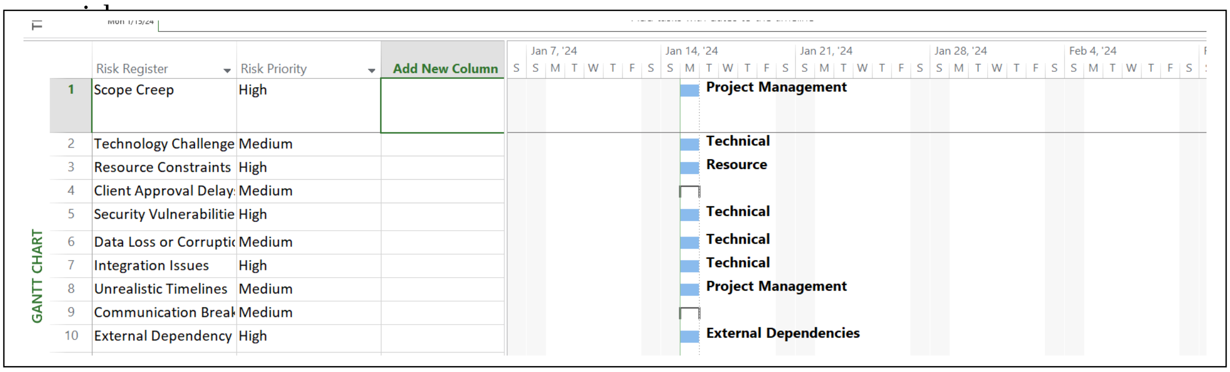
- Add a new column named 'Risk Category' and classify each risk (e.g., technical, external).



	Risk Register	Risk Category	Add New Column
1	Scope Creep	Project Management	
2	Technology Challenges	Technical	
3	Resource Constraints	Resource	
4	Client Approval Delays	Stakeholder	
5	Security Vulnerabilities	Technical	
6	Data Loss or Corruption	Technical	
7	Integration Issues	Technical	
8	Unrealistic Timelines	Project Manageme	
9	Communication Breakdown	Communication	
10	External Dependency Risks	External Depen	

Step 6: Assign Initial Priorities

- Add another column named 'Priority' and assign a level (High, Medium, Low) to each



	Risk Register	Risk Priority	Add New Column	Gantt label
1	Scope Creep	High		Project Management
2	Technology Challenge	Medium		Technical
3	Resource Constraints	High		Resource
4	Client Approval Delay	Medium		
5	Security Vulnerabilitie	High		Technical
6	Data Loss or Corruptic	Medium		Technical
7	Integration Issues	High		Technical
8	Unrealistic Timelines	Medium		Project Management
9	Communication Breal	Medium		
10	External Dependency	High		External Dependencies

Step 7: Discuss and Refine

- Review the risk register with your peers or instructor for feedback.

Step 8: Save the Project File

- Click on 'File' then 'Save As,' choose a location, and save your project file.

Activity 2:

Qualitative Risk Analysis:

Step 1: Open Your Project File

- Click on 'File,' then 'Open,' and select your project file.

Step 2: Create Impact and Probability Columns

- Insert two new columns named 'Impact' and 'Probability' in the task list.

Step 3: Assess Each Risk

- For each risk, enter an impact and probability value on a scale of 1-10.

	Risk Register	Risk Category	Risk Priority	Impact	Probability	Add New Column
/15/24	Scope Creep	Project Management	High	8	7	
/15/24	Technology Challenge	Technical	Medium	6	6	
/15/24	Resource Constraints	Resource	High	7	8	
/15/24	Client Approval Delay:	Stakeholder	Medium	5	6	
/15/24	Security Vulnerabilitie	Technical	High	9	7	
/15/24	Data Loss or Corruptic	Technical	High	9	6	
/15/24	Integration Issues	Technical	Medium	6	5	
/15/24	Unrealistic Timelines	Project Management	High	8	7	
/15/24	Communication Breal	Communication	Medium	5	5	
/15/24	External Dependency	External Dependencie	Medium	7	6	

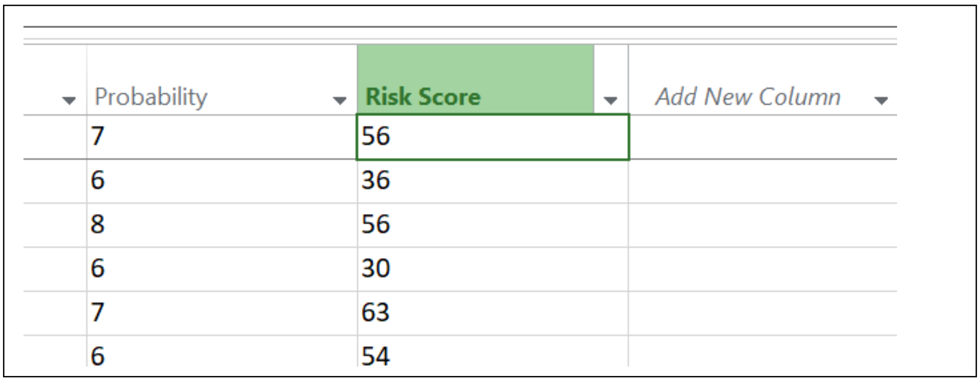
Step 4: Determine Risk Scores

- Add a new column for 'Risk Score' and calculate it by multiplying impact and probability.

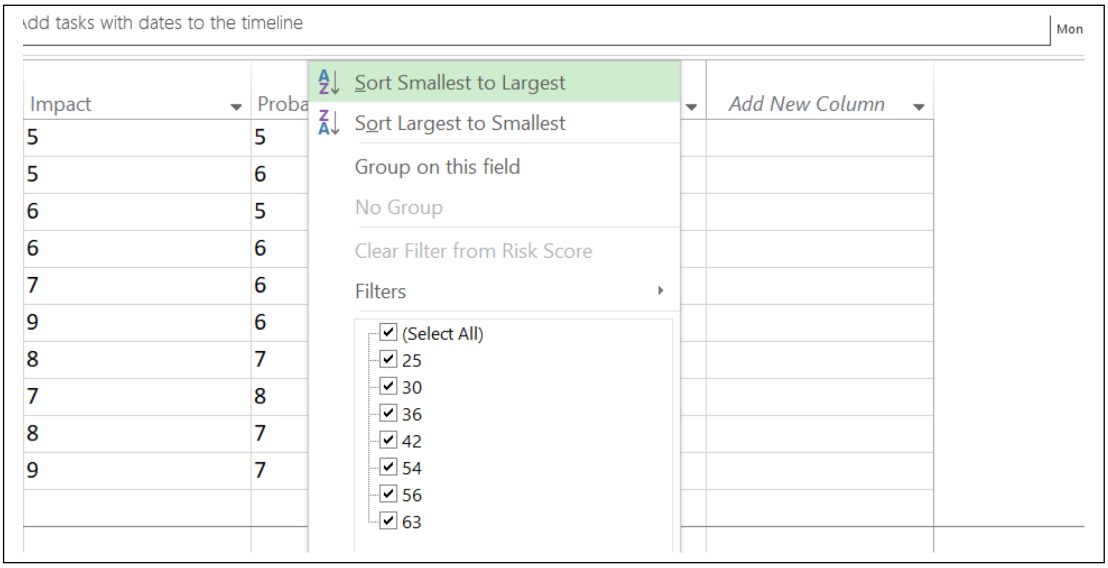
Risk Register	Risk Category	Risk Priority	Impact	Probability	Risk Score	Add New
Scope Creep	Project Management	High	8	7	56	
Technology Challenge	Technical	Medium	6	6	36	
Resource Constraints	Resource	High	7	8	56	
Client Approval Delay:	Stakeholder	Medium	5	6	30	
Security Vulnerabilitie	Technical	High	9	7	63	
Data Loss or Corruptic	Technical	High	9	6	54	
Integration Issues	Technical	Medium	6	5	30	
Unrealistic Timelines	Project Management	High	8	7	56	
Communication Breal	Communication	Medium	5	5	25	
External Dependency	External Dependencie	Medium	7	6	42	

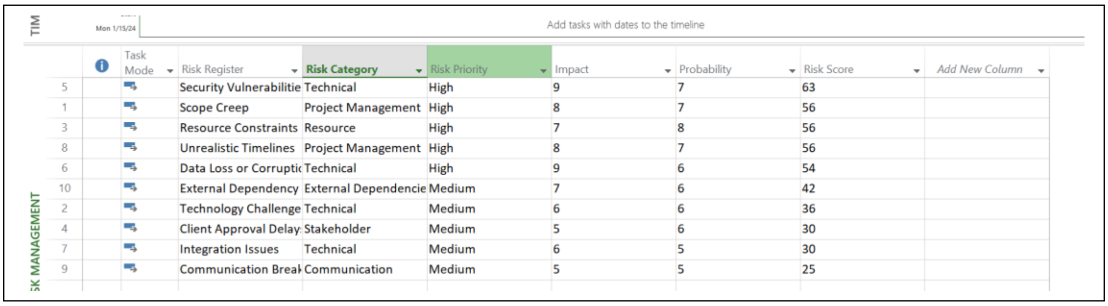
Step 5: Rank the Risks

- Sort the risks in the 'Risk Score' column in descending order.



Probability	Risk Score	Add New Column
7	56	
6	36	
8	56	
6	30	
7	63	
6	54	





	Task Mode	Risk Register	Risk Category	Risk Priority	Impact	Probability	Risk Score	Add New Column
5		Security Vulnerabilitie	Technical	High	9	7	63	
1		Scope Creep	Project Management	High	8	7	56	
3		Resource Constraints	Resource	High	7	8	56	
8		Unrealistic Timelines	Project Management	High	8	7	56	
6		Data Loss or Corruptic	Technical	High	9	6	54	
10		External Dependency	External Dependencie	Medium	7	6	42	
2		Technology Challenge	Technical	Medium	6	6	36	
4		Client Approval Delay	Stakeholder	Medium	5	6	30	
7		Integration Issues	Technical	Medium	6	5	30	
9		Communication Breal	Communication	Medium	5	5	25	

Step 6: Document Assessment

- Write a brief justification for each risk's score.

Step 7: Review with Instructor

- Discuss your risk ranking with the instructor for input.

Step 8: Update and Save

- Make any adjustments based on feedback and save your project file.

Activity 3:

Quantitative Risk Analysis

Step 1: Set Up for Analysis

- Open your project file that contains the ranked risks.

Step 2: Choose Risks for Quantitative Analysis

- Select the highest-ranking risks for further detailed analysis.

Step 3: Develop Risk Scenarios

- Create scenarios for each selected risk in new columns for 'Best Case,' 'Worst Case,' and 'Most Likely Case.'

Task Mode	Risk Register	Risk Category	Risk Priority	Impact	Probability	Risk Score	Best Case	Most Likely	Worst Case	Add New Column
	Security Vulnerabilitie	Technical	High	9	7	63	The security meas	Some minor vulnerab	Significant security vu	
	Scope Creep	Project Management	High	8	7	56	Project scope is we	Some minor changes	Uncontrolled changes	
	Resource Constraints	Resource	High	7	8	56	Adequate resourc	Minor resource const	Severe resource short	
	Unrealistic Timelines	Project Management	High	8	7	56	The project adhere	Minor adjustments to	Unrealistic timelines l	
	Data Loss or Corruptic	Technical	High	9	6	54	Robust backup sys	Regular backups and	Critical data is lost or	

Step 4: Assign Values to Scenarios

- Estimate the cost and time implications for each scenario.

Step 5: Run Simulations

- Use MS Project or external tools to simulate the impact of each scenario on the project.

Step 6: Analyze Results

- Assess how each scenario could affect the project's timeline and budget.

Step 7: Document Findings

- Record the outcomes of your quantitative analysis.

Step 8: Update Project Plan

- Adjust your project plan based on the analysis and save the file.

Activity 4:

Risk Response Planning

Step 1: Review High-Priority Risks

- Open your project file and focus on the high-priority risks.

Step 2: Develop Response Strategies

- For each high-priority risk, brainstorm potential response strategies.

Step 3: Assign Responses in MS Project

- Link each risk to its response strategy in the project plan.

Step 4: Estimate Response Costs and Times

- Assign estimated costs and durations to each response action.

Step 5: Review Plan with Instructor

- Discuss your response plans with the instructor for refinement.

Step 6: Document Strategies

- Record each strategy and its rationale in the project file.

Step 7: Update Risk Register

- Reflect these strategies in your risk register.

		Task Mode	Risk Register	Risk Priority	Impact	Probability	Risk Score	Risk Response
5			Security Vulnerabilitie	High	9	7	63	Mitigation
1			Scope Creep	High	8	7	56	Avoidance
3			Resource Constraints	High	7	8	56	Mitigation
8			Unrealistic Timelines	High	8	7	56	Avoidance
6			Data Loss or Corruptic	High	9	6	54	Avoidance
10			External Dependency	Medium	7	6	42	Transfer
2			Technology Challenge	Medium	6	6	36	Mitigation
4			Client Approval Delay:	Medium	5	6	30	Transfer
7			Integration Issues	Medium	6	5	30	Mitigation

Step 8: Save Your Work

- Save all changes in your project file.

Activity 5:

Risk Monitoring and Control

Step 1: Set Up Risk Monitoring

- Open your project file that includes the risk response strategies.

Step 2: Create Monitoring Columns

- Add new columns for 'Monitoring Status' and 'Control Actions.'

Task Mode	Risk Register	Risk Priority	Impact	Probability	Risk Score	Risk Response	Monitoring Status	Controlling Action	Ad
	Security Vulnerabilitie	High	9	7	63	Mitigation	Ongoing	Implement and updat	
	Scope Creep	High	8	7	56	Avoidance	Ongoing	Change control proce:	
	Resource Constraints	High	7	8	56	Mitigation	Ongoing	Cross-train team mem	
	Unrealistic Timelines	High	8	7	56	Avoidance	Ongoing	Regularly review proje	
	Data Loss or Corruptic	High	9	6	54	Avoidance	Ongoing	Implement and regula	
	External Dependency	Medium	7	6	42	Transfer	Ongoing	Identify and commun	
	Technology Challenge	Medium	6	6	36	Mitigation	Ongoing	Conduct technology a	
	Client Approval Delay:	Medium	5	6	30	Transfer	Ongoing	Collaborate closely wi	
	Integration Issues	Medium	6	5	30	Mitigation	Ongoing	Plan and test integrati	
	Communication Breal	Medium	5	5	25	Mitigation	Ongoing	Establish effective cor	

Step 3: Assign Monitoring Responsibilities

- Allocate team members to monitor each risk.

Step 4: Develop Monitoring Plans

- Outline specific actions for tracking each risk.

Step 5: Implement Control Measures

- Detail actions to be taken if risks materialize.

Step 6: Simulate Monitoring Process

- Role-play the monitoring process for a set of risks.

Step 7: Document Observations

- Record the effectiveness of the monitoring and control measures.

Step 8: Finalize and Save

- Review your work, make necessary adjustments, and save the project file.

Graded Lab Task:

Note: The instructor can design graded lab activities according to the level of difficulty and complexity of the solved lab activities. The lab tasks assigned by the instructor should be evaluated in the same lab.

Task 1:

Develop a risk management plan for a segment of your FYP, identifying and analyzing risks using MS Project. Conduct a qualitative risk analysis for a hypothetical project scenario, documenting your findings in MS Project. Perform a quantitative risk analysis on a critical aspect of your FYP, using MS Project or external tools to simulate risk impacts.

Task 2:

Create a comprehensive risk response and monitoring plan for your FYP, integrating it into your project schedule in MS Project.

Lab 13

Setting-up Resources for Project Human Resource Management

Objective:

The objective of this lab is to make students understand the fundamental concepts of human resource management in a project setting, focusing on team roles, responsibilities, and skills. It also Familiarize students with MS Project Interface which helps them learn to navigate through the MS Project interface, focusing on features relevant to HR management.

Activity Outcomes:

After successful completion of this lab, the students will be able to:

1. Apply theoretical HR concepts practically using MS Project.
2. Gain hands-on experience with MS Project, enhancing software proficiency.
3. Develop skills to manage and allocate human resources effectively in project settings.
4. Analyze and improve team performance through MS Project's reporting tools.
5. Develop problem-solving skills by tackling real-world HR management scenarios in the lab.

Instructor Note:

As pre lab activity read chapter 9 “Project Human resource Management” from PMBOK. Also, ensure that students have prior basic knowledge of MS Project. Encourage active participation and problem-solving during the lab session.

Useful Concepts

- Project Human Resource Management is a critical function within project management that focuses on effectively organizing, managing, and leading the project team. It involves processes for planning, acquiring, developing, and managing the human resources necessary to complete a project successfully. Key activities include identifying and documenting project roles and responsibilities, recruiting team members with the required skills, and creating a project team structure. It also encompasses training and development initiatives to enhance team competencies, performance evaluations, and implementing strategies to motivate and retain talent. Effective human resource management ensures that the right people are in the right roles, fostering a collaborative and productive work environment. By aligning the skills and motivations of the team with the project goals, HR Management contributes significantly to achieving project objectives, maintaining team morale, and ensuring a smooth project execution.

2) **Solved Lab Activities**

Sr.No	*Allocated Time*	*Level of Complexity*	*CLO Mapping*
1	*50*	*Medium*	*CLO-5*
2	*50*	*Medium*	*CLO-5*
3	*30*	*Medium*	*CLO-5*

Activity 1:

How to allocate Resources in MS Project

<u>Solution</u>

Step 1: Launch MS Project

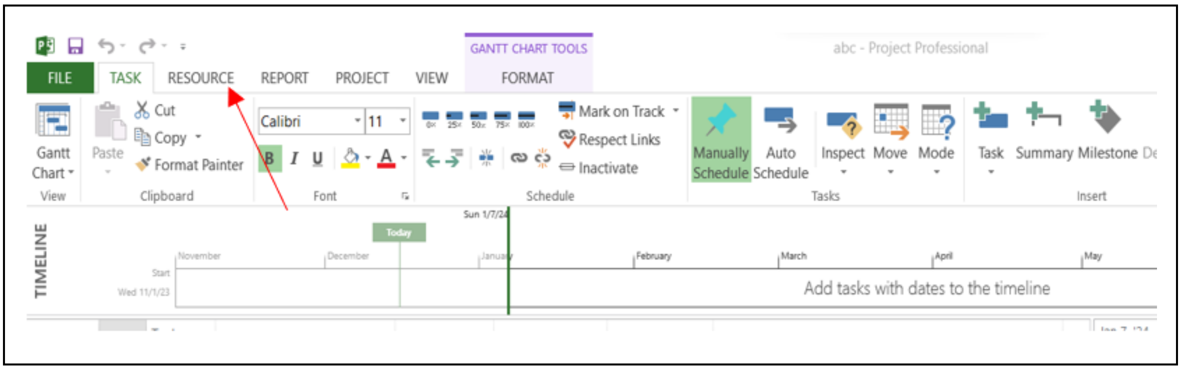
- On your laptop, click on the Windows Start menu or search bar.
- Type "Microsoft Project" and select it from the search results to open the application.

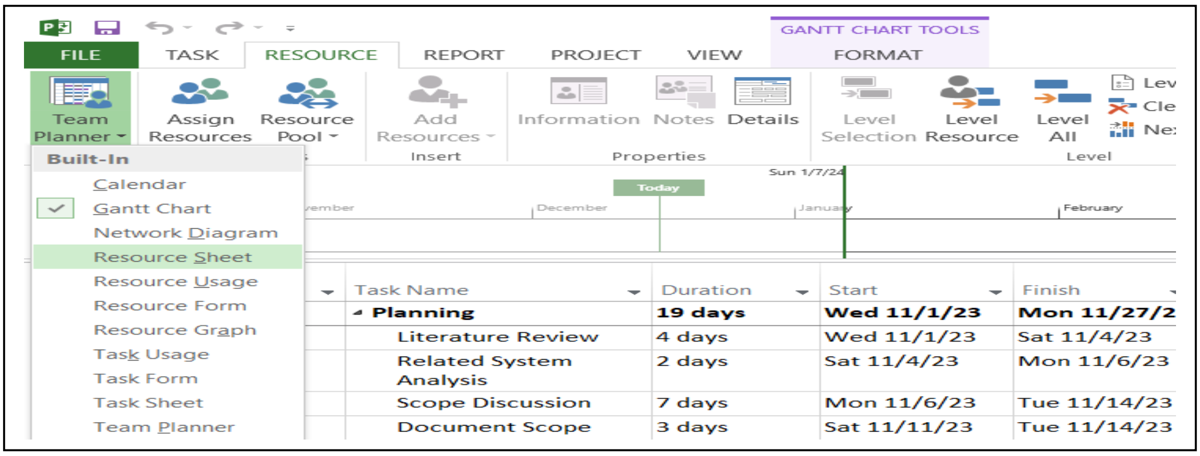
Step 2: Create a New Project

- In MS Project, go to the "File" menu and select "New."
- Choose "Blank Project" to start with a clean slate.

Step 3: Define Project Resources

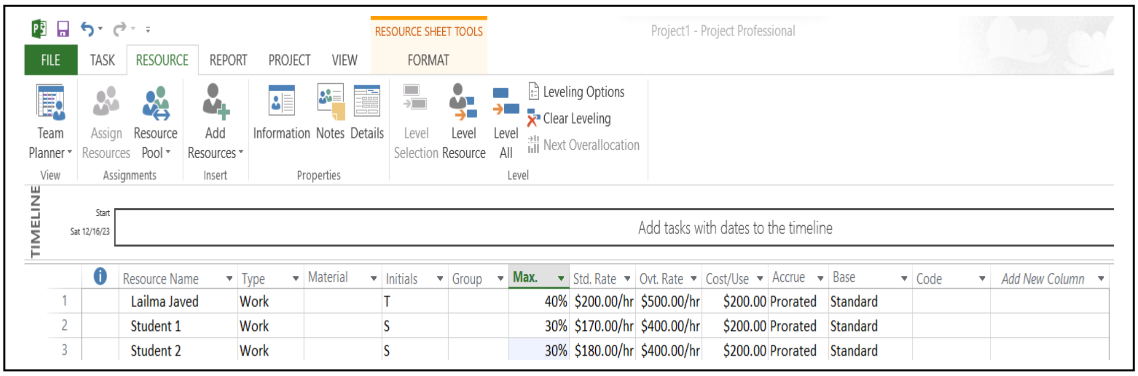
- Go to the "Resource" tab on the top ribbon.





- Click on "Resource Sheet" in the Team Planner section.

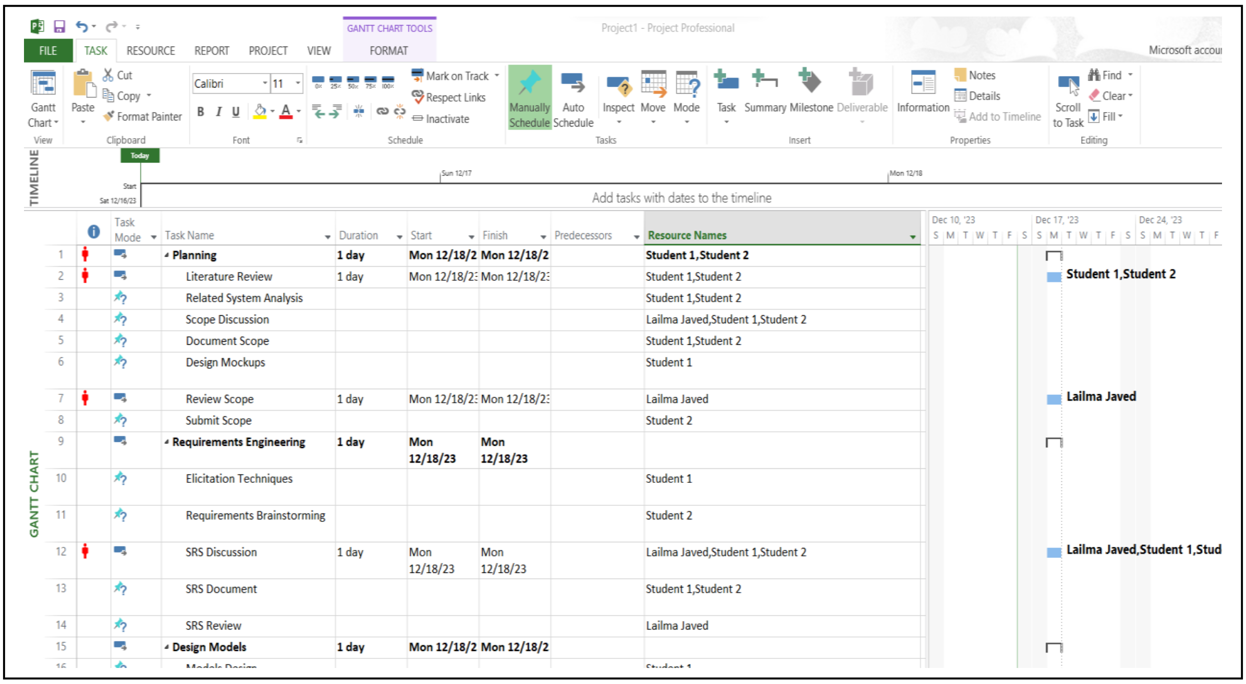
- Enter the names of the team members in the "Resource Name" column.
- Specify the type (usually "Work"), their availability, and other relevant details.



Step 4: Assign Resources to Tasks

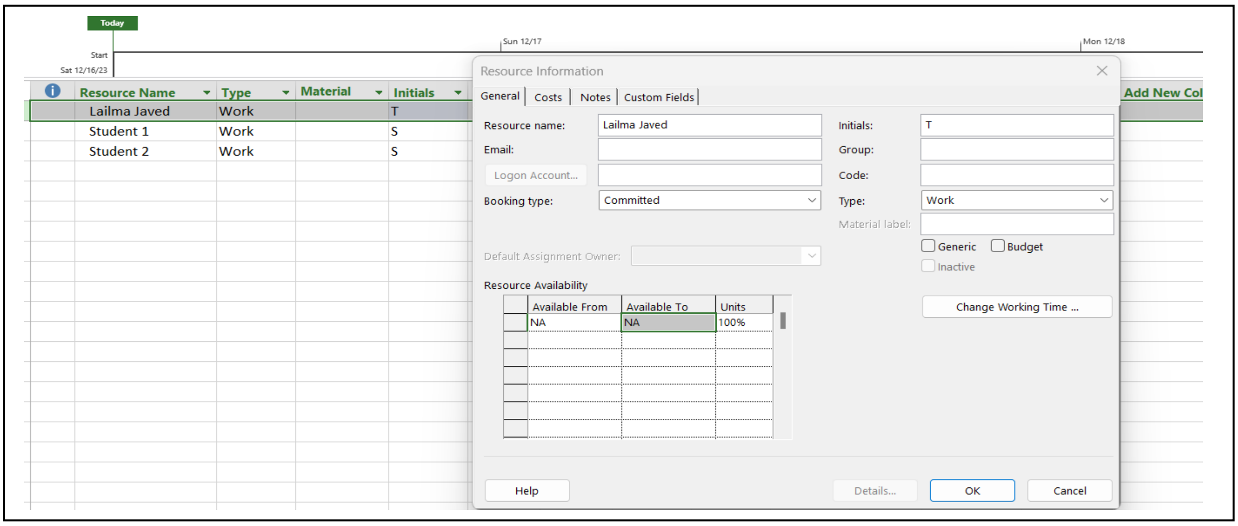
- Switch to the "Gantt Chart" view under the "Task" tab.
- Enter the tasks required for your project in the "Task Name" column.

- In the "Resource Names" column, assign the appropriate team members to each task.

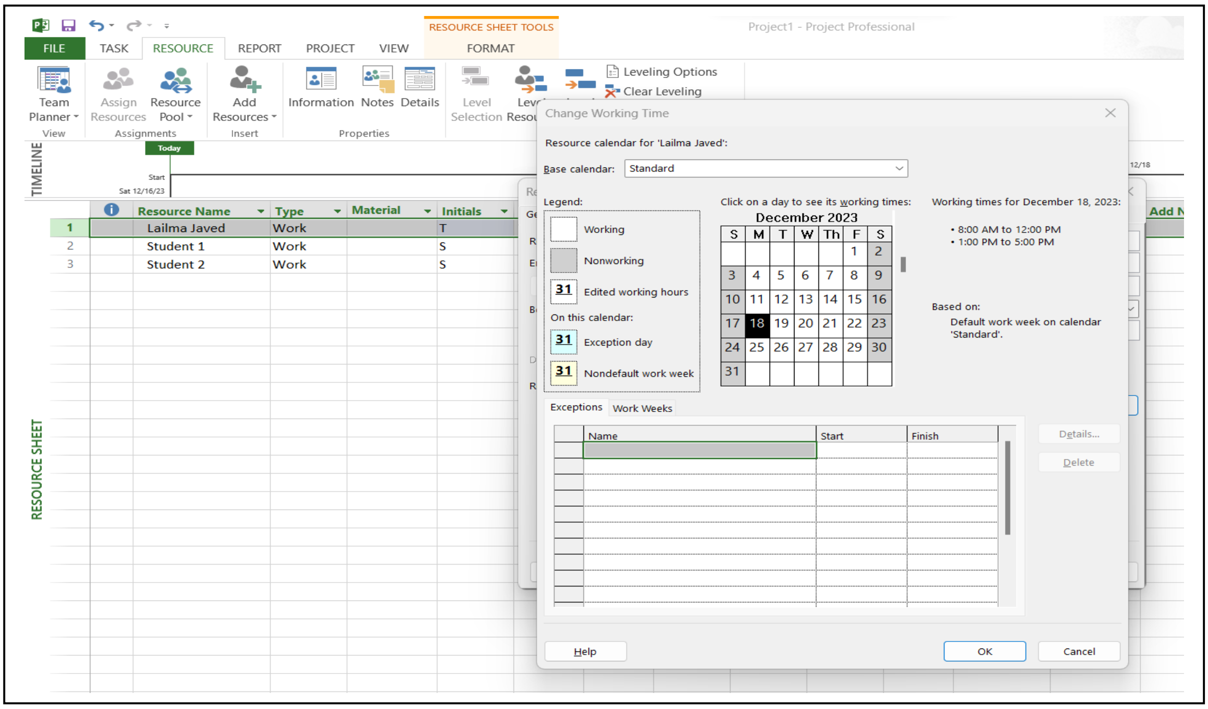


Step 5: Adjust Resource Calendars

- Go back to the "Resource Sheet" view.
- Double-click on a resource name to open the "Resource Information" dialog.

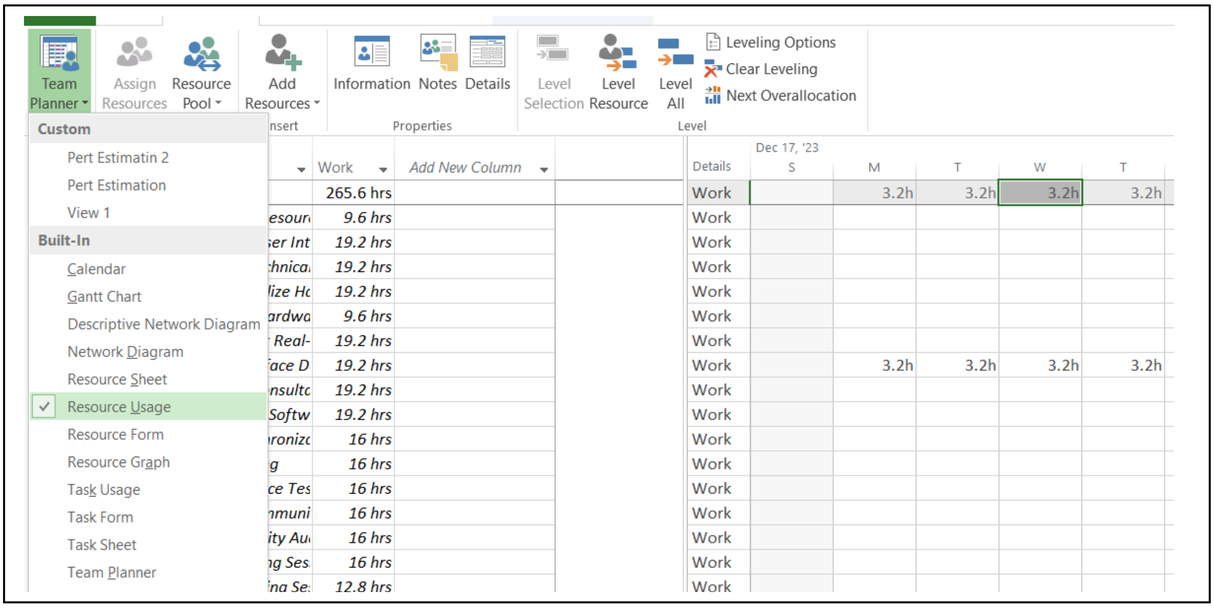


- Under the "General" tab, click on the "Change Working Time" button to set custom working times and holidays.

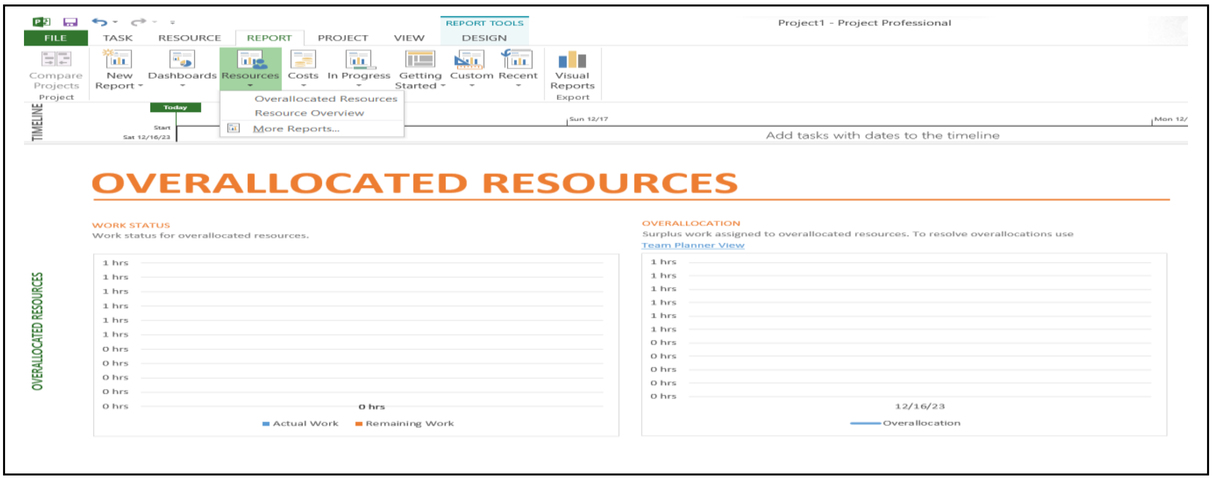


Step 6: Analyze Resource Allocation

- Switch to the "Resource Usage" view under the "View" tab.
- Review the allocation for each resource, looking for over-allocations or imbalances.



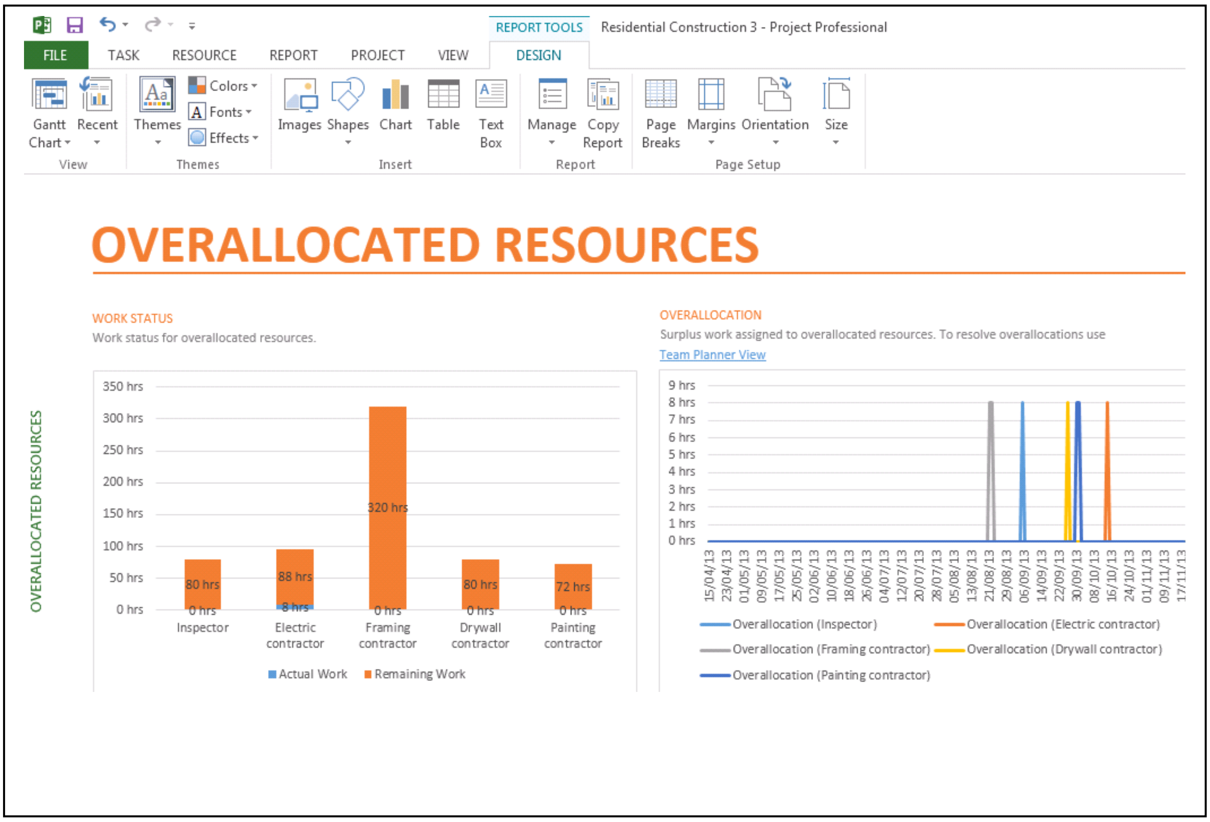
Step 7: Balance Resource Load



- Go on report, click on resource view the overallocated resource

If over-allocation is detected, go back to the "Gantt Chart" view.

- Adjust task assignments or durations to evenly distribute the workload



Step 8: Review and Document

- Discuss the resource allocation in the project with the instructor or peers.

- Make notes of any adjustments and the rationale behind them.

Step 9: Save Your Work

- Go to the "File" menu and select "Save As."
- Choose a location and save your project file.

Activity 2:

"Team Performance Tracking Simulation"

Solution

Step 1: Open MS Project

- Access MS Project from your computer's search bar or Start menu.

Step 2: Set Up Project Tracking

- In MS Project, open your existing project file or create a new one.
- Go to the "Project" tab and click on "Project Information" to set the start date of your project.

Step 3: Enter Project Tasks

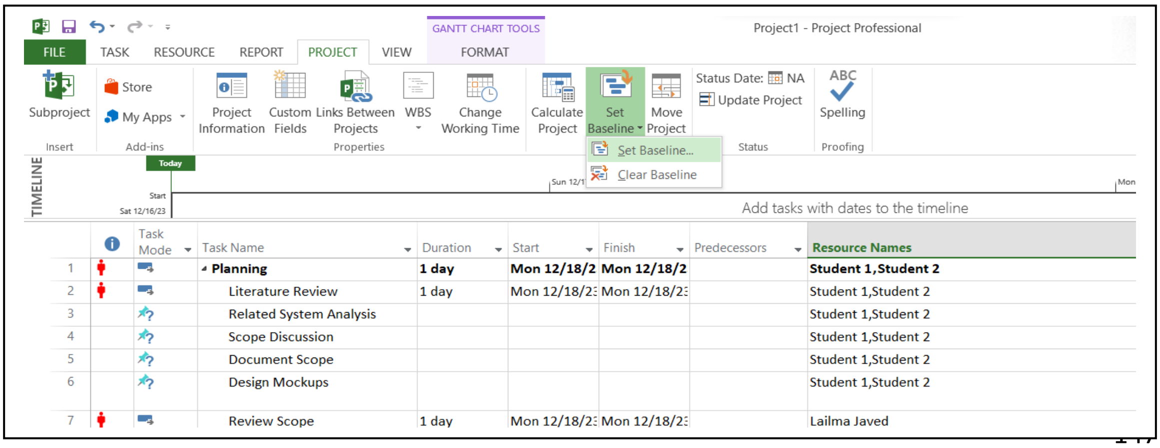
- Switch to the "Gantt Chart" view.
- Enter all the tasks related to your project in the "Task Name" column.

Step 4: Assign Resources to Tasks

- Assign team members to each task in the "Resource Names" column, as previously detailed in the Resource Allocation Exercise.

Step 5: Set Baseline for Tracking

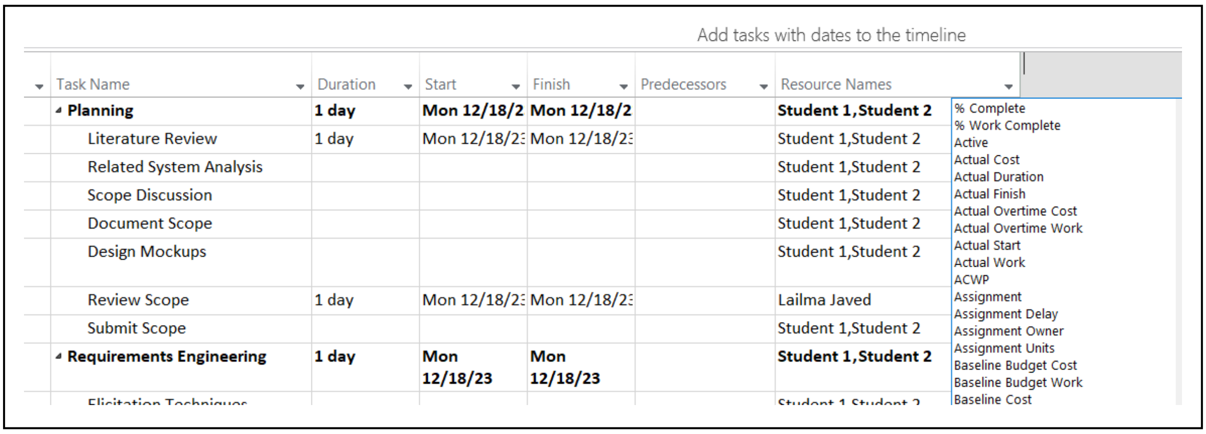
- Go to the "Project" tab.
- Click on "Set Baseline" and then "Set Baseline" again in the dropdown. Choose



"Entire Project" to set your baseline.

Step 6: Track Progress

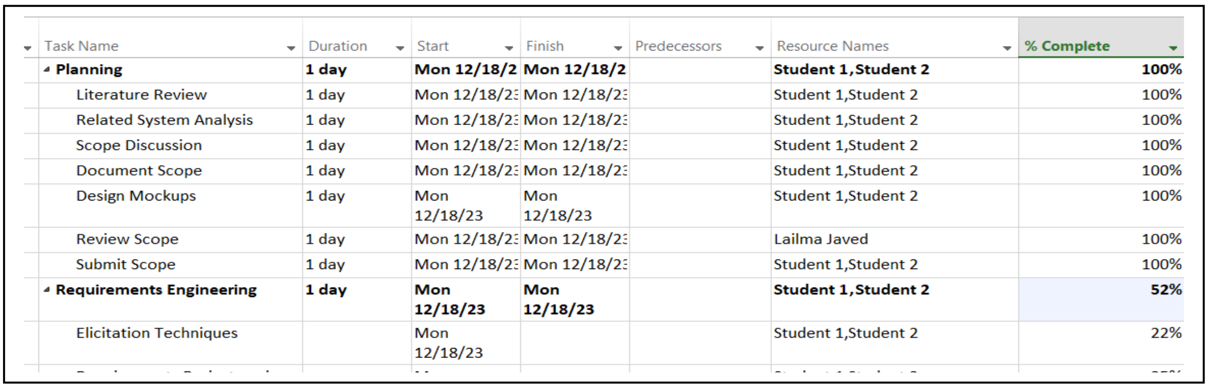
- As your project progresses, update the status of each task.
- Add new column name % Complete .
- Enter the percentage of work completed for each task in the "% Complete" column.



Add tasks with dates to the timeline

Task Name	Duration	Start	Finish	Predecessors	Resource Names
Planning	**1 day**	**Mon 12/18/2**	**Mon 12/18/2**		**Student 1,Student 2**
Literature Review	1 day	Mon 12/18/2:	Mon 12/18/2:		Student 1,Student 2
Related System Analysis					Student 1,Student 2
Scope Discussion					Student 1,Student 2
Document Scope					Student 1,Student 2
Design Mockups					Student 1,Student 2
Review Scope	1 day	Mon 12/18/2:	Mon 12/18/2:		Lailma Javed
Submit Scope					Student 1,Student 2
Requirements Engineering	**1 day**	**Mon 12/18/23**	**Mon 12/18/23**		**Student 1,Student 2**
Elicitation Techniques					Student 1,Student 2

After Selecting:



Task Name	Duration	Start	Finish	Predecessors	Resource Names	% Complete
Planning	**1 day**	**Mon 12/18/2**	**Mon 12/18/2**		**Student 1,Student 2**	**100%**
Literature Review	1 day	Mon 12/18/2:	Mon 12/18/2:		Student 1,Student 2	100%
Related System Analysis	1 day	Mon 12/18/2:	Mon 12/18/2:		Student 1,Student 2	100%
Scope Discussion	1 day	Mon 12/18/2:	Mon 12/18/2:		Student 1,Student 2	100%
Document Scope	1 day	Mon 12/18/2:	Mon 12/18/2:		Student 1,Student 2	100%
Design Mockups	1 day	Mon 12/18/23	Mon 12/18/23		Student 1,Student 2	100%
Review Scope	1 day	Mon 12/18/2:	Mon 12/18/2:		Lailma Javed	100%
Submit Scope	1 day	Mon 12/18/2:	Mon 12/18/2:		Student 1,Student 2	100%
Requirements Engineering	**1 day**	**Mon 12/18/23**	**Mon 12/18/23**		**Student 1,Student 2**	**52%**
Elicitation Techniques		Mon 12/18/23			Student 1,Student 2	22%

Step 7: Analyze Performance

- Use the "Tracking Gantt" view to compare planned vs. actual progress.
- Identify any delays or ahead-of-schedule tasks.

Step 8: Document Observations

- Note any discrepancies between planned and actual progress.
- Discuss potential reasons and corrective actions with peers or the instructor.

Step 9: Save Your Tracking Data

- Regularly save your project file to avoid losing any data.

Activity 3

“Conflict Resolution Scenario Analysis”

<u>Solution</u>

Step 1: Open MS Project

- Launch MS Project from your computer.

Step 2: Create a New Project for Simulation

- Go to "File" > "New" and select "Blank Project."

Step 3: Enter Project Tasks

- Add hypothetical tasks that could lead to team conflicts in the "Task Name" column.

	Task Mode	Task Name
		Ambiguous Project Goals
		Unequal Distribution of \
		Tight and Unrealistic Dea
		Lack of Clear Communica
		Insufficient Resources
		Differing Work Styles
		Inadequate Training
		Role Ambiguity
		Conflicting Priorities
		Poorly Defined Responsi
		Unresolved Past Issues
		Disagreements on Projec
		Lack of Recognition for Ir
		Unclear Decision-Making
		Limited Access to Inform
		Incompatible Technolog

Step 4: Define Team Resources

- Add team members as resources with varying skills and availability.

		Task Mode	Task Name	Resource Names	Skills	Availability	Add New Column
1			Ambiguous Project Goals	Alex	Project Management, Leadership,	Full-time	
2			Unequal Distribution of \	Emily	Software Developmer	Part-time (20 hours/w	
3			Tight and Unrealistic Dea	Jordan	Graphic Design, Adob	Full-time	
4			Lack of Clear Communica	Ryan	Data Analysis, Excel, S	Full-time	
5			Insufficient Resources	Morgan	Marketing, Social Mec	Part-time (15 hours/w	
6			Differing Work Styles	Taylor	Customer Support, Cc	Full-time	
7			Inadequate Training	Casey	Quality Assurance, Te	Part-time (25 hours/w	
8			Role Ambiguity	Riley	Copywriting, Editing, (	Full-time	

T CHART

Step 5: Simulate Conflict Scenarios

- Create scenarios where resources are over-allocated, indicating potential conflicts.

Step 6: Apply Conflict Resolution Techniques

- Adjust task assignments and durations to resolve conflicts.
- Discuss alternative conflict resolution strategies.

Step 7: Evaluate the Outcome

- Assess the effectiveness of conflict resolution strategies.

Step 8: Document the Process

- Make detailed notes on how each conflict was resolved and the rationale behind the decisions.

Step 9: Save and Share Your Work

- Save your project file and be prepared to share your findings in the next class.

Graded Lab Task:

Note: The instructor can design graded lab activities according to the level of difficulty and complexity of the solved lab activities. The lab tasks assigned by the instructor should be evaluated in the same lab.

Lab Task 1

Develop a Resource Management Plan for Your FYP.

Create a detailed resource management plan for your Final Year Project using MS Project.

Requirements

1. Identify all the tasks required for your FYP.
2. Define the resources (team members, equipment, etc.) needed for these tasks.
3. Assign these resources to tasks, ensuring a balanced workload.
4. Create a custom calendar for each resource, considering their availability and other commitments.
5. Prepare a report summarizing your resource management plan, including how it will assist in the successful completion of your FYP.

- Submission: Submit the MS Project file along with the resource management report.